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Fused imidazopyridines as reversible inhibitors of Brutons Tyrosine Kinase (BTK)

Abstract

The present invention relates to fused imidazo pyridine compounds, and pharmaceutically acceptable compositions thereof, useful as BTK inhibitors.

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Background/Summary

RELATED APPLICATIONS (1) This application is a US national stage application of PCT international application PCT/EP19/72660, filed on Aug. 26, 2019, which claims the benefit of U.S. Provisional Application 62/723,648, filed Aug. 28, 2018. The entire contents of the aforementioned applications are incorporated herein by reference.

TECHNICAL FIELD OF THE INVENTION

(1) The present invention relates to fused imidazo pyridine compounds that are useful as inhibitors of Bruton's Tyrosine Kinase (BTK). The invention also provides pharmaceutically acceptable compositions comprising compounds of the present invention and methods of using said compositions in the treatment of various disorders.

BACKGROUND OF THE INVENTION

(2) Protein kinases constitute one of the largest families of human enzymes and regulate many different signaling processes by adding phosphate groups to proteins (T. Hunter, *Cell* 1987 50:823-829). Specifically, tyrosine kinases phosphorylate proteins on the phenolic moiety of tyrosine residues. The tyrosine kinase family includes members that control cell growth, migration, and differentiation. Abnormal kinase activity has been implicated in a variety of human diseases including cancers, autoimmune and inflammatory diseases. Since protein kinases are among the key regulators of cell signaling, they provide a target to modulate cellular function with small molecular kinase inhibitors and thus make good drug targets. In addition to treatment of kinase-mediated disease processes, selective and efficacious inhibitors of kinase activity are also useful for investigation of cell signaling processes and identification of other cellular targets of therapeutic interest.

(3) There is good evidence that B-cells play a key role in the pathogenesis of autoimmune and/or inflammatory disease. Protein-based therapeutics that deplete B cells such as Rituxan are effective against autoantibody-driven inflammatory diseases such as rheumatoid arthritis (Rastetter et al. *Annu Rev Med* 2004 55:477). Therefore inhibitors of the protein kinases that play a role in B-cell activation should be useful therapeutics for B-cell mediated disease pathology, such as autoantibody production.

(4) Signaling through the B-cell receptor (BCR) controls a range of B-cell responses including proliferation and differentiation into mature antibody producing cells. The BCR is a key regulatory point for B-cell activity and aberrant signaling can cause deregulated B-cell proliferation and formation of pathogenic autoantibodies that lead to multiple autoimmune and/or inflammatory diseases. Bruton's Tyrosine Kinase (BTK) is a non-BCR associated kinase that is membrane proximal and immediately downstream from BCR. Lack of BTK has been shown to block BCR signaling and therefore inhibition of BTK could be a useful therapeutic approach to block B-cell mediated disease processes. Also, BTK has been reported to play a role in apoptosis (Islam and Smith *Immunol. Rev.* 2000 178:49) and thus BTK inhibitors would be useful for the treatment of certain B-cell lymphomas and leukemias (Feldhahn et al. *J. Exp. Med.* 2005 201:1837).

(5) BTK is a member of the Tec family of tyrosine kinases, and has been shown to be a critical regulator of early B-cell development and mature B-cell activation and survival (Khan et al. *Immunity* 1995 3:283; Ellmeier et al. *J. Exp. Med.* 2000 192:1611). Mutation of BTK in humans leads to the condition X-linked agammaglobulinemia (XLA) (reviewed in Rosen et al. *New Eng. J. Med.* 1995 333:431 and Lindvall et al. *Immunol. Rev.* 2005 203:200). These patients are immune-compromised and show impaired maturation of B-cells, decreased immunoglobulin and peripheral B-cell levels, diminished T-cell independent immune responses as well as attenuated calcium mobilization following BCR stimulation.

(6) Evidence for a role for BTK in autoimmune and inflammatory diseases has also been provided by BTK-deficient mouse models. In preclinical murine models of systemic lupus erythematosus (SLE), BTK-deficient mice show marked amelioration of disease progression. In addition, BTK-deficient mice are resistant to collagen-induced arthritis (Jansson and Holmdahl *Clin. Exp. Immunol.* 1993 94:459). A selective BTK inhibitor has demonstrated dose-dependent efficacy in a mouse arthritis model (Z. Pan et al., *Chem. Med Chem.* 2007 2:58-61).

(7) BTK is also expressed by cells other than B-cells that may be involved in disease processes. BTK is key component of Fc-gamma signaling in myeloid cells. For example, BTK is expressed by

mast cells and BTK-deficient bone marrow derived mast cells demonstrate impaired antigen induced degranulation (Iwaki et al. J. Biol. Chem. 2005 280:40261). This shows BTK could be useful to treat pathological mast cells responses such as allergy and asthma. Also monocytes from XLA patients, in which BTK activity is absent, show decreased TNF alpha production following stimulation (Horwood et al. J Exp Med 197:1603, 2003). Therefore TNF alpha mediated inflammation could be modulated by small molecular BTK inhibitors.

SUMMARY OF THE INVENTION

(8) It has now been found that compounds of this invention, and pharmaceutically acceptable compositions thereof, are effective as inhibitors of BTK. Such compounds have general formula I:

(9) ##STR00001##

or a tautomer, or a pharmaceutically acceptable salt, hydrate, solvate, or prodrug thereof, wherein each of X, Y, ring Z, R.sup.1, R.sup.a, n, and p, is as defined and described in embodiments herein.

(10) Compounds of the present invention, and pharmaceutically acceptable compositions thereof, are useful for treating a variety of diseases, disorders or conditions, associated with BTK. Such diseases, disorders, or conditions include those described herein.

Description

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

1. General Description of Compounds of the Invention

(1) In certain aspects, the present invention provides for inhibitors of BTK. In some embodiments, such compounds include those of the formulae described herein, or a pharmaceutically acceptable salt thereof, wherein each variable is as defined and described herein.

2. Compounds and Definitions

(2) Compounds of this invention include those described generally above, and are further illustrated by the classes, subclasses, and species disclosed herein. As used herein, the following definitions shall apply unless otherwise indicated. For purposes of this invention, the chemical elements are identified in accordance with the Periodic Table of the Elements, CAS version, Handbook of Chemistry and Physics, 75th Ed. Additionally, general principles of organic chemistry are described in "Organic Chemistry", Thomas Sorrell, University Science Books, Sausalito: 1999, and "March's Advanced Organic Chemistry", 5th Ed., Ed.: Smith, M. B. and March, J., John Wiley & Sons, New York. 2001, the entire contents of which are hereby incorporated by reference.

(3) The term "aliphatic" or "aliphatic group", as used herein, means a straight-chain (i.e., unbranched) or branched, substituted or unsubstituted hydrocarbon chain that is completely saturated or that contains one or more units of unsaturation, or a monocyclic hydrocarbon or bicyclic hydrocarbon that is completely saturated or that contains one or more units of unsaturation, but which is not aromatic (also referred to herein as "carbocycle" "cycloaliphatic" or "cycloalkyl"), that has a single point of attachment to the rest of the molecule. Unless otherwise specified, aliphatic groups contain 1-6 aliphatic carbon atoms. In some embodiments, aliphatic groups contain 1-5 aliphatic carbon atoms. In other embodiments, aliphatic groups contain 1-4 aliphatic carbon atoms. In still other embodiments, aliphatic groups contain 1-3 aliphatic carbon atoms, and in yet other embodiments, aliphatic groups contain 1-2 aliphatic carbon atoms. In some embodiments, "cycloaliphatic" (or "carbocycle" or "cycloalkyl") refers to a monocyclic C₃-C₆ hydrocarbon that is completely saturated or that contains one or more units of unsaturation, but which is not aromatic, that has a single point of attachment to the rest of the molecule. Exemplary aliphatic groups are linear or branched, substituted or unsubstituted C₁-C₈ alkyl, C₂-C₈ alkenyl, C₂-C₈ alkynyl groups and hybrids thereof such as (cycloalkyl)alkyl, (cycloalkenyl)alkyl or (cycloalkyl)alkenyl.

(4) The term "lower alkyl" refers to a C₁-4 straight or branched alkyl group. Exemplary lower

alkyl groups are methyl, ethyl, propyl, isopropyl, butyl, isobutyl, and tert-butyl.

(5) The term “lower haloalkyl” refers to a C.sub.1-4 straight or branched alkyl group that is substituted with one or more halogen atoms.

(6) The term “heteroatom” means one or more of oxygen, sulfur, nitrogen, or phosphorus (including, any oxidized form of nitrogen, sulfur, or phosphorus; the quaternized form of any basic nitrogen or; a substitutable nitrogen of a heterocyclic ring, for example N (as in 3,4-dihydro-2H-pyrrolyl), NH (as in pyrrolidinyl) or NR^{sup.+} (as in N-substituted pyrrolidinyl)).

(7) The term “unsaturated”, as used herein, means that a moiety has one or more units of unsaturation.

(8) As used herein, the term “bivalent C.sub.1-8 (or C.sub.1-6) saturated or unsaturated, straight or branched, hydrocarbon chain”, refers to bivalent alkylene, alkenylene, and alkynylene chains that are straight or branched as defined herein.

(9) The term “alkylene” refers to a bivalent alkyl group. An “alkylene chain” is a polymethylene group, i.e., —(CH_{sub.2})_{sub.n}—, wherein n is a positive integer, preferably from 1 to 6, from 1 to 4, from 1 to 3, from 1 to 2, or from 2 to 3. A substituted alkylene chain is a polymethylene group in which one or more methylene hydrogen atoms are replaced with a substituent. Suitable substituents include those described below for a substituted aliphatic group.

(10) The term “alkenylene” refers to a bivalent alkenyl group. A substituted alkenylene chain is a polymethylene group containing at least one double bond in which one or more hydrogen atoms are replaced with a substituent. Suitable substituents include those described below for a substituted aliphatic group.

(11) The term “halogen” means F, Cl, Br, or I.

(12) The term “aryl” used alone or as part of a larger moiety as in “aralkyl”, “aralkoxy”, or “aryloxyalkyl”, refers to monocyclic and bicyclic ring systems having a total of five to fourteen ring members, wherein at least one ring in the system is aromatic and wherein each ring in the system contains three to seven ring members. The term “aryl” is used interchangeably with the term “aryl ring”. In certain embodiments of the present invention, “aryl” refers to an aromatic ring system. Exemplary aryl groups are phenyl, biphenyl, naphthyl, anthracyl and the like, which optionally includes one or more substituents. Also included within the scope of the term “aryl”, as it is used herein, is a group in which an aromatic ring is fused to one or more non-aromatic rings, such as indanyl, phthalimidyl, naphthimidyl, phenanthridinyl, or tetrahydronaphthyl, and the like.

(13) The terms “heteroaryl” and “heteroar-”, used alone or as part of a larger moiety, e.g., “heteroaralkyl”, or “heteroaralkoxy”, refer to groups having 5 to 10 ring atoms, preferably 5, 6, or 9 ring atoms; having 6, 10, or 14 π electrons shared in a cyclic array; and having, in addition to carbon atoms, from one to five heteroatoms. The term “heteroatom” refers to nitrogen, oxygen, or sulfur, and includes any oxidized form of nitrogen or sulfur, and any quaternized form of a basic nitrogen. Heteroaryl groups include, without limitation, thienyl, furanyl, pyrrolyl, imidazolyl, pyrazolyl, triazolyl, tetrazolyl, oxazolyl, isoxazolyl, oxadiazolyl, thiazolyl, isothiazolyl, thiadiazolyl, pyridyl, pyridazinyl, pyrimidinyl, pyrazinyl, indolizinyl, purinyl, naphthyridinyl, and pteridinyl. The terms “heteroaryl” and “heteroar-”, as used herein, also include groups in which a heteroaromatic ring is fused to one or more aryl, cycloaliphatic, or heterocyclyl rings, where the radical or point of attachment is on the heteroaromatic ring. Nonlimiting examples include indolyl, isoindolyl, benzothienyl, benzofuranyl, dibenzofuranyl, indazolyl, benzimidazolyl, benzthiazolyl, quinolyl, isoquinolyl, cinnolinyl, phthalazinyl, quinazolinyl, quinoxalinyl, 4H-quinolizinyl, carbazolyl, acridinyl, phenazinyl, phenothiazinyl, phenoxazinyl, tetrahydroquinolizinyl, tetrahydroisoquinolizinyl, and pyrido[2,3-b]-1,4-oxazin-3(4H)-one. A heteroaryl group is optionally mono- or bicyclic. The term “heteroaryl” is used interchangeably with the terms “heteroaryl ring”, “heteroaryl group”, or “heteroaromatic”, any of which terms include rings that are optionally substituted. The term “heteroaralkyl” refers to an alkyl group substituted by a heteroaryl, wherein the alkyl and heteroaryl portions independently are optionally substituted.

(14) As used herein, the terms “heterocycle”, “heterocyclyl”, “heterocyclic radical”, and “heterocyclic ring” are used interchangeably and refer to a stable 5- to 7-membered monocyclic or 7-10-membered bicyclic heterocyclic moiety that is either saturated or partially unsaturated, and having, in addition to carbon atoms, one or more, preferably one to four, heteroatoms, as defined above. When used in reference to a ring atom of a heterocycle, the term “nitrogen” includes a substituted nitrogen. As an example, in a saturated or partially unsaturated ring having 0-3 heteroatoms selected from oxygen, sulfur or nitrogen, the nitrogen is N (as in 3,4-dihydro-2H-pyrrolyl), NH (as in pyrrolidinyl), or .sup.+NR (as in N-substituted pyrrolidinyl).

(15) A heterocyclic ring can be attached to its pendant group at any heteroatom or carbon atom that results in a stable structure and any of the ring atoms can be optionally substituted. Examples of such saturated or partially unsaturated heterocyclic radicals include, without limitation, tetrahydrofuranyl, tetrahydrothiophenyl pyrrolidinyl, piperidinyl, pyrrolinyl, tetrahydroquinolinyl, tetrahydroisoquinolinyl, decahydroquinolinyl, oxazolidinyl, piperazinyl, dioxanyl, dioxolanyl, diazepinyl, oxazepinyl, thiazepinyl, morpholinyl, and quinuclidinyl. The terms “heterocycle”, “heterocyclyl”, “heterocyclyl ring”, “heterocyclic group”, “heterocyclic moiety”, and “heterocyclic radical”, are used interchangeably herein, and also include groups in which a heterocyclyl ring is fused to one or more aryl, heteroaryl, or cycloaliphatic rings, such as indolinyl, 3H-indolyl, chromanyl, phenanthridinyl, or tetrahydroquinolinyl, where the radical or point of attachment is on the heterocyclyl ring. A heterocyclyl group is optionally mono- or bicyclic. The term “heterocyclylalkyl” refers to an alkyl group substituted by a heterocyclyl, wherein the alkyl and heterocyclyl portions independently are optionally substituted.

(16) As used herein, the term “partially unsaturated” refers to a ring moiety that includes at least one double or triple bond. The term “partially unsaturated” is intended to encompass rings having multiple sites of unsaturation, but is not intended to include aryl or heteroaryl moieties, as herein defined.

(17) As described herein, certain compounds of the invention contain “optionally substituted” moieties. In general, the term “substituted”, whether preceded by the term “optionally” or not, means that one or more hydrogens of the designated moiety are replaced with a suitable substituent. “Substituted” applies to one or more hydrogens that are either explicit or implicit from the structure (e.g.,

(18) ##STR00002##

refers to at least

(19) ##STR00003##

refers to at least

(20) ##STR00004##

Unless otherwise indicated, an “optionally substituted” group has a suitable substituent at each substitutable position of the group, and when more than one position in any given structure is substituted with more than one substituent selected from a specified group, the substituent is either the same or different at every position. Combinations of substituents envisioned by this invention are preferably those that result in the formation of stable or chemically feasible compounds. The term “stable”, as used herein, refers to compounds that are not substantially altered when subjected to conditions to allow for their production, detection, and, in certain embodiments, their recovery, purification, and use for one or more of the purposes disclosed herein.

(21) Suitable monovalent substituents on a substitutable carbon atom of an “optionally substituted” group are independently deuterium; halogen; —(CH.sub.2).sub.0-4R.sup.°; —(CH.sub.2).sub.0-4OR.sup.°; —O(CH.sub.2).sub.0-4R.sup.°, —O—(CH.sub.2).sub.0-4C(O)OR.sup.°; —(CH.sub.2).sub.0-4CH(OR.sup.°).sub.2; —(CH.sub.2).sub.0-4SR.sup.°; —(CH.sub.2).sub.0-4Ph, which are optionally substituted with R.sup.°; —(CH.sub.2).sub.0-4O(CH.sub.2).sub.0-1Ph which is optionally substituted with R.sup.°; —CH=CHPh, which is optionally substituted with R.sup.°; —(CH.sub.2).sub.0-4O(CH.sub.2).sub.0-1-pyridyl which is optionally substituted with R.sup.°; —

NO.sub.2; —CN; —N.sub.3; —(CH.sub.2).sub.0-4N(R.sup.°).sub.2; —(CH.sub.2).sub.0-4N(R.sup.°)C(O)R.sup.°; —N(R.sup.°)C(S)R.sup.°; —(CH.sub.2).sub.0-4N(R.sup.°)C(O)NR.sup.° .sub.2; —N(R.sup.°)C(S)NR.sup.° .sub.2; —(CH.sub.2).sub.0-4N(R.sup.°)C(O)OR.sup.°; —N(R.sup.°)N(R.sup.°)C(O)R.sup.°; —N(R.sup.°)N(R.sup.°)C(O)NR.sup.° .sub.2; —N(R.sup.°)N(R.sup.°)C(O)OR.sup.°; —(CH.sub.2).sub.0-4C(O)R.sup.°; —C(S)R.sup.°; —(CH.sub.2).sub.0-4C(O)OR.sup.°; —(CH.sub.2).sub.0-4C(O)SR.sup.°; —(CH.sub.2).sub.0-4C(O)OSiR.sup.° .sub.3; —(CH.sub.2).sub.0-4OC(O)R.sup.°; —OC(O)(CH.sub.2).sub.0-4SR.sup.°, SC(S)SR.sup.°; —(CH.sub.2).sub.0-4SC(O)R.sup.°; —(CH.sub.2).sub.0-4C(O)NR.sup.° .sub.2; —C(S)NR.sup.° .sub.2; —C(S)SR.sup.°; —SC(S)SR.sup.°, —(CH.sub.2).sub.0-4OC(O)NR.sup.° .sub.2; —C(O)N(OR.sup.°)R.sup.°; —C(O)C(O)R.sup.°; —C(O)CH.sub.2C(O)R.sup.°; —C(NOR.sup.°)R.sup.°; —(CH.sub.2).sub.0-4SSR.sup.°; —(CH.sub.2).sub.0-4S(O).sub.2R.sup.°; —(CH.sub.2).sub.0-4S(O).sub.20R.sup.°; —(CH.sub.2).sub.0-4OS(O).sub.2R.sup.°; —S(O).sub.2NR.sup.° .sub.2; —(CH.sub.2).sub.0-4S(O)R.sup.°; —N(R.sup.°)S(O).sub.2NR.sup.° .sub.2; —N(R.sup.°)S(O).sub.2R.sup.°; —N(OR.sup.°)R.sup.°; —C(NH)NR.sup.° .sub.2; —P(O).sub.2R.sup.°; —P(O)R.sup.° .sub.2; —OP(O)R.sup.° .sub.2; —OP(O)(OR.sup.°).sub.2; SiR.sup.° .sub.3; —(C.sub.1-4 straight or branched alkylene)O—N(R.sup.°).sub.2; or —(C.sub.1-4 straight or branched alkylene)C(O)O—N(R.sup.°).sub.2, wherein each R.sup.° is optionally substituted as defined below and is independently hydrogen, C.sub.1-6 aliphatic, —CH.sub.2Ph, —O(CH.sub.2).sub.0-1Ph, —CH.sub.2-(5-6 membered heteroaryl ring), or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, or, notwithstanding the definition above, two independent occurrences of R.sup.°, taken together with their intervening atom(s), form a 3-12-membered saturated, partially unsaturated, or aryl mono- or bicyclic ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, which is optionally substituted as defined below.

(22) Suitable monovalent substituents on R.sup.° (or the ring formed by taking two independent occurrences of R.sup.° together with their intervening atoms), are independently deuterium, halogen, —(CH.sub.2).sub.0-2R.sup.° .circle-solid., —(haloR.sup.° .circle-solid.), —(CH.sub.2).sub.0-2OH, —(CH.sub.2).sub.0-2OR.sup.° .circle-solid., —(CH.sub.2).sub.0-2CH(OR.sup.° .circle-solid.).sub.2; —O(haloR.sup.° .circle-solid.), —CN, —N.sub.3, —(CH.sub.2).sub.0-2C(O)R.sup.° .circle-solid., —(CH.sub.2).sub.0-2C(O)OH, —(CH.sub.2).sub.0-2C(O)OR.sup.° .circle-solid., —(CH.sub.2).sub.0-2SR.sup.° .circle-solid., —(CH.sub.2).sub.0-2SH, —(CH.sub.2).sub.0-2NH.sub.2, —(CH.sub.2).sub.0-2NHR.sup.° .circle-solid., —(CH.sub.2).sub.0-2NR.sup.° .circle-solid..sub.2, —NO.sub.2, —SiR.sup.° .circle-solid..sub.3, —OSiR.sup.° .circle-solid..sub.3, —C(O)SR.sup.° .circle-solid., —(C.sub.1-4 straight or branched alkylene)C(O)OR.sup.° .circle-solid., or —SSR.sup.° .circle-solid. wherein each R.sup.° .circle-solid. is unsubstituted or where preceded by “halo” is substituted only with one or more halogens, and is independently selected from C.sub.1-4 aliphatic, —CH.sub.2Ph, —O(CH.sub.2).sub.0-1Ph, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. Suitable divalent substituents on a saturated carbon atom of R.sup.° include =O and =S.

(23) Suitable divalent substituents on a saturated carbon atom of an “optionally substituted” group include the following: =O, =S, =NNR*.sub.2, =NNHC(O)R*, =NNHC(O)OR*, =NNHS(O).sub.2R*, =NR*, =NOR*, —O(C(R*.sub.2)).sub.2-3O—, or —S(C(R*.sub.2)).sub.2-3S—, wherein each independent occurrence of R* is selected from hydrogen, C.sub.1-6 aliphatic which is substituted as defined below, or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. Suitable divalent substituents that are bound to vicinal substitutable carbons of an

“optionally substituted” group include: $\text{—O}(\text{CR}^*.\text{sub.2}).\text{sub.2-3O—}$, wherein each independent occurrence of R^* is selected from hydrogen, C.sub.1-6 aliphatic which is optionally substituted as defined below, or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

(24) Suitable substituents on the aliphatic group of R^* include halogen, $\text{—R}.\text{sup.}\text{circle-solid.}$, $\text{—(haloR}.\text{sup.}\text{circle-solid.})}$, —OH , $\text{—OR}.\text{sup.}\text{circle-solid.}$, $\text{—O(haloR}.\text{sup.}\text{circle-solid.})}$, —CN , —C(O)OH , $\text{—C(O)OR}.\text{sup.}\text{circle-solid.}$, $\text{—NH}.\text{sub.2}$, $\text{—NHR}.\text{sup.}\text{circle-solid.}$, $\text{—NR}.\text{sup.}\text{circle-solid.}\text{sub.2}$, or $\text{—NO}.\text{sub.2}$, wherein each $\text{R}.\text{sup.}\text{circle-solid.}$ is unsubstituted or where preceded by “halo” is substituted only with one or more halogens, and is independently C.sub.1-4 aliphatic, $\text{—CH}.\text{sub.2Ph}$, $\text{—O(CH}.\text{sub.2}).\text{sub.0-1Ph}$, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

(25) Suitable substituents on a substitutable nitrogen of an “optionally substituted” group include $\text{—R}.\text{sup.}\dagger$, $\text{—NR}.\text{sup.}\dagger.\text{sub.2}$, $\text{—C(O)R}.\text{sup.}\dagger$, $\text{—C(O)OR}.\text{sup.}\dagger$, $\text{—C(O)C(O)R}.\text{sup.}\dagger$, $\text{—C(O)CH}.\text{sub.2C(O)R}.\text{sup.}\dagger$, $\text{—S(O)}.\text{sub.2R}.\text{sup.}\dagger$, $\text{—S(O)}.\text{sub.2NR}.\text{sup.}\dagger.\text{sub.2}$, $\text{—C(S)NR}.\text{sup.}\dagger.\text{sub.2}$, $\text{—C(NH)NR}.\text{sup.}\dagger.\text{sub.2}$, or $\text{—N(R}.\text{sup.}\dagger)\text{S(O)}.\text{sub.2R}.\text{sup.}\dagger$; wherein each $\text{R}.\text{sup.}\dagger$ is independently hydrogen, C.sub.1-6 aliphatic which is optionally substituted as defined below, unsubstituted —OPh , or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, or, notwithstanding the definition above, two independent occurrences of R , taken together with their intervening atom(s) form an unsubstituted 3-12-membered saturated, partially unsaturated, or aryl mono- or bicyclic ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

(26) Suitable substituents on the aliphatic group of $\text{R}.\text{sup.}\dagger$ are independently halogen, $\text{—R}.\text{sup.}\text{circle-solid.}$, $\text{—(haloR}.\text{sup.}\text{circle-solid.})}$, —OH , $\text{—OR}.\text{sup.}\text{circle-solid.}$, $\text{—O(haloR}.\text{sup.}\text{circle-solid.})}$, —CN , —C(O)OH , $\text{—C(O)OR}.\text{sup.}\text{circle-solid.}$, $\text{—NH}.\text{sub.2}$, $\text{—NHR}.\text{sup.}\text{circle-solid.}$, $\text{—NR}.\text{sup.}\text{circle-solid.}\text{sub.2}$, or $\text{—NO}.\text{sub.2}$, wherein each $\text{R}.\text{sup.}\text{circle-solid.}$ is unsubstituted or where preceded by “halo” is substituted only with one or more halogens, and is independently C.sub.1-4 aliphatic, $\text{—CH}.\text{sub.2Ph}$, $\text{—O(CH}.\text{sub.2}).\text{sub.0-1Ph}$, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

(27) In certain embodiments, the terms “optionally substituted”, “optionally substituted alkyl”, “optionally substituted “optionally substituted alkenyl”, “optionally substituted alkynyl”, “optionally substituted carbocyclic”, “optionally substituted aryl”, “optionally substituted heteroaryl”, “optionally substituted heterocyclic”, and any other optionally substituted group as used herein, refer to groups that are substituted or unsubstituted by independent replacement of one, two, or three or more of the hydrogen atoms thereon with typical substituents including, but not limited to: —F , —Cl , —Br , —I , deuterium, —OH , protected hydroxy, alkoxy, oxo, thiooxo, $\text{—NO}.\text{sub.2}$, —CN , $\text{CF}.\text{sub.3}$, $\text{N}.\text{sub.3}$, $\text{—NH}.\text{sub.2}$, protected amino, —NH alkyl, —NH alkenyl, —NH alkynyl, —NH cycloalkyl, —NH -aryl, —NH -heteroaryl, —NH -heterocyclic, -dialkylamino, -diarylamino, -diheteroarylamino, —O— alkyl, —O— alkenyl, —O— alkynyl, —O— cycloalkyl, —O— aryl, —O— heteroaryl, —O— heterocyclic, —C(O)— alkyl, —C(O)— alkenyl, —C(O)— alkynyl, —C(O)— carbocyclyl, —C(O)— aryl, —C(O)— heteroaryl, —C(O)— heterocyclyl, $\text{—CONH}.\text{sub.2}$, —CONH— alkyl, —CONH— alkenyl, —CONH— alkynyl, —CONH— carbocyclyl, —CONH— aryl, —CONH— heteroaryl, —CONH— heterocyclyl, $\text{—OCO}.\text{sub.2—}$ alkyl, $\text{—OCO}.\text{sub.2—}$ alkenyl, $\text{—OCO}.\text{sub.2—}$ alkynyl, $\text{—OCO}.\text{sub.2—}$ carbocyclyl, $\text{—OCO}.\text{sub.2—}$ aryl, $\text{—OCO}.\text{sub.2—}$ heteroaryl, $\text{—OCO}.\text{sub.2—}$ heterocyclyl, $\text{—OCONH}.\text{sub.2}$, —OCONH— alkyl, —OCONH— alkenyl, —OCONH— alkynyl, —OCONH— carbocyclyl, —OCONH— aryl, —OCONH— heteroaryl, —OCONH— heterocyclyl, —NHC(O)— alkyl, —NHC(O)— alkenyl, —NHC(O)— alkynyl, —NHC(O)— carbocyclyl, —NHC(O)— aryl, —NHC(O)— heteroaryl, —NHC(O)— heterocyclyl, $\text{—NHCO}.\text{sub.2—}$ alkyl, $\text{—NHCO}.\text{sub.2—}$ alkenyl, $\text{—NHCO}.\text{sub.2—}$

alkynyl, —NHC(O).sub.2— carbocyclyl, —NHC(O).sub.2— aryl, —NHC(O).sub.2— heteroaryl, —NHC(O).sub.2—heterocyclyl, —NHC(O)NH.sub.2, —NHC(O)NH— alkyl, —NHC(O)NH— alkenyl, —NHC(O)NH— alkenyl, —NHC(O)NH— carbocyclyl, —NHC(O)NH-aryl, —NHC(O)NH-heteroaryl, —NHC(O)NH— heterocyclyl, NHC(S)NH.sub.2, —NHC(S)NH— alkyl, —NHC(S)NH— alkenyl, —NHC(S)NH— alkynyl, —NHC(S)NH— carbocyclyl, —NHC(S)NH-aryl, —NHC(S)NH-heteroaryl, —NHC(S)NH-heterocyclyl, —NHC(NH)NH.sub.2, —NHC(NH)NH— alkyl, —NHC(NH)NH— alkenyl, —NHC(NH)NH— alkenyl, —NHC(NH)NH— carbocyclyl, —NHC(NH)NH-aryl, —NHC(NH)NH-heteroaryl, —NHC(NH)NH— heterocyclyl, —NHC(NH)— alkyl, —NHC(NH)— alkenyl, —NHC(NH)— alkenyl, —NHC(NH)— carbocyclyl, —NHC(NH)-aryl, —NHC(NH)-heteroaryl, —NHC(NH)-heterocyclyl, —C(NH)NH— alkyl, —C(NH)NH— alkenyl, —C(NH)NH— alkynyl, —C(NH)NH— carbocyclyl, —C(NH)NH-aryl, —C(NH)NH-heteroaryl, —C(NH)NH-heterocyclyl, —S(O)— alkyl, —S(O)— alkenyl, —S(O)— alkynyl, —S(O)— carbocyclyl, —S(O)-aryl, —S(O)-heteroaryl, —S(O)-heterocyclyl —SO.sub.2NH.sub.2, —SO.sub.2NH— alkyl, —SO.sub.2NH— alkenyl, —SO.sub.2NH— alkynyl, —SO.sub.2NH— carbocyclyl, —SO.sub.2NH— aryl, —SO.sub.2NH— heteroaryl, —SO.sub.2NH— heterocyclyl, —NHSO.sub.2— alkyl, —NHSO.sub.2— alkenyl, —NHSO.sub.2— alkynyl, —NHSO.sub.2— carbocyclyl, —NHSO.sub.2-aryl, —NHSO.sub.2-heteroaryl, —NHSO.sub.2-heterocyclyl, —CH.sub.2NH.sub.2, —CH.sub.2SO.sub.2CH.sub.3, -mono-, di-, or tri-alkyl silyl, -alkyl, -alkenyl, -alkynyl, -aryl, -arylalkyl, -heteroaryl, -heteroarylalkyl, -heterocycloalkyl, -cycloalkyl, -carbocyclic, -heterocyclic, polyalkoxyalkyl, polyalkoxy, -methoxymethoxy, -methoxyethoxy, —SH, —S— alkyl, —S— alkenyl, —S— alkynyl, —S— carbocyclyl, —S-aryl, —S-heteroaryl, —S-heterocyclyl, or methylthiomethyl.

(28) As used herein, the term “pharmaceutically acceptable salt” refers to those salts which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and lower animals without undue toxicity, irritation, allergic response and the like, and are commensurate with a reasonable benefit/risk ratio. Pharmaceutically acceptable salts are well known in the art. For example, S. M. Berge et al., describe pharmaceutically acceptable salts in detail in *J. Pharmaceutical Sciences*, 1977, 66, 1-19, incorporated herein by reference.

Pharmaceutically acceptable salts of the compounds of this invention include those derived from suitable inorganic and organic acids and bases. Examples of pharmaceutically acceptable, nontoxic acid addition salts are salts of an amino group formed with inorganic acids such as hydrochloric acid, hydrobromic acid, phosphoric acid, sulfuric acid and perchloric acid or with organic acids such as acetic acid, oxalic acid, maleic acid, tartaric acid, citric acid, succinic acid or malonic acid or by using other methods used in the art such as ion exchange. Other pharmaceutically acceptable salts include adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bisulfate, borate, butyrate, camphorate, camphorsulfonate, citrate, cyclopentanepropionate, digluconate, dodecylsulfate, ethanesulfonate, formate, fumarate, glucoheptonate, glycerophosphate, gluconate, hemisulfate, heptanoate, hexanoate, hydroiodide, 2-hydroxy-ethanesulfonate, lactobionate, lactate, laurate, lauryl sulfate, malate, maleate, malonate, methanesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, oleate, oxalate, palmitate, pamoate, pectinate, persulfate, 3-phenylpropionate, phosphate, pivalate, propionate, stearate, succinate, sulfate, tartrate, thiocyanate, p-toluenesulfonate, undecanoate, valerate salts, and the like.

(29) Salts derived from appropriate bases include alkali metal, alkaline earth metal, ammonium and N.sup.+(C.sub.1-4alkyl).sub.4 salts. Representative alkali or alkaline earth metal salts include sodium, lithium, potassium, calcium, magnesium, and the like. Further pharmaceutically acceptable salts include, when appropriate, nontoxic ammonium, quaternary ammonium, and amine cations formed using counterions such as halide, hydroxide, carboxylate, sulfate, phosphate, nitrate, loweralkyl sulfonate and aryl sulfonate.

(30) Unless otherwise stated, structures depicted herein are also meant to include all isomeric (e.g., enantiomeric, diastereomeric, and geometric (or conformational)) forms of the structure; for

example, the R and S configurations for each asymmetric center, Z and E double bond isomers, and Z and E conformational isomers. Therefore, single stereochemical isomers as well as enantiomeric, diastereomeric, and geometric (or conformational) mixtures of the present compounds are within the scope of the invention. Unless otherwise stated, all tautomeric forms of the compounds of the invention are within the scope of the invention.

(31) As used herein, the term “tautomer” means each of two or more isomers of a compound that exist together in equilibrium, and are readily interchanged by migration of an atom or group within the molecule. Tautomers are constitutional isomers of organic compounds that readily interconvert by a chemical reaction called tautomerization. This reaction commonly results in the formal migration of a hydrogen atom or proton, accompanied by a switch of a single bond and adjacent double bond. For example, see

(32) ##STR00005##

(33) Additionally, unless otherwise stated, structures depicted herein are also meant to include compounds that differ only in the presence of one or more isotopically enriched atoms. For example, compounds having the present structures including the replacement of hydrogen by deuterium or tritium, or the replacement of a carbon by a ¹³C- or ¹⁴C-enriched carbon are within the scope of this invention. In some embodiments, the group comprises one or more deuterium atoms.

(34) There is furthermore intended that a compound of the formula I includes isotope-labeled forms thereof. An isotope-labeled form of a compound of the formula I is identical to this compound apart from the fact that one or more atoms of the compound have been replaced by an atom or atoms having an atomic mass or mass number which differs from the atomic mass or mass number of the atom which usually occurs naturally. Examples of isotopes which are readily commercially available and which can be incorporated into a compound of the formula I by well-known methods include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorus, fluorine and chlorine, for example ²H, ³H, ¹³C, ¹⁴C, ¹⁵N, ¹⁸O, ¹⁷O, ³¹P, ³²P, ³⁵S, ¹⁸F and ³⁶Cl, respectively. A compound of the formula I, a prodrug, thereof or a pharmaceutically acceptable salt of either which contains one or more of the above-mentioned isotopes and/or other isotopes of other atoms is intended to be part of the present invention. An isotope-labeled compound of the formula I can be used in a number of beneficial ways. For example, an isotope-labeled compound of the formula I into which, for example, a radioisotope, such as ³H or ¹⁴C, has been incorporated, is suitable for medicament and/or substrate tissue distribution assays. These radioisotopes, i.e. tritium (³H) and carbon-14 (¹⁴C), are particularly preferred owing to simple preparation and excellent detectability. Incorporation of heavier isotopes, for example deuterium (²H), into a compound of the formula I has therapeutic advantages owing to the higher metabolic stability of this isotope-labeled compound. Higher metabolic stability translates directly into an increased in vivo half-life or lower dosages, which under most circumstances would represent a preferred embodiment of the present invention. An isotope-labeled compound of the formula I can usually be prepared by carrying out the procedures disclosed in the synthesis schemes and the related description, in the example part and in the preparation part in the present text, replacing a non-isotope-labeled reactant by a readily available isotope-labeled reactant.

(35) Deuterium (²H) can also be incorporated into a compound of the formula I for the purpose in order to manipulate the oxidative metabolism of the compound by way of the primary kinetic isotope effect. The primary kinetic isotope effect is a change of the rate for a chemical reaction that results from exchange of isotopic nuclei, which in turn is caused by the change in ground state energies necessary for covalent bond formation after this isotopic exchange. Exchange of a heavier isotope usually results in a lowering of the ground state energy for a chemical bond and thus causes a reduction in the rate in rate-limiting bond breakage. If the bond breakage occurs in or in the vicinity of a saddle-point region along the coordinate of a multi-product reaction, the

product distribution ratios can be altered substantially. For explanation: if deuterium is bonded to a carbon atom at a non-exchangeable position, rate differences of $k_{\text{sub.M}}/k_{\text{sub.D}}=2-7$ are typical. If this rate difference is successfully applied to a compound of the formula I that is susceptible to oxidation, the profile of this compound in vivo can be drastically modified and result in improved pharmacokinetic properties.

(36) When discovering and developing therapeutic agents, the person skilled in the art is able to optimize pharmacokinetic parameters while retaining desirable in vitro properties. It is reasonable to assume that many compounds with poor pharmacokinetic profiles are susceptible to oxidative metabolism. In vitro liver microsomal assays currently available provide valuable information on the course of oxidative metabolism of this type, which in turn permits the rational design of deuterated compounds of the formula I with improved stability through resistance to such oxidative metabolism. Significant improvements in the pharmacokinetic profiles of compounds of the formula I are thereby obtained, and can be expressed quantitatively in terms of increases in the in vivo half-life ($t/2$), concentration at maximum therapeutic effect ($C_{\text{sub.max}}$), area under the dose response curve (AUC), and F; and in terms of reduced clearance, dose and materials costs.

(37) The following is intended to illustrate the above: a compound of the formula I which has multiple potential sites of attack for oxidative metabolism, for example benzylic hydrogen atoms and hydrogen atoms bonded to a nitrogen atom, is prepared as a series of analogues in which various combinations of hydrogen atoms are replaced by deuterium atoms, so that some, most or all of these hydrogen atoms have been replaced by deuterium atoms. Half-life determinations enable favorable and accurate determination of the extent to which the improvement in resistance to oxidative metabolism has improved. In this way, it is determined that the half-life of the parent compound can be extended by up to 100% as the result of deuterium-hydrogen exchange of this type.

(38) Deuterium-hydrogen exchange in a compound of the formula I can also be used to achieve a favorable modification of the metabolite spectrum of the starting compound in order to diminish or eliminate undesired toxic metabolites. For example, if a toxic metabolite arises through oxidative carbon-hydrogen (C—H) bond cleavage, it can reasonably be assumed that the deuterated analogue will greatly diminish or eliminate production of the unwanted metabolite, even if the particular oxidation is not a rate-determining step. Further information on the state of the art with respect to deuterium-hydrogen exchange may be found, for example in Hanzlik et al., J. Org. Chem. 55, 3992-3997, 1990, Reider et al., J. Org. Chem. 52, 3326-3334, 1987, Foster, Adv. Drug Res. 14, 1-40, 1985, Gillette et al, Biochemistry 33(10) 2927-2937, 1994, and Jarman et al. Carcinogenesis 16(4), 683-688, 1993.

(39) As used herein, the term “modulator” is defined as a compound that binds to and/or inhibits the target with measurable affinity. In certain embodiments, a modulator has an IC_{50} and/or binding constant of less than about 50 μM , less than about 5 μM , less than about 1 μM , less than about 500 nM, less than about 100 nM, or less than about 10 nM.

(40) The terms “measurable affinity” and “measurably inhibit,” as used herein, means a measurable change in BTK activity between a sample comprising a compound of the present invention, or composition thereof, and BTK, and an equivalent sample comprising BTK, in the absence of said compound, or composition thereof.

(41) Combinations of substituents and variables envisioned by this invention are only those that result in the formation of stable compounds. The term “stable”, as used herein, refers to compounds which possess stability sufficient to allow manufacture and which maintains the integrity of the compound for a sufficient period of time to be useful for the purposes detailed herein (e.g., therapeutic or prophylactic administration to a subject).

(42) The recitation of a listing of chemical groups in any definition of a variable herein includes definitions of that variable as any single group or combination of listed groups. The recitation of an embodiment for a variable herein includes that embodiment as any single embodiment or in

combination with any other embodiments or portions thereof.

3. Description of Exemplary Compounds

(43) According to one aspect, the present invention provides a compound of formula I,

(44) ##STR00006## or a tautomer, or a pharmaceutically acceptable salt, hydrate, solvate, or prodrug thereof, wherein: X is C(R).sub.2 or O; Y is —O—, —S—, —SO.sub.2—, —SO—, —C(O)—, —CO.sub.2—, —C(O)N(R).sup.2—, —NR.sup.2C(O)—, —NR.sup.2C(O)N(R)—, —NR.sup.2SO.sub.2—, or —N(R).sup.2—; R.sup.1 is C.sub.5-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted; Ring Z is C.sub.5-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, a 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted; each R.sup.a is independently —R, halogen, —OR, —SR, —CN, —NO.sub.2, —SO.sub.2R, —SOR, —C(O)R, —CO.sub.2R, —C(O)N(R).sub.2, —NRC(O)R, —NRC(O)N(R).sub.2, —NRSO.sub.2R, or —N(R).sub.2; or two R.sup.a groups, together with the atoms to which each is attached, forms a C.sub.5-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, a 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted; each R is independently hydrogen, C.sub.1-6 aliphatic, C.sub.5-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, a 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted; or two R groups on the same atom are taken together with the atom to which they are attached to form a C.sub.5-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, a 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; a 10-15 membered fused aryl ring; a 6-15 membered saturated or partially unsaturated fused carbocyclic ring; a 10-15 membered fused heteroaryl ring; or a 6-15 membered saturated or partially unsaturated fused heterocyclic ring; each of which is optionally substituted; n is 0, 1, or 2; and p is 0, 1, 2, or 3.

(45) In certain embodiments, X is CH.sub.2. In certain embodiments, X is O.

(46) In certain embodiments, X is CH.sub.2 and n is 1.

(47) In certain embodiments, X is CH.sub.2 and n is 2.

(48) In certain embodiments, X is O and n is 1.

(49) In certain embodiments, X is O and n is 2.

(50) In certain embodiments, Y is —C(O)—, —CO.sub.2—, —C(O)N(R).sup.2—, —NR.sup.2C(O)—, —NR.sup.2C(O)N(R)—, —NR.sup.2SO.sub.2—, or —N(R).sup.2—.

(51) In certain embodiments, Y is —C(O)N(R).sup.2—, —NR.sup.2C(O)—, —NR.sup.2C(O)N(R)—, or —NR.sup.2SO.sub.2—.

(52) In certain embodiments, Y is —NR.sup.2C(O)—.

(53) In certain embodiments, Y is —C(O)N(R).sup.2—.

(54) In certain embodiments, R.sup.1 is an optionally substituted 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or R.sup.1 is an optionally substituted 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

(55) In certain embodiments, R.sup.1 is an optionally substituted 3-7 membered heterocyclic ring

having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

(56) In certain embodiments, R.sup.1 is an optionally substituted 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

(57) In certain embodiments, R.sup.1 is azocinyl, 2H,6H-1,5,2-dithiazinyl, dihydrofuro [2,3-b]tetrahydrofuran, furanyl, furazanyl, imidazolidinyl, imidazolyl, imidazolyl, isothiazolyl, isoxazolyl, morpholinyl, oxadiazolyl, 1,2,3-oxadiazolyl, 1,2,4-oxadiazolyl; 1,2,5oxadiazolyl, 1,3,4-oxadiazolyl, oxazolidinyl, oxazolyl, oxazolidinyl, pyrimidinyl, piperazinyl, piperidinyl, piperidine-one, pyranyl, pyrazinyl, pyrazolidinyl, pyrazolinyl, pyrazolyl, pyridazinyl, pyridinyl, pyridyl, pyrimidinyl, pyrrolidinyl, pyrrolinyl, 2H-pyrrolyl, pyrrolyl, tetrahydrofuranyl, tetrahydropyridine, tetrahydropyran, 6H-1,2,5-thiadiazinyl, 1,2,3-thiadiazolyl, 1,2,4-thiadiazolyl, 1,2,5-thiadiazolyl, 1,3,4thiadiazolyl, thiazolyl, thienyl, triazinyl, 1,2,3-triazolyl, 1,2,4-triazolyl, 1,2,5-triazolyl, 1,3,4-triazolyl, oxetanyl, or azetidiny; each of which is optionally substituted.

(58) In certain embodiments, R.sup.1 is imidazolidinyl, imidazolyl, imidazolyl, isoxazolyl, oxadiazolyl, 1,2,3-oxadiazolyl, 1,2,4-oxadiazolyl; 1,2,5oxadiazolyl, 1,3,4-oxadiazolyl, oxazolidinyl, oxazolyl, oxazolidinyl, pyranyl, pyrazinyl, pyrazolidinyl, pyrazolinyl, pyrazolyl, pyridazinyl, pyridinyl, pyridyl, pyrimidinyl, pyrrolidinyl, pyrrolinyl, 2H-pyrrolyl, pyrrolyl, tetrahydrofuranyl, tetrahydropyridine, tetrahydropyran, 6H-1,2,5-thiadiazinyl, 1,2,3-thiadiazolyl, 1,2,4-thiadiazolyl, 1,2,5-thiadiazolyl, 1,3,4thiadiazolyl, thiazolyl, thienyl, triazinyl, 1,2,3-triazolyl, 1,2,4-triazolyl, 1,2,5-triazolyl, 1,3,4-triazolyl, oxetanyl, or azetidiny; each of which is optionally substituted.

(59) In certain embodiments, R.sup.1 is selected from:

(60) ##STR00007##

(61) In certain embodiments, Ring Z is a 3-8 membered saturated or partially unsaturated carbocyclic ring or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted.

(62) In certain embodiments, Ring Z is cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, imidazolyl, pyrazolyl, triazolyl, oxazolyl, isoxazolyl, oxadiazolyl, thiazolyl, or isothiazolyl; each of which is optionally substituted.

(63) In certain embodiments, Ring Z is cyclopropyl, imidazolyl, pyrazolyl, or pyridyl; each of which is optionally substituted.

(64) In certain embodiments, Ring Z is

(65) ##STR00008##

(66) In certain embodiments, Ring Z is

(67) ##STR00009## ##STR00010##

(68) In certain embodiments, each of X, Y, Ring Z, R.sup.1, R.sup.a, n, and p, is as defined above and described in embodiments, classes and subclasses above and herein, singly or in combination.

(69) In certain embodiments, the present invention provides a compound of formula II,

(70) ##STR00011## or a pharmaceutically acceptable salt thereof, wherein each of Ring Z, R.sup.1, R.sup.a, and p, is as defined above and described in embodiments, classes and subclasses above and herein, singly or in combination.

(71) In certain embodiments, the present invention provides a compound of formula III,

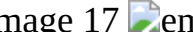
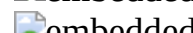
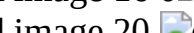
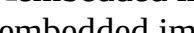
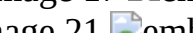
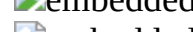
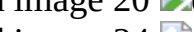
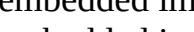
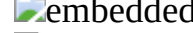
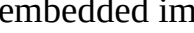
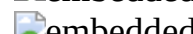
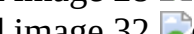
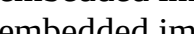
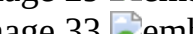
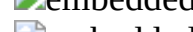
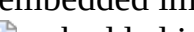
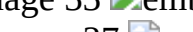
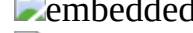
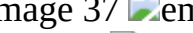
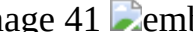
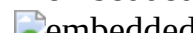
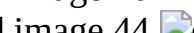
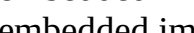
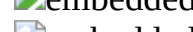
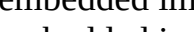
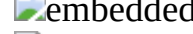
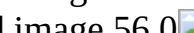
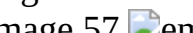
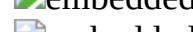
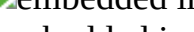
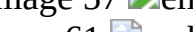
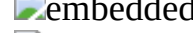
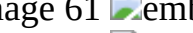
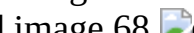
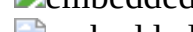
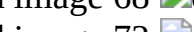
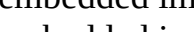
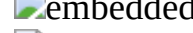
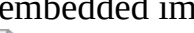
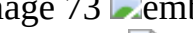
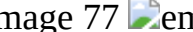
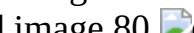
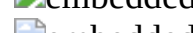
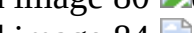
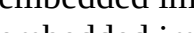
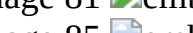
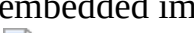
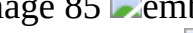
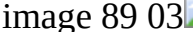
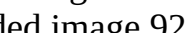
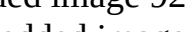
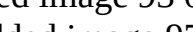
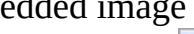
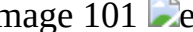
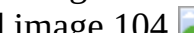
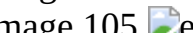
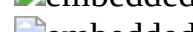
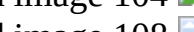
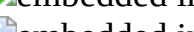
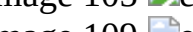
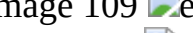
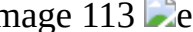
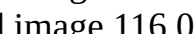
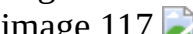
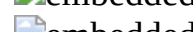
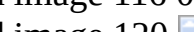
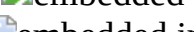
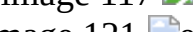
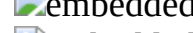
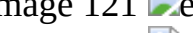
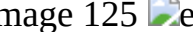
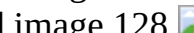
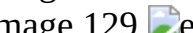
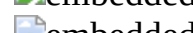
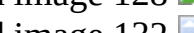
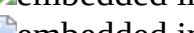
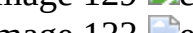
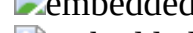
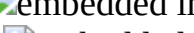
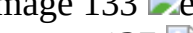
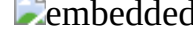
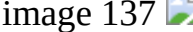
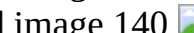
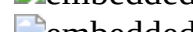
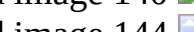
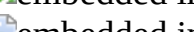
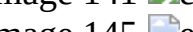
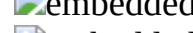
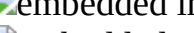
(72) ##STR00012## or a pharmaceutically acceptable salt thereof, wherein each of Ring Z, R.sup.1, R.sup.a, and p, is as defined above and described in embodiments, classes and subclasses above and herein, singly or in combination.

(73) In certain embodiments, the present invention provides a compound of formula IV,

(74) ##STR00013## or a pharmaceutically acceptable salt thereof, wherein each of Ring Z, R.sup.1, R.sup.a, and p, is as defined above and described in embodiments, classes and subclasses above and herein, singly or in combination.

(75) In certain embodiments, the invention provides a compound selected from Table 1:

(76) TABLE-US-00001   
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 embedded image 188  embedded image 189  embedded image 190

(77) In some embodiments, the present invention provides a compound selected from those depicted above, or a pharmaceutically acceptable salt thereof.

(78) Various structural depictions may show a heteroatom without an attached group, radical, charge, or counterion. Those of ordinary skill in the art are aware that such depictions are meant to indicate that the heteroatom is attached to hydrogen (e.g.,

(79) ##STR00204##

is understood to be

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(81) In certain embodiments, the compounds of the invention were synthesized in accordance with the schemes provided in the Examples below.

4. Uses, Formulation and Administration

(82) Pharmaceutically Acceptable Compositions

(83) According to another embodiment, the invention provides a composition comprising a compound of this invention or a pharmaceutically acceptable derivative thereof and a pharmaceutically acceptable carrier, adjuvant, or vehicle. The amount of compound in compositions of this invention is such that is effective to measurably inhibit BTK, or a mutant thereof, in a biological sample or in a patient. In certain embodiments, the amount of compound in compositions of this invention is such that is effective to measurably inhibit BTK, or a mutant thereof, in a biological sample or in a patient. In certain embodiments, a composition of this invention is formulated for administration to a patient in need of such composition.

(84) The term "patient" or "subject", as used herein, means an animal, preferably a mammal, and most preferably a human.

(85) The term "pharmaceutically acceptable carrier, adjuvant, or vehicle" refers to a non-toxic carrier, adjuvant, or vehicle that does not destroy the pharmacological activity of the compound with which it is formulated. Pharmaceutically acceptable carriers, adjuvants or vehicles that are used in the compositions of this invention include, but are not limited to, ion exchangers, alumina, aluminum stearate, lecithin, serum proteins, such as human serum albumin, buffer substances such as phosphates, glycine, sorbic acid, potassium sorbate, partial glyceride mixtures of saturated vegetable fatty acids, water, salts or electrolytes, such as protamine sulfate, disodium hydrogen phosphate, potassium hydrogen phosphate, sodium chloride, zinc salts, colloidal silica, magnesium trisilicate, polyvinyl pyrrolidone, cellulose-based substances, polyethylene glycol, sodium carboxymethylcellulose, polyacrylates, waxes, polyethylene-polyoxypropylene-block polymers, polyethylene glycol and wool fat.

(86) A "pharmaceutically acceptable derivative" means any non-toxic salt, ester, salt of an ester or other derivative of a compound of this invention that, upon administration to a recipient, is capable of providing, either directly or indirectly, a compound of this invention or an inhibitorily active metabolite or residue thereof.

(87) Compositions of the present invention are administered orally, parenterally, by inhalation spray, topically, rectally, nasally, buccally, vaginally or via an implanted reservoir. The term "parenteral" as used herein includes subcutaneous, intravenous, intramuscular, intra-articular, intra-synovial, intrasternal, intrathecal, intrahepatic, intralesional and intracranial injection or infusion techniques. Preferably, the compositions are administered orally, intraperitoneally or intravenously. Sterile injectable forms of the compositions of this invention include aqueous or oleaginous suspension. These suspensions are formulated according to techniques known in the art using suitable dispersing or wetting agents and suspending agents. The sterile injectable preparation may also be a sterile injectable solution or suspension in a non-toxic parenterally acceptable diluent or solvent, for example as a solution in 1,3-butanediol. Among the acceptable vehicles and solvents that are employed are water, Ringer's solution and isotonic sodium chloride solution. In addition, sterile, fixed oils are conventionally employed as a solvent or suspending medium.

(88) For this purpose, any bland fixed oil employed includes synthetic mono- or diglycerides. Fatty acids, such as oleic acid and its glyceride derivatives are useful in the preparation of injectables, as are natural pharmaceutically-acceptable oils, such as olive oil or castor oil, especially in their polyoxyethylated versions. These oil solutions or suspensions also contain a long-chain alcohol diluent or dispersant, such as carboxymethyl cellulose or similar dispersing agents that are commonly used in the formulation of pharmaceutically acceptable dosage forms including emulsions and suspensions. Other commonly used surfactants, such as Tweens, Spans and other

emulsifying agents or bioavailability enhancers which are commonly used in the manufacture of pharmaceutically acceptable solid, liquid, or other dosage forms are also be used for the purposes of formulation.

(89) Pharmaceutically acceptable compositions of this invention are orally administered in any orally acceptable dosage form. Exemplary oral dosage forms are capsules, tablets, aqueous suspensions or solutions. In the case of tablets for oral use, carriers commonly used include lactose and corn starch. Lubricating agents, such as magnesium stearate, are also typically added. For oral administration in a capsule form, useful diluents include lactose and dried cornstarch. When aqueous suspensions are required for oral use, the active ingredient is combined with emulsifying and suspending agents. If desired, certain sweetening, flavoring or coloring agents are optionally also added.

(90) Alternatively, pharmaceutically acceptable compositions of this invention are administered in the form of suppositories for rectal administration. These can be prepared by mixing the agent with a suitable non-irritating excipient that is solid at room temperature but liquid at rectal temperature and therefore will melt in the rectum to release the drug. Such materials include cocoa butter, beeswax and polyethylene glycols.

(91) Pharmaceutically acceptable compositions of this invention are also administered topically, especially when the target of treatment includes areas or organs readily accessible by topical application, including diseases of the eye, the skin, or the lower intestinal tract. Suitable topical formulations are readily prepared for each of these areas or organs.

(92) Topical application for the lower intestinal tract can be effected in a rectal suppository formulation (see above) or in a suitable enema formulation. Topically-transdermal patches are also used.

(93) For topical applications, provided pharmaceutically acceptable compositions are formulated in a suitable ointment containing the active component suspended or dissolved in one or more carriers. Exemplary carriers for topical administration of compounds of this are mineral oil, liquid petrolatum, white petrolatum, propylene glycol, polyoxyethylene, polyoxypropylene compound, emulsifying wax and water. Alternatively, provided pharmaceutically acceptable compositions can be formulated in a suitable lotion or cream containing the active components suspended or dissolved in one or more pharmaceutically acceptable carriers. Suitable carriers include, but are not limited to, mineral oil, sorbitan monostearate, polysorbate 60, cetyl esters wax, cetearyl alcohol, 2-octyldodecanol, benzyl alcohol and water.

(94) Pharmaceutically acceptable compositions of this invention are optionally administered by nasal aerosol or inhalation. Such compositions are prepared according to techniques well-known in the art of pharmaceutical formulation and are prepared as solutions in saline, employing benzyl alcohol or other suitable preservatives, absorption promoters to enhance bioavailability, fluorocarbons, and/or other conventional solubilizing or dispersing agents.

(95) Most preferably, pharmaceutically acceptable compositions of this invention are formulated for oral administration. Such formulations may be administered with or without food. In some embodiments, pharmaceutically acceptable compositions of this invention are administered without food. In other embodiments, pharmaceutically acceptable compositions of this invention are administered with food.

(96) The amount of compounds of the present invention that are optionally combined with the carrier materials to produce a composition in a single dosage form will vary depending upon the host treated, the particular mode of administration. Preferably, provided compositions should be formulated so that a dosage of between 0.01-100 mg/kg body weight/day of the compound can be administered to a patient receiving these compositions.

(97) It should also be understood that a specific dosage and treatment regimen for any particular patient will depend upon a variety of factors, including the activity of the specific compound employed, the age, body weight, general health, sex, diet, time of administration, rate of excretion,

drug combination, and the judgment of the treating physician and the severity of the particular disease being treated. The amount of a compound of the present invention in the composition will also depend upon the particular compound in the composition.

(98) Uses of Compounds and Pharmaceutically Acceptable Compositions

(99) In certain embodiments, the invention provides a method for inhibiting BTK, or a mutant thereof, in a patient or in a biological sample comprising the step of administering to said patient or contacting said biological sample with a compound according to the invention.

(100) In certain embodiments, the invention is directed to the use of compounds of the invention and/or physiologically acceptable salts thereof, for modulating or inhibiting a BTK enzyme. The term “modulation” denotes any change in BTK-mediated signal transduction, which is based on the action of the specific inventive compounds capable to interact with the BTK target in such a manner that makes recognition, binding and activating possible. The compounds are characterized by such a high affinity to BTK, which ensures a reliable binding of BTK. In certain embodiments, the substances are highly selective for BTK over most other kinases in order to guarantee an exclusive and directed recognition with the single BTK target. In the context of the present invention, the term “recognition”—without being limited thereto—relates to any type of interaction between the specific compounds and the target, particularly covalent or non-covalent binding or association, such as a covalent bond, hydrophobic/hydrophilic interactions, van der Waals forces, ion pairs, hydrogen bonds, ligand-receptor (enzyme-inhibitor) interactions, and the like. Such association may also encompass the presence of other molecules such as peptides, proteins or nucleotide sequences. The present protein/ligand(enzyme-inhibitor)-interaction is characterized by high affinity, high selectivity and minimal or even lacking cross-reactivity to other target molecules to exclude unhealthy and harmful impacts to the treated subject.

(101) In certain embodiments, the present invention relates to a method for inhibiting a BTK enzyme, with at least one compound of formula (I) according to the invention and/or physiologically acceptable salts thereof, under conditions such that said BTK enzyme is inhibited. In certain embodiments, the system is a cellular system. In other embodiments, the system is an in-vitro translation which is based on the protein synthesis without living cells. The cellular system is defined to be any subject provided that the subject comprises cells. Hence, the cellular system can be selected from the group of single cells, cell cultures, tissues, organs and animals. In certain embodiments, the method for modulating a BTK enzyme is performed in-vitro. The prior teaching of the present specification concerning the compounds of formula (I), including any embodiments thereof, is valid and applicable without restrictions to the compounds according to formula (I) and their salts when used in the method for inhibiting BTK. The prior teaching of the present specification concerning the compounds of formula (I), including any embodiments thereof, is valid and applicable without restrictions to the compounds according to formula (I) and their salts when used in the method for inhibiting BTK.

(102) Patients with mutations in BTK have a profound block in B cell development, resulting in the almost complete absence of mature B lymphocytes and plasma cells, severely reduced Ig levels and a profound inhibition of humoral response to recall antigens (reviewed in Vihinen et al *Frontiers in Bioscience* 5: d917-928). Mice deficient in BTK also have a reduced number of peripheral B cells and greatly decreased serum levels of IgM and IgG3. BTK deletion in mice has a profound effect on B cell proliferation induced by anti-IgM, and inhibits immune responses to thymus-independent type II antigens (Ellmeier et al, *J Exp Med* 192: 1611-1623 (2000)). BTK also plays a crucial role in mast cell activation through the high-affinity IgE receptor (Fc epsilon RI). BTK deficient murine mast cells have reduced degranulation and decreased production of proinflammatory cytokines following Fc epsilon RI cross-linking (Kawakami et al. *Journal of Leukocyte Biology* 65: 286-290).

(103) Provided compounds are inhibitors of BTK and are therefore useful for treating one or more disorders associated with activity of BTK. Thus, in some embodiments, the present invention

provides a method for treating a BTK-mediated disorder comprising the step of administering to a patient in need thereof a compound of the present invention, or pharmaceutically acceptable composition thereof.

(104) As used herein, the term “BTK-mediated” disorders or conditions as used herein means any disease or other deleterious condition in which BTK, or a mutant thereof, is known to play a role. Accordingly, another embodiment of the present invention relates to treating or lessening the severity of one or more diseases in which BTK, or a mutant thereof, is known to play a role. Specifically, the present invention relates to a method of treating or lessening the severity of a disease or condition selected from a proliferative disorder or an autoimmune disorder, wherein said method comprises administering to a patient in need thereof a compound or composition according to the present invention.

(105) In some embodiments, the present invention provides a method for treating or lessening the severity of one or more diseases and conditions associated with BTK. In some embodiments, the disease or condition is an autoimmune disease, e.g., inflammatory bowel disease, arthritis, systemic lupus erythematosus (SLE or lupus), lupus nephritis, vasculitis, idiopathic thrombocytopenic purpura (ITP), rheumatoid arthritis (RA), psoriatic arthritis, osteoarthritis, Still's disease, juvenile arthritis, diabetes, myasthenia gravis, Hashimoto's thyroiditis, Ord's thyroiditis, Graves' disease, autoimmune thyroiditis, Sjogren's syndrome, multiple sclerosis (MS), systemic sclerosis, Lyme neuroborreliosis, Guillain-Barre syndrome, acute disseminated encephalomyelitis, Addison's disease, opsoclonus-myoclonus syndrome, ankylosing spondylosis, antiphospholipid antibody syndrome, aplastic anemia, autoimmune hepatitis, autoimmune gastritis, pernicious anemia, celiac disease, Goodpasture's syndrome, idiopathic thrombocytopenic purpura, optic neuritis, scleroderma, primary biliary cirrhosis, Reiter's syndrome, Takayasu's arteritis, temporal arteritis, warm autoimmune hemolytic anemia, Wegener's granulomatosis, psoriasis, alopecia universalis, Behcet's disease, chronic fatigue, dysautonomia, membranous glomerulonephropathy, endometriosis, interstitial cystitis, pemphigus vulgaris, bullous pemphigoid, neuromyotonia, scleroderma, or vulvodynia. In certain embodiments, the disease or condition is systemic lupus erythematosus (SLE or lupus) or lupus nephritis. In certain embodiments, the disease or condition is RA. In certain embodiments, the disease or condition is MS.

(106) In some embodiments, the disease or condition is a hyperproliferative disease or immunologically-mediated diseases including rejection of transplanted organs or tissues and Acquired Immunodeficiency Syndrome (AIDS, also known as HIV).

(107) In some embodiments, the present invention provides a method for treating or lessening the severity of one or more diseases and conditions associated with BTK, wherein the disease or condition is selected from heteroimmune conditions or diseases, which include, but are not limited to graft versus host disease, transplantation, transfusion, anaphylaxis, allergies (e.g., allergies to plant pollens, latex, drugs, foods, insect poisons, animal hair, animal dander, dust mites, or cockroach calyx), type I hypersensitivity, allergic conjunctivitis, allergic rhinitis, and atopic dermatitis.

(108) In some embodiments, the present invention provides a method for treating or lessening the severity of one or more diseases and conditions associated with BTK, wherein the disease or condition is selected from an inflammatory disease, e.g., asthma, appendicitis, atopic dermatitis, asthma, allergy, blepharitis, bronchiolitis, bronchitis, bursitis, cervicitis, cholangitis, cholecystitis, chronic graft rejection, colitis, conjunctivitis, Crohn's disease, cystitis, dacryoadenitis, dermatitis, dermatomyositis, encephalitis, endocarditis, endometritis, enteritis, enterocolitis, epicondylitis, epididymitis, fasciitis, fibrositis, gastritis, gastroenteritis, Henoch-Schonlein purpura, hepatitis, hidradenitis suppurativa, immunoglobulin A nephropathy, interstitial lung disease, laryngitis, mastitis, meningitis, myelitis myocarditis, myositis, nephritis, oophoritis, orchitis, osteitis, otitis, pancreatitis, parotitis, pericarditis, peritonitis, pharyngitis, pleuritis, phlebitis, pneumonitis, pneumonia, polymyositis, proctitis, prostatitis, pyelonephritis, rhinitis, salpingitis, sinusitis,

stomatitis, synovitis, tendonitis, tonsillitis, ulcerative colitis, uveitis, vaginitis, vasculitis, or vulvitis.

(109) In some embodiments, the present invention provides a method for treating or lessening the severity of one or more diseases and conditions associated with BTK, wherein the disease or condition is cancer. In one embodiment, the cancer is a B-cell proliferative disorder, e.g., diffuse large B cell lymphoma, follicular lymphoma, chronic lymphocytic lymphoma, chronic lymphocytic leukemia, acute lymphocytic leukemia, B-cell prolymphocytic leukemia, lymphoplasmacytic lymphoma/Waldenstrom macroglobulinemia, splenic marginal zone lymphoma, multiple myeloma (also known as plasma cell myeloma), non-Hodgkin's lymphoma, Hodgkin's lymphoma, plasmacytoma, extranodal marginal zone B cell lymphoma, nodal marginal zone B cell lymphoma, mantle cell lymphoma, mediastinal (thymic) large B cell lymphoma, intravascular large B cell lymphoma, primary effusion lymphoma, Burkitt lymphoma/leukemia, or lymphomatoid granulomatosis. In some embodiments, the cancer is breast cancer, prostate cancer, or cancer of the mast cells (e.g., mastocytoma, mast cell leukemia, mast cell sarcoma, systemic mastocytosis). In one embodiment, the cancer is bone cancer. In another embodiment, the cancer is of other primary origin and metastasizes to the bone. In certain embodiments, the cancer is colorectal cancer or pancreatic cancer.

(110) In some embodiments, the present invention provides a method for treating or lessening the severity of one or more diseases or conditions associated with BTK including diseases of the bone and joints including, without limitation, rheumatoid arthritis, seronegative spondyloarthropathies (including ankylosing spondylitis, psoriatic arthritis and Reiter's disease), Behcet's disease, Sjogren's syndrome, systemic sclerosis, osteoporosis, bone cancer, and bone metastasis.

(111) In some embodiments, the present invention provides a method for treating or lessening the severity of one or more diseases and conditions associated with BTK, wherein the disease or condition is selected from a thromboembolic disorder or cardiovascular disorder, e.g., myocardial infarct, angina pectoris, reocclusion after angioplasty, restenosis after angioplasty, reocclusion after aortocoronary bypass, restenosis after aortocoronary bypass, stroke, transitory ischemia, a peripheral arterial occlusive disorder, pulmonary embolism, or deep venous thrombosis. In certain embodiments, the present invention provides an anti-thrombotic agent because Btk is also involved in the activation of platelets.

(112) In some embodiments, the present invention provides a method for treating or lessening the severity of one or more diseases and conditions associated with BTK, including infectious and noninfectious inflammatory events and autoimmune and other inflammatory diseases. These autoimmune and inflammatory diseases, disorders, and syndromes include inflammatory pelvic disease, urethritis, skin sunburn, sinusitis, pneumonitis, encephalitis, meningitis, myocarditis, nephritis, osteomyelitis, myositis, hepatitis, gastritis, enteritis, dermatitis, gingivitis, appendicitis, pancreatitis, cholecystitis, agammaglobulinemia, psoriasis, allergy, Crohn's disease, irritable bowel syndrome, ulcerative colitis, Sjogren's disease, tissue graft rejection, hyperacute rejection of transplanted organs, asthma, allergic rhinitis, chronic obstructive pulmonary disease (COPD), autoimmune polyglandular disease (also known as autoimmune polyglandular syndrome), autoimmune alopecia, pernicious anemia, glomerulonephritis, dermatomyositis, multiple sclerosis, scleroderma, vasculitis, autoimmune hemolytic and thrombocytopenic states, Goodpasture's syndrome, atherosclerosis, Addison's disease, Parkinson's disease, Alzheimer's disease, diabetes, septic shock, systemic lupus erythematosus (SLE), rheumatoid arthritis, psoriatic arthritis, juvenile arthritis, osteoarthritis, chronic idiopathic thrombocytopenic purpura, Waldenstrom macroglobulinemia, myasthenia gravis, Hashimoto's thyroiditis, atopic dermatitis, degenerative joint disease, vitiligo, autoimmune hypopituitarism, Guillain-Barre syndrome, Behcet's disease, scleroderma, mycosis fungoides, acute inflammatory responses (such as acute respiratory distress syndrome and ischemia/reperfusion injury), and Graves' disease. In certain embodiments, the diabetes is type I diabetes.

(113) In some embodiments, the present invention provides a method for treating or lessening the severity of one or more diseases and conditions associated with BTK, selected from rheumatoid arthritis, multiple sclerosis, B-cell chronic lymphocytic leukemia, acute lymphocytic leukemia, hairy cell leukemia, non-Hodgkin's lymphoma, Hodgkin's lymphoma, multiple myeloma, bone cancer, bone metastasis, osteoporosis, diabetes (e.g. type I diabetes), irritable bowel syndrome, Crohn's disease, lupus and renal transplant.

(114) It is another object of the invention to provide a method for treating diseases that are caused, mediated and/or propagated by BTK activity, wherein at least one compound of formula (I) according to the invention and/or physiologically acceptable salts thereof is administered to a mammal in need of such treatment. In certain embodiments, the invention provides a method for treating lupus, wherein at least one compound of formula (I) according to the invention and/or physiologically acceptable salts thereof is administered to a mammal in need of such treatment. In certain embodiments, the compound is administered in an effective amount as defined above. In certain embodiments, the treatment is an oral administration.

(115) The method of the invention can be performed either in-vitro or in-vivo. The susceptibility of a particular cell to treatment with the compounds according to the invention can be particularly determined by in-vitro tests, whether in the course of research or clinical application. Typically, a culture of the cell is combined with a compound according to the invention at various concentrations for a period of time which is sufficient to allow the active agents to inhibit BTK activity, usually between about one hour and one week. In-vitro treatment can be carried out using cultivated cells from a biopsy sample or cell line.

(116) The host or patient can belong to any mammalian species, for example a primate species, particularly humans; rodents, including mice, rats and hamsters; rabbits; horses, cows, dogs, cats, etc. Animal models are of interest for experimental investigations, providing a model for treatment of human disease.

(117) For identification of a signal transduction pathway and for detection of interactions between various signal transduction pathways, various scientists have developed suitable models or model systems, for example cell culture models and models of transgenic animals. For the determination of certain stages in the signal transduction cascade, interacting compounds can be utilized in order to modulate the signal. The compounds according to the invention can also be used as reagents for testing BTK-dependent signal transduction pathways in animals and/or cell culture models or in the clinical diseases mentioned in this application.

(118) Moreover, the subsequent teaching of the present specification concerning the use of the compounds according to formula (I) and its derivatives for the production of a medicament for the prophylactic or therapeutic treatment and/or monitoring is considered as valid and applicable without restrictions to the use of the compound for the inhibition of BTK activity if expedient.

(119) The invention also relates to the use of compounds according to formula (I) and/or physiologically acceptable salts thereof for the prophylactic or therapeutic treatment and/or monitoring of diseases that are caused, mediated and/or propagated by BTK activity. Furthermore, the invention relates to the use of compounds according to formula (I) and/or physiologically acceptable salts thereof for the production of a medicament for the prophylactic or therapeutic treatment and/or monitoring of diseases that are caused, mediated and/or propagated by BTK activity. In certain embodiments, the invention provides the use of a compound according to formula I or physiologically acceptable salts thereof, for the production of a medicament for the prophylactic or therapeutic treatment of a BTK-mediated disorder.

(120) Compounds of formula (I) and/or a physiologically acceptable salt thereof can furthermore be employed as intermediate for the preparation of further medicament active ingredients. The medicament is preferably prepared in a non-chemical manner, e.g. by combining the active ingredient with at least one solid, fluid and/or semi-fluid carrier or excipient, and optionally in conjunction with a single or more other active substances in an appropriate dosage form.

(121) Another object of the present invention are compounds of formula (I) according to the invention and/or physiologically acceptable salts thereof for use in the prophylactic or therapeutic treatment and/or monitoring of diseases that are caused, mediated and/or propagated by BTK activity. Another preferred object of the invention concerns compounds of formula (I) according to the invention and/or physiologically acceptable salts thereof for use in the prophylactic or therapeutic treatment and/or monitoring of lupus. The prior teaching of the present specification concerning the compounds of formula (I), including any preferred embodiment thereof, is valid and applicable without restrictions to the compounds according to formula (I) and their salts for use in the prophylactic or therapeutic treatment and/or monitoring of lupus.

(122) The compounds of formula (I) according to the invention can be administered before or following an onset of disease once or several times acting as therapy. The aforementioned compounds and medical products of the inventive use are particularly used for the therapeutic treatment. A therapeutically relevant effect relieves to some extent one or more symptoms of a disorder, or returns to normality, either partially or completely, one or more physiological or biochemical parameters associated with or causative of a disease or pathological condition. Monitoring is considered as a kind of treatment provided that the compounds are administered in distinct intervals, e.g. in order to boost the response and eradicate the pathogens and/or symptoms of the disease completely. Either the identical compound or different compounds can be applied. The methods of the invention can also be used to reduce the likelihood of developing a disorder or even prevent the initiation of disorders associated with BTK activity in advance or to treat the arising and continuing symptoms.

(123) In the meaning of the invention, prophylactic treatment is advisable if the subject possesses any preconditions for the aforementioned physiological or pathological conditions, such as a familial disposition, a genetic defect, or a previously incurred disease.

(124) The invention furthermore relates to a medicament comprising at least one compound according to the invention and/or pharmaceutically usable derivatives, salts, solvates and stereoisomers thereof, including mixtures thereof in all ratios. In certain embodiments, the invention relates to a medicament comprising at least one compound according to the invention and/or physiologically acceptable salts thereof.

(125) A “medicament” in the meaning of the invention is any agent in the field of medicine, which comprises one or more compounds of formula (I) or preparations thereof (e.g. a pharmaceutical composition or pharmaceutical formulation) and can be used in prophylaxis, therapy, follow-up or aftercare of patients who suffer from diseases, which are associated with BTK activity, in such a way that a pathogenic modification of their overall condition or of the condition of particular regions of the organism could establish at least temporarily.

(126) In various embodiments, the active ingredient may be administered alone or in combination with other treatments. A synergistic effect may be achieved by using more than one compound in the pharmaceutical composition, i.e. the compound of formula (I) is combined with at least another agent as active ingredient, which is either another compound of formula (I) or a compound of different structural scaffold. The active ingredients can be used either simultaneously or sequentially.

(127) Included herein are methods of treatment in which at least one chemical entity provided herein is administered in combination with an anti-inflammatory agent. Anti-inflammatory agents include but are not limited to NSAIDs, non-specific and COX-2 specific cyclooxygenase enzyme inhibitors, gold compounds, corticosteroids, methotrexate, tumor necrosis factor (TNF) antagonists, immunosuppressants and methotrexate.

(128) Examples of NSAIDs include, but are not limited to, ibuprofen, flurbiprofen, naproxen and naproxen sodium, diclofenac, combinations of diclofenac sodium and misoprostol, sulindac, oxaprozin, diflunisal, piroxicam, indomethacin, etodolac, fenoprofen calcium, ketoprofen, sodium nabumetone, sulfasalazine, tolmetin sodium, and hydroxychloroquine. Examples of NSAIDs also

include COX-2 specific inhibitors such as celecoxib, valdecoxib, lumiracoxib and/or etoricoxib.

(129) In some embodiments, the anti-inflammatory agent is a salicylate. Salicylates include but are not limited to acetylsalicylic acid or aspirin, sodium salicylate, and choline and magnesium salicylates.

(130) The anti-inflammatory agent may also be a corticosteroid. For example, the corticosteroid may be cortisone, dexamethasone, methylprednisolone, prednisolone, prednisolone sodium phosphate, or prednisone.

(131) In additional embodiments the anti-inflammatory agent is a gold compound such as gold sodium thiomalate or auranofin.

(132) The invention also includes embodiments in which the anti-inflammatory agent is a metabolic inhibitor such as a dihydrofolate reductase inhibitor, such as methotrexate or a dihydroorotate dehydrogenase inhibitor, such as leflunomide.

(133) Other embodiments of the invention pertain to combinations in which at least one anti-inflammatory compound is an anti-monoclonal antibody (such as eculizumab or pexelizumab), a TNF antagonist, such as etanercept, or infliximab, which is an anti-TNF alpha monoclonal antibody.

(134) Still other embodiments of the invention pertain to combinations in which at least one active agent is an immunosuppressant compound such as an immunosuppressant compound chosen from methotrexate, leflunomide, cyclosporine, tacrolimus, azathioprine, and mycophenolate mofetil.

(135) B-cells and B-cell precursors expressing BTK have been implicated in the pathology of B-cell malignancies, including, but not limited to, B-cell lymphoma, lymphoma (including Hodgkin's and non-Hodgkin's lymphoma), hairy cell lymphoma, multiple myeloma, chronic and acute myelogenous leukemia and chronic and acute lymphocytic leukemia.

(136) BTK has been shown to be an inhibitor of the Fas/APO-1 (CD-95) death inducing signaling complex (DISC) in B-lineage lymphoid cells. The fate of leukemia/lymphoma cells may reside in the balance between the opposing proapoptotic effects of caspases activated by DISC and an upstream anti-apoptotic regulatory mechanism involving BTK and/or its substrates (Vassilev et al., J. Biol. Chem. 1998, 274, 1646-1656).

(137) It has also been discovered that BTK inhibitors are useful as chemosensitizing agents, and, thus, are useful in combination with other chemotherapeutic drugs, in particular, drugs that induce apoptosis. Examples of other chemotherapeutic drugs that can be used in combination with chemosensitizing BTK inhibitors include topoisomerase I inhibitors (camptothecin or topotecan), topoisomerase II inhibitors (e.g. daunomycin and etoposide), alkylating agents (e.g. cyclophosphamide, melphalan and BCNU), tubulin directed agents (e.g. taxol and vinblastine), and biological agents (e.g. antibodies such as anti CD20 antibody, IDEC 8, immunotoxins, and cytokines).

(138) The disclosed compounds of the formula I can be administered in combination with other known therapeutic agents, including anticancer agents. As used here, the term "anticancer agent" relates to any agent which is administered to a patient with cancer for the purposes of treating the cancer.

(139) The anti-cancer treatment defined above may be applied as a monotherapy or may involve, in addition to the herein disclosed compounds of formula I, conventional surgery or radiotherapy or medicinal therapy. Such medicinal therapy, e.g. a chemotherapy or a targeted therapy, may include one or more, but preferably one, of the following anti-tumor agents: Alkylating agents: such as altretamine, bendamustine, busulfan, carmustine, chlorambucil, chlormethine, cyclophosphamide, dacarbazine, ifosfamide, improsulfan, tosilate, lomustine, melphalan, mitobronitol, mitolactol, nimustine, ranimustine, temozolomide, thiotepe, treosulfan, mechlorethamine, carboquone; apaziquone, fotemustine, glufosfamide, palifosfamide, pipobroman, trofosfamide, uramustine, TH-302.sup.4, VAL-083.sup.4; Platinum Compounds: such as carboplatin, cisplatin, eptaplatin, miriplatine hydrate, oxaliplatin, lobaplatin, nedaplatin, picoplatin, satraplatin; lobaplatin,

nedaplatin, picoplatin, satraplatin; DNA altering agents: such as amrubicin, bisantrene, decitabine, mitoxantrone, procarbazine, trabectedin, clofarabine; amsacrine, brostallicin, pixantrone, laromustine.sup.1,3; Topoisomerase Inhibitors: such as etoposide, irinotecan, razoxane, sobuzoxane, teniposide, topotecan; amonafide, belotecan, elliptinium acetate, voreloxin; Microtubule modifiers: such as cabazitaxel, docetaxel, eribulin, ixabepilone, paclitaxel, vinblastine, vincristine, vinorelbine, vindesine, vinflunine; fosbretabulin, tesetaxel; Antimetabolites: such as asparaginase.sup.3, azacitidine, calcium levofolinate, capecitabine, cladribine, cytarabine, enocitabine, floxuridine, fludarabine, fluorouracil, gemcitabine, mercaptopurine, methotrexate, nelarabine, pemetrexed, pralatrexate, azathioprine, thioguanine, carmofur; doxifluridine, elacytarabine, raltitrexed, sapacitabine, tegafur.sup.2,3, trimetrexate; Anticancer antibiotics: such as bleomycin, dactinomycin, doxorubicin, epirubicin, idarubicin, levamisole, miltefosine, mitomycin C, romidepsin, streptozocin, valrubicin, zinostatin, zorubicin, daunorubicin, plicamycin; aclarubicin, peplomycin, pirarubicin; Hormones/Antagonists: such as abarelix, abiraterone, bicalutamide, buserelin, calusterone, chlorotrianisene, degarelix, dexamethasone, estradiol, fluocortolone fluoxymesterone, flutamide, fulvestrant, goserelin, histrelin, leuprorelin, megestrol, mitotane, nafarelin, nandrolone, nilutamide, octreotide, prednisolone, raloxifene, tamoxifen, thyrotropin alfa, toremifene, trilostane, triptorelin, diethylstilbestrol; acolbifene, danazol, deslorelin, epitiostanol, orteronel, enzalutamide.sup.1,3; Aromatase inhibitors: such as aminoglutethimide, anastrozole, exemestane, fadrozole, letrozole, testolactone; formestane; Small molecule kinase inhibitors: such as crizotinib, dasatinib, erlotinib, imatinib, lapatinib, nilotinib, pazopanib, regorafenib, ruxolitinib, sorafenib, sunitinib, vandetanib, vemurafenib, bosutinib, gefitinib, axitinib; afatinib, alisertib, dabrafenib, dacomitinib, dinaciclib, dovitinib, enzastaurin, nintedanib, lenvatinib, linifanib, linsitinib, masitinib, midostaurin, motesanib, neratinib, orantinib, perifosine, ponatinib, radotinib, rigosertib, tipifarnib, tivantinib, tivozanib, trametinib, pimasertib, brivanib alaninate, cediranib, apatinib.sup.4, cabozantinib S-malate.sup.1,3, ibrutinib.sup.1,3, icotinib.sup.4, buparlisib.sup.2, cipatinib.sup.4, cobimetinibi.sup.1,3, idelalisibi.sup.1,3, fedratinibi, XL-647.sup.4; Photosensitizers: such as methoxsalen.sup.3; porfimer sodium, talaporfin, temoporfin; Antibodies: such as alemtuzumab, besilesomab, brentuximab vedotin, cetuximab, denosumab, ipilimumab, ofatumumab, panitumumab, rituximab, tositumomab, trastuzumab, bevacizumab, pertuzumab.sup.2,3; catumaxomab, elotuzumab, epratuzumab, farletuzumab, mogamulizumab, necitumumab, nimotuzumab, obinutuzumab, ocaratuzumab, oregovomab, ramucirumab, rilotumumab, siltuximab, tocilizumab, zalutumumab, zanolimumab, matuzumab, dalotuzumab.sup.1,2,3 onartuzumab.sup.1,3 racotumomabi, tabalumab.sup.1,3 EMD-525797.sup.4, nivolumab.sup.1,3. Cytokines: such as aldesleukin, interferon alfa.sup.2, interferon alfa2a.sup.3, interferon alfa2b.sup.2,3; celmoleukin, tasonermin, teceleukin, oprelvekin.sup.1,3, recombinant interferon beta-1a.sup.4; Drug Conjugates: such as denileukin diftitox, ibritumomab tiuxetan, iobenguane I123, prednimustine, trastuzumab emtansine, estramustine, gemtuzumab, ozogamicin, aflibercept; cintredekin besudotox, edotreotide, inotuzumab ozogamicin, naptumomab estafenatox, oportuzumab monatox, technetium (99mTc) arcitumomabi.sup.1,3, vintafolide.sup.1,3; Vaccines: such as sipuleucel.sup.3; vitespen.sup.3, emepepimut-S.sup.3, oncoVAX.sup.4, rindopepimut.sup.3, troVax.sup.4, MGN-1601.sup.4, MGN-1703.sup.4; and Miscellaneous: alitretinoin, bexarotene, bortezomib, everolimus, ibandronic acid, imiquimod, lenalidomide, lentinan, metirosine, mifamurtide, pamidronic acid, pegaspargase, pentostatin, sipuleucel.sup.3, sizofiran, tamibarotene, temsirolimus, thalidomide, tretinoin, vismodegib, zoledronic acid, vorinostat; celecoxib, cilengitide, entinostat, etanidazole, ganetespib, idronoxil, iniparib, ixazomib, lonidamine, nimorazole, panobinostat, peretinoin, plitidepsin, pomalidomide, procodazol, ridaforolimus, tasquinimod, telotristat, thymalfasin, tirapazamine, tosedostat, trabedersen, ubenimex, valsopodar, gendicine.sup.4, picibanil.sup.4, reolysin.sup.4, retaspimycin hydrochloride.sup.1,3, trebananib.sup.2,3, virulizin.sup.4, carfilzomibi.sup.3, endostatin.sup.4, immucothel.sup.4, belinostat.sup.3, MGN-1703.sup.4. (.sup.1Prop. INN (Proposed International

Nonproprietary Name); .sup.2 Rec. INN (Recommended International Nonproprietary Names); .sup.3 USAN (United States Adopted Name); .sup.4 no INN).

(140) In another aspect, the invention provides for a kit consisting of separate packs of an effective amount of a compound according to the invention and/or pharmaceutically acceptable salts, derivatives, solvates and stereoisomers thereof, including mixtures thereof in all ratios, and optionally, an effective amount of a further active ingredient. The kit comprises suitable containers, such as boxes, individual bottles, bags or ampoules. The kit may, for example, comprise separate ampoules, each containing an effective amount of a compound according to the invention and/or pharmaceutically acceptable salts, derivatives, solvates and stereoisomers thereof, including mixtures thereof in all ratios, and an effective amount of a further active ingredient in dissolved or lyophilized form.

(141) As used herein, the terms "treatment," "treat," and "treating" refer to reversing, alleviating, delaying the onset of, or inhibiting the progress of a disease or disorder, or one or more symptoms thereof, as described herein. In some embodiments, treatment is administered after one or more symptoms have developed. In other embodiments, treatment is administered in the absence of symptoms. For example, treatment is administered to a susceptible individual prior to the onset of symptoms (e.g., in light of a history of symptoms and/or in light of genetic or other susceptibility factors). Treatment is also continued after symptoms have resolved, for example to prevent or delay their recurrence.

(142) The compounds and compositions, according to the method of the present invention, are administered using any amount and any route of administration effective for treating or lessening the severity of a disorder provided above. The exact amount required will vary from subject to subject, depending on the species, age, and general condition of the subject, the severity of the infection, the particular agent, its mode of administration, and the like. Compounds of the invention are preferably formulated in dosage unit form for ease of administration and uniformity of dosage. The expression "dosage unit form" as used herein refers to a physically discrete unit of agent appropriate for the patient to be treated. It will be understood, however, that the total daily usage of the compounds and compositions of the present invention will be decided by the attending physician within the scope of sound medical judgment. The specific effective dose level for any particular patient or organism will depend upon a variety of factors including the disorder being treated and the severity of the disorder; the activity of the specific compound employed; the specific composition employed; the age, body weight, general health, sex and diet of the patient; the time of administration, route of administration, and rate of excretion of the specific compound employed; the duration of the treatment; drugs used in combination or coincidental with the specific compound employed, and like factors well known in the medical arts.

(143) Pharmaceutically acceptable compositions of this invention can be administered to humans and other animals orally, rectally, parenterally, intracisternally, intravaginally, intraperitoneally, topically (as by powders, ointments, or drops), buccally, as an oral or nasal spray, or the like, depending on the severity of the infection being treated. In certain embodiments, the compounds of the invention are administered orally or parenterally at dosage levels of about 0.01 mg/kg to about 100 mg/kg and preferably from about 1 mg/kg to about 50 mg/kg, of subject body weight per day, one or more times a day, to obtain the desired therapeutic effect.

(144) Liquid dosage forms for oral administration include, but are not limited to, pharmaceutically acceptable emulsions, microemulsions, solutions, suspensions, syrups and elixirs. In addition to the active compounds, the liquid dosage forms optionally contain inert diluents commonly used in the art such as, for example, water or other solvents, solubilizing agents and emulsifiers such as ethyl alcohol, isopropyl alcohol, ethyl carbonate, ethyl acetate, benzyl alcohol, benzyl benzoate, propylene glycol, 1,3-butylene glycol, dimethylformamide, oils (in particular, cottonseed, groundnut, corn, germ, olive, castor, and sesame oils), glycerol, tetrahydrofurfuryl alcohol, polyethylene glycols and fatty acid esters of sorbitan, and mixtures thereof. Besides inert diluents,

the oral compositions can also include adjuvants such as wetting agents, emulsifying and suspending agents, sweetening, flavoring, and perfuming agents.

(145) Injectable preparations, for example, sterile injectable aqueous or oleaginous suspensions are formulated according to the known art using suitable dispersing or wetting agents and suspending agents. The sterile injectable preparation are also a sterile injectable solution, suspension or emulsion in a nontoxic parenterally acceptable diluent or solvent, for example, as a solution in 1,3-butanediol. Among the acceptable vehicles and solvents that may be employed are water, Ringer's solution, U.S.P. and isotonic sodium chloride solution. In addition, sterile, fixed oils are conventionally employed as a solvent or suspending medium. For this purpose any bland fixed oil can be employed including synthetic mono- or diglycerides. In addition, fatty acids such as oleic acid are used in the preparation of injectables.

(146) Injectable formulations can be sterilized, for example, by filtration through a bacterial-retaining filter, or by incorporating sterilizing agents in the form of sterile solid compositions which can be dissolved or dispersed in sterile water or other sterile injectable medium prior to use.

(147) In order to prolong the effect of a compound of the present invention, it is often desirable to slow the absorption of the compound from subcutaneous or intramuscular injection. This is accomplished by the use of a liquid suspension of crystalline or amorphous material with poor water solubility. The rate of absorption of the compound then depends upon its rate of dissolution that, in turn, may depend upon crystal size and crystalline form. Alternatively, delayed absorption of a parenterally administered compound form is accomplished by dissolving or suspending the compound in an oil vehicle. Injectable depot forms are made by forming microencapsule matrices of the compound in biodegradable polymers such as polylactide-polyglycolide. Depending upon the ratio of compound to polymer and the nature of the particular polymer employed, the rate of compound release can be controlled. Examples of other biodegradable polymers include poly(orthoesters) and poly(anhydrides). Depot injectable formulations are also prepared by entrapping the compound in liposomes or microemulsions that are compatible with body tissues.

(148) Compositions for rectal or vaginal administration are preferably suppositories which can be prepared by mixing the compounds of this invention with suitable non-irritating excipients or carriers such as cocoa butter, polyethylene glycol or a suppository wax which are solid at ambient temperature but liquid at body temperature and therefore melt in the rectum or vaginal cavity and release the active compound.

(149) Solid dosage forms for oral administration include capsules, tablets, pills, powders, and granules. In such solid dosage forms, the active compound is mixed with at least one inert, pharmaceutically acceptable excipient or carrier such as sodium citrate or dicalcium phosphate and/or a) fillers or extenders such as starches, lactose, sucrose, glucose, mannitol, and silicic acid, b) binders such as, for example, carboxymethylcellulose, alginates, gelatin, polyvinylpyrrolidone, sucrose, and acacia, c) humectants such as glycerol, d) disintegrating agents such as agar-agar, calcium carbonate, potato or tapioca starch, alginic acid, certain silicates, and sodium carbonate, e) solution retarding agents such as paraffin, f) absorption accelerators such as quaternary ammonium compounds, g) wetting agents such as, for example, cetyl alcohol and glycerol monostearate, h) absorbents such as kaolin and bentonite clay, and i) lubricants such as talc, calcium stearate, magnesium stearate, solid polyethylene glycols, sodium lauryl sulfate, and mixtures thereof. In the case of capsules, tablets and pills, the dosage form also optionally comprises buffering agents.

(150) Solid compositions of a similar type are also employed as fillers in soft and hard-filled gelatin capsules using such excipients as lactose or milk sugar as well as high molecular weight polyethylene glycols and the like. The solid dosage forms of tablets, dragees, capsules, pills, and granules can be prepared with coatings and shells such as enteric coatings and other coatings well known in the pharmaceutical formulating art. They optionally contain opacifying agents and can also be of a composition that they release the active ingredient(s) only, or preferentially, in a certain part of the intestinal tract, optionally, in a delayed manner. Examples of embedding compositions

that can be used include polymeric substances and waxes. Solid compositions of a similar type are also employed as fillers in soft and hard-filled gelatin capsules using such excipients as lactose or milk sugar as well as high molecular weight polyethylene glycols and the like.

(151) The active compounds can also be in micro-encapsulated form with one or more excipients as noted above. The solid dosage forms of tablets, dragees, capsules, pills, and granules can be prepared with coatings and shells such as enteric coatings, release controlling coatings and other coatings well known in the pharmaceutical formulating art. In such solid dosage forms the active compound may be admixed with at least one inert diluent such as sucrose, lactose or starch. Such dosage forms also comprise, as is normal practice, additional substances other than inert diluents, e.g., tableting lubricants and other tableting aids such as magnesium stearate and microcrystalline cellulose. In the case of capsules, tablets and pills, the dosage forms optionally also comprise buffering agents. They optionally contain opacifying agents and can also be of a composition that they release the active ingredient(s) only, or preferentially, in a certain part of the intestinal tract, optionally, in a delayed manner. Examples of embedding compositions that can be used include polymeric substances and waxes.

(152) Dosage forms for topical or transdermal administration of a compound of this invention include ointments, pastes, creams, lotions, gels, powders, solutions, sprays, inhalants or patches. The active component is admixed under sterile conditions with a pharmaceutically acceptable carrier and any needed preservatives or buffers as required. Ophthalmic formulation, ear drops, and eye drops are also contemplated as being within the scope of this invention. Additionally, the present invention contemplates the use of transdermal patches, which have the added advantage of providing controlled delivery of a compound to the body. Such dosage forms can be made by dissolving or dispersing the compound in the proper medium. Absorption enhancers can also be used to increase the flux of the compound across the skin. The rate can be controlled by either providing a rate controlling membrane or by dispersing the compound in a polymer matrix or gel.

(153) According to one embodiment, the invention relates to a method of inhibiting BTK activity in a biological sample comprising the step of contacting said biological sample with a compound of this invention, or a composition comprising said compound.

(154) According to another embodiment, the invention relates to a method of inhibiting BTK, or a mutant thereof, activity in a biological sample in a positive manner, comprising the step of contacting said biological sample with a compound of this invention, or a composition comprising said compound.

(155) The compounds of the invention are useful in-vitro as unique tools for understanding the biological role of BTK, including the evaluation of the many factors thought to influence, and be influenced by, the production of BTK and the interaction of BTK. The present compounds are also useful in the development of other compounds that interact with BTK since the present compounds provide important structure-activity relationship (SAR) information that facilitate that development. Compounds of the present invention that bind to BTK can be used as reagents for detecting BTK in living cells, fixed cells, in biological fluids, in tissue homogenates, in purified, natural biological materials, etc. For example, by labeling such compounds, one can identify cells expressing BTK. In addition, based on their ability to bind BTK, compounds of the present invention can be used in in-situ staining, FACS (fluorescence-activated cell sorting), sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE), ELISA (enzyme-linked immunoadsorbent assay), etc., enzyme purification, or in purifying cells expressing BTK inside permeabilized cells. The compounds of the invention can also be utilized as commercial research reagents for various medical research and diagnostic uses. Such uses can include but are not limited to: use as a calibration standard for quantifying the activities of candidate BTK inhibitors in a variety of functional assays; use as blocking reagents in random compound screening, i.e. in looking for new families of BTK ligands, the compounds can be used to block recovery of the presently claimed BTK compounds; use in the co-crystallization with BTK enzyme, i.e. the

compounds of the present invention will allow formation of crystals of the compound bound to BTK, enabling the determination of enzyme/compound structure by x-ray crystallography; other research and diagnostic applications, wherein BTK is preferably activated or such activation is conveniently calibrated against a known quantity of an BTKinhibitor, etc.; use in assays as probes for determining the expression of BTK in cells; and developing assays for detecting compounds which bind to the same site as the BTK binding ligands.

(156) The compounds of the invention can be applied either themselves and/or in combination with physical measurements for diagnostics of treatment effectiveness. Pharmaceutical compositions containing said compounds and the use of said compounds to treat BTK-mediated conditions is a promising, novel approach for a broad spectrum of therapies causing a direct and immediate improvement in the state of health, whether in human or animal. The orally bioavailable and active new chemical entities of the invention improve convenience for patients and compliance for physicians.

(157) The compounds of formula (I), their salts, isomers, tautomers, enantiomeric forms, diastereomers, racemates, derivatives, prodrugs and/or metabolites are characterized by a high specificity and stability, low manufacturing costs and convenient handling. These features form the basis for a reproducible action, wherein the lack of cross-reactivity is included, and for a reliable and safe interaction with the target structure.

(158) The term “biological sample”, as used herein, includes, without limitation, cell cultures or extracts thereof; biopsied material obtained from a mammal or extracts thereof; and blood, saliva, urine, feces, semen, tears, or other body fluids or extracts thereof.

(159) Modulation of BTK, or a mutant thereof, activity in a biological sample is useful for a variety of purposes that are known to one of skill in the art. Examples of such purposes include, but are not limited to, blood transfusion, organ transplantation, biological specimen storage, and biological assays.

Exemplification

General Conditions and Analytical Methods

(160) As depicted in the Examples below, in certain exemplary embodiments, compounds were prepared according to the following general procedures. It will be appreciated that, although the general methods depict the synthesis of certain compounds of the present invention, the following general methods, and other methods known to one of ordinary skill in the art, can be applied to all compounds and subclasses and species of each of these compounds, as described herein.

(161) The symbols and conventions used in the following descriptions of processes, schemes, and examples are consistent with those used in the contemporary scientific literature, for example, the Journal of the American Chemical Society or the Journal of Biological Chemistry.

(162) Unless otherwise indicated, all temperatures are expressed in ° C. (degrees Centigrade). All reactions were conducted at room temperature unless otherwise noted. All compounds of the present invention were synthesized by processes developed by the inventors.

(163) ¹H-NMR spectra were recorded on a Bruker Avance III 400 MHz. Chemical shifts are expressed in parts per million (ppm, δ units). Coupling constants are in units of Hertz (Hz). Splitting patterns describe apparent multiplicities and are designated as s (singlet), d (doublet), t (triplet), q (quartet), m (multiplet), or br (broad).

(164) Mass spectra were obtained on Waters Acquity UPLC/MS systems, using either Atmospheric Chemical Ionization (APCI) or Electrospray Ionization (ESI). Column: XBridge C8, 3.5 μ m, 4.6 $\times$ 50 mm; Solvent A: water+0.1% FA; Solvent B: CAN; Flow: 0.8 ml/min; Gradient: 15-95% B over 1.2 minutes min.

(165) HPLC data were obtained using Agilent 1100 series HPLC from Agilent technologies using XBridge column (C8, 3.5 μ m, 4.6 $\times$ 50 mm). Solvent A: water+0.1% TFA; Solvent B: ACN; Flow: 2 ml/min; Gradient: 0 min: 5% B, 8 min: 100% B, 8.1 min: 100% B, 8.5 min: 5% B, 10 min 5% B.

(166) The microwave reactions were conducted using Biotage Initiator Microwave Synthesizer

using standard protocols that are known in the art.

(167) Some abbreviations that may appear in this application are as follows:

(168) TABLE-US-00002 Aq. Aqueous BBFO Broad band fluorine observation BrettPhos 2-(Dicyclohexylphosphino)3,6-dimethoxy-2',4',6'- triisopropyl-1,1'-biphenyl δ chemical shift d deuterium or doublet dd doublet of doublets DCM dichloromethane DIPEA diisopropylethylamine DMF dimethylformamide DMSO dimethylsulfoxide dppf 1,1'-bis(diphenylphosphino)ferrocene eq. equivalent ES electrospray h hour .sup.1H proton HPLC high pressure liquid chromatography IR infrared J coupling constant K kelvin LC liquid chromatography m Multiplet or meta M molecular ion Me methyl MHz Megahertz min minute mL milliliter MS mass spectrometry m/z mass-to-charge ratio N Normality (equivalent per liter) NMR nuclear magnetic resonance Pet petroleum RBF Round Bottom Flask RT room temperature RuPhos 2-Dicyclohexylphosphino-2',6'-diisopropoxybiphenyl s singlet TFA Trifluoroacetic acid THF tetrahydrofuran TLC thin layer chromatography UFLC Ultra fast liquid chromatography UV ultraviolet V Volume VT Variable temperature

(169) Compound numbers utilized in the Examples below correspond to compound numbers set forth supra.

(170) ##STR00206## ##STR00207##

(171) ##STR00208##

(172) (4E)-5-(3-bromophenyl)pent-4-enoic acid (Step 1): In a 250-mL round bottom flask purged and maintained with an inert atmosphere of nitrogen, to a solution of (3-carboxypropyl)triphenylphosphonium bromide (23.26 g, 51.5 mmol, 1.00 equiv) in DMSO (200 mL) was added sodium hydride (60% in oil, 5.4 g, 135.0 mmol, 2.63 equiv) at 0° C. The resulting mixture was stirred for 1 h at 25° C., and then was added by 3-bromobenzaldehyde (10 g, 51.4 mmol, 1.00 equiv) in batches. The reaction mixture was allowed to warm up to 30° C. and stirred for another 16 h. When the reaction was done, it was quenched by the addition of water (200 mL) and the pH value of the resulting mixture was adjusted to ~1 with hydrogen chloride solution (2 N). The mixture was then extracted with ethyl acetate (3×100 mL), and the combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified in a silica gel column eluting with ethyl acetate in petroleum ether (10% to 80% gradient) to afford (4E)-5-(3-bromophenyl)pent-4-enoic acid as yellow oil (13.8 g, 90%). MS: m/z=253.1 [M-H].sup.-

(173) 5-(3-Bromophenyl)pentanoic acid (Step 2): In a 500-mL round bottom flask, to a solution of (4E)-5-(3-bromophenyl)pent-4-enoic acid (16.2 g, 63.5 mmol, 1.00 equiv) and AcOH (2 mL, 34.9 mmol, 0.82 equiv) in ethyl acetate (200 mL) was added Pd/C (10%, 6 g) under nitrogen atmosphere. The reaction flask was vacuumed and flushed with hydrogen. The reaction mixture was hydrogenated at room temperature for 16 hours under hydrogen atmosphere using a hydrogen balloon. After the reaction was done, the reaction mixture was filtered through a Celite pad and the filtrate was concentrated under reduced pressure to afford 5-(3-bromophenyl)pentanoic acid as light brown oil (16 g, 99%).

(174) 2-Bromo-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-one (Step 3): In a 250-mL round bottom flask, a mixture of 5-(3-bromophenyl)pentanoic acid (5.4 g, 21.0 mol, 1 equiv) in polyphosphoric acid (60 mL) was stirred overnight at 130° C. After the reaction was done, the reaction mixture was cooled to 0° C. and quenched by the addition of sat. NaHCO.sub.3 solution slowly. The resulting mixture was extracted with EtOAc (3×100 mL), and the combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by silica gel column eluting with ethyl acetate in petroleum ether (1% to 10% gradient) to afford 2-bromo-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-one as brown oil (3 g, 53.8%). m/z=238.9 [M+H].sup.+

(175) 2-Bromo-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine (Step 4): In a 250-mL round bottom flask purged and maintained with an inert atmosphere of nitrogen, 2-bromo-6,7,8,9-

tetrahydro-5H-benzo[7]annulen-5-one (2.7 g, 11.3 mol, 1 equiv), NH₄sub.4OAc (17.46 g, 226.5 mmol, 20.0 equiv) and NaBH₄sub.3CN (3.6 g, 57.3 mmol, 5.07 equiv) were mixed in i-PrOH (60 mL) at room temperature. The resulting mixture was stirred overnight at 80° C. When the reaction was done, it was quenched by the addition of NaOH solution (1 N, 100 mL) and the mixture was extracted with DCM (3×100 mL). The combined organic layers were washed by brine, dried over sodium sulfate and concentrated under reduced pressure to afford 2-bromo-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine as brown oil (2.7 g, 90%). m/z=240.0 [M+H]⁺.sup.+

(176) tert-butyl N-[2-bromo-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (Step 5): In a 250-mL round bottom flask, to a solution of 2-bromo-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine (3 g, 11.2 mmol, 1 equiv) and TEA (1.27 g, 11.3 mmol, 1.00 equiv) in dichloromethane (30 mL) was added (Boc)₂O (2.74 g, 11.3 mmol, 1.0 equiv) at 0° C. The resulting solution was stirred for 16 h at 25° C. When the reaction was done, it was quenched by the addition of water (10 mL) and the mixture was extracted with dichloromethane (3×20 mL). The combined organic layers were washed by brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified in a silica gel column eluting with dichloromethane in petroleum ether (5% to 50% gradient) to afford tert-butyl N-[2-bromo-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate as brown solid (2.5 g, 58.8%). MS: m/z=284.0 [M+H-56]⁺.sup.-

(177) tert-butyl N-[2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (Step 6): In a 250-mL round bottom flask purged and maintained with an inert atmosphere of nitrogen, to a stirring solution of tert-butyl N-[2-bromo-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (2.25 g, 6.6 mol, 1 equiv) and 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (2.8 g, 10.5 mmol, 1.58 equiv) in dioxane (40 mL) were added Pd(dppf)Cl₂sub.2 CH₂sub.2Cl₂sub.2 (0.60 g, 0.73 mmol, 1.58 equiv) and KOAc (1.45 g, 14.0 mmol, 2.12 equiv) at room temperature. The resulting mixture was stirred at 100° C. for 2 h. After the reaction was done, the reaction mixture was filtered to remove insoluble solids and the filtrate was concentrated under reduced pressure to give tert-butyl N-[2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate as black oil (2.84 g, crude). MS: m/z=388.0 [M+H]⁺.sup.+

(178) tert-butyl N-[2-(2-amino-3-nitropyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (Step 7): In a 250-mL round bottom flask purged and maintained with an inert atmosphere of nitrogen, to a solution of tert-butyl N-[2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (2.56 g, 6.60 mol, 1 equiv) and 4-chloro-3-nitropyridin-2-amine (1.27 g, 6.95 mol, 1.05 equiv) in dioxane (30 mL) and water (3 mL) were added Pd(PPh₃)₃sub.2Cl₂sub.2 (0.52 g, 0.70 mmol, 0.11 equiv) and Na₂sub.2CO₃sub.3 (2.34 g, 20.97 mmol, 3.18 equiv) at room temperature. The resulting mixture was stirred at 100° C. for 16 h. After the reaction was done, the reaction mixture was filtered through a Celite pad and the filtrate was concentrated under reduced pressure. The residue was purified by silica gel column eluting with ethyl acetate in petroleum ether (5% to 30% gradient) to afford tert-butyl N-[2-(2-amino-3-nitropyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate as brown oil (3.5 g, 93.2%). MS: m/z=399.1 [M+H]⁺.sup.+

(179) tert-butyl N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (Step 8): To a solution of tert-butyl N-[2-(2-amino-3-nitropyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (5 g, 11.29 mol, 1 equiv, 90%) in 200 mL MeOH was added Pd/C (10%, 2.5 g) under nitrogen atmosphere. The mixture was hydrogenated at room temperature for 16 hours under hydrogen atmosphere using a hydrogen balloon, filtered through a Celite pad and concentrated under reduced pressure to give tert-butyl N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate as a brown oil (4.6 g, 77.38%). MS: m/z=369.1 [M+H]⁺.sup.+

(180) tert-butyl N-(2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl)carbamate (Step 9): In a 250-mL

round bottom flask purged and maintained with an inert atmosphere of nitrogen, to a stirring solution of tert-butyl N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (2.73 g, 7.4 mmol, 1 equiv) and 3,5-dimethyl-1-(propan-2-yl)-1H-pyrazole-4-carbaldehyde (1.23 g, 7.0 mmol, 0.95 equiv) in DMF (75 mL) was added 4-methylbenzene-1-sulfonic acid (0.18 g, 0.99 mmol, 0.13 equiv) at room temperature. The resulting mixture was stirred for 16 h at 100° C. After the reaction was done, the reaction mixture was concentrated under reduced pressure. The residue was purified by silica gel column eluting with methanol in DCM (1% to 10% gradient) to afford tert-butyl N-(2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl)carbamate as brown oil (2.16 g, 56.6%). MS: m/z=515.1[M+H].sup.+.

(181) 2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine (Step 10): At room temperature, tert-butyl N-(2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl)carbamate (4.8 g, 8.14 mmol, 1 equiv) was added to a solution of HCl in MeOH (4 M, 40 mL, 160 mmol). The resulting mixture was stirred for 1 h at room temperature, and then was concentrated under reduced pressure. The residue was purified by reverse phase flash chromatography eluting with MeOH in water (10% to 50% gradient in 60 min) to afford 2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine (2.5 g, 66.68%) as a brown solid. MS: m/z=415.6 [M+H].sup.+.

(182) 3-tert-butyl-N-[(5R)-2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2,4-oxadiazole-5-carboxamide (Step 11): In a 100-mL round bottom flask, to a solution of 12-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine (1.8 g, 4.3 mmol, 1 equiv) and TEA (2.44 g, 22.9 mmol, 5.28 equiv) in MeOH (30 mL) was added methyl 3-tert-butyl-1,2,4-oxadiazole-5-carboxylate (4.0 g, 21.7 mmol, 5.00 equiv) at room temperature. The resulting solution was stirred for 16 h at 50° C. When the reaction was done, it was quenched by the addition of water (20 mL) and the mixture was extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was first purified in a silica gel column eluting with ethyl acetate in petroleum ether (10% to 90% gradient) to afford the product as a mixture of two enantiomers, which was further separated and purified by chiral-HPLC with the following conditions: column, CHIRALPAK IC-33×100 mm 3 um; mobile phase, methanol (with 20 mM NH₃·H₂O); detector, 220/254 nm. 3-tert-butyl-N-[(5R)-2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2,4-oxadiazole-5-carboxamide was obtained as white solid (549.3 mg, 49.7%). HPLC: 98.9% purity. MS: m/z=567.4 [M+H].sup.+ .sup.1H-NMR (300 MHz, DMSO-d₆, ppm): δ 12.78 (s, 1H), 9.99 (d, J=8.1 Hz, 1H), 8.31-8.25 (m, 2H), 8.18 (d, J=8.0 Hz, 1H), 7.53 (d, J=5.3 Hz, 1H), 7.38 (d, J=8.1 Hz, 1H), 5.32-5.27 (m, 1H), 4.60-4.56 (m, 1H), 2.99-2.98 (m, 2H), 2.57 (s, 3H), 2.44 (s, 3H), 1.98-1.81 (m, 6H), 1.42-1.40 (m, 15H).

(183) ##STR00209## ##STR00210##

Example 2

(184) ##STR00211##

(185) 3,5-dimethyl-1-(propan-2-yl)-1H-pyrazole-4-carbaldehyde (Step 1): In a 500-mL round-bottom flask, to a solution of 3,5-dimethyl-1H-pyrazole-4-carbaldehyde (8 g, 58.00 mmol, 1.00 equiv) and 2-iodopropane (14.17 g, 75.02 mmol, 1.29 equiv) in MeCN (160 mL) was added Cs₂CO₃ (27.17 g, 75.05 mmol, 1.29 equiv) at room temperature. The resulting solution was stirred overnight at 50° C. After the reaction was done, the reaction mixture was filtered to remove insoluble solids and the filtrate was concentrated under reduced pressure. The residue was purified in a silica gel column eluting with ethyl acetate in petroleum ether (5% to 25% gradient) to

give 3,5-dimethyl-1-(propan-2-yl)-1H-pyrazole-4-carbaldehyde as a light yellow solid (9.5 g, 89%). MS: $m/z=167.0$ $[M+H].sup.+$

(186) 4-[5-amino-6,7,8,9-tetrahydro-5H-benzo[7]annulen-2-yl]pyridine-2,3-diamine (Step 2): At room temperature, tert-butyl N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (1.0 g, 2.44 mmol, 1 equiv) was added to a solution of HCl in MeOH (4M, 10 mL). The resulting mixture was stirred for 1 h at room temperature. After the reaction was done, the reaction mixture was concentrated under reduced pressure to give 4-[5-amino-6,7,8,9-tetrahydro-5H-benzo[7]annulen-2-yl]pyridine-2,3-diamine, HCl salt as brown oil (720 mg, crude) MS: $m/z=269.0$ $[M+H].sup.+$.

(187) N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-3-(1-methylcyclopropyl)-1,2,4-oxadiazole-5-carboxamide (Step 3): In a 50-mL round-bottom flask, to a solution of 4-[5-amino-6,7,8,9-tetrahydro-5H-benzo[7]annulen-2-yl]pyridine-2,3-diamine, HCl salt (700 mg, ~2.35 mmol, 1 equiv) and TEA (1.20 g, 11.75 mmol, 5.02 equiv) in MeOH (10 mL) was added methyl 3-(1-methylcyclopropyl)-1,2,4-oxadiazole-5-carboxylate (1.28 g, 7.05 mmol, 3.01 equiv) at room temperature. The resulting solution was stirred for 16 h at 50° C. When the reaction was done, it was quenched by the addition of water (20 mL) and the mixture was extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified in a silica gel column eluting with methanol in dichloromethane (5% to 10% gradient) to afford N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-3-(1-methylcyclopropyl)-1,2,4-oxadiazole-5-carboxamide as brown oil (1.0 g, 87% for two steps). MS: $m/z=419.2$ $[M+H].sup.+$

(188) N-[(5S)-2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-3-(1-methylcyclopropyl)-1,2,4-oxadiazole-5-carboxamide (Step 4) In a 50-mL round-bottom flask, to a solution of N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-3-(1-methylcyclopropyl)-1,2,4-oxadiazole-5-carboxamide (1 g, 2.15 mmol, 1 equiv) and 3,5-dimethyl-1-(propan-2-yl)-1H-pyrazole-4-carbaldehyde (398 mg, 2.15 mmol, 1.00 equiv) in DMF (10.0 mL) was added p-toluenesulfonic acid (41 mg, 0.21 mmol, 0.10 equiv) at room temperature. The resulting solution was stirred for 16 h at 100° C. After the reaction was done, the reaction mixture was concentrated under reduced pressure and the residue was first purified by Prep-HPLC with the following conditions: column, XBridge Shield RP18 OBD Column, 30×150 mm, 5 μ m; mobile phase, ACN in water (with 10 mM NH_4HCO_3 and 0.1% NH_4OH), 50% to 62% gradient in 8 min; detector, uv 220/254 nm. The obtained product is a racemic mixture of two enantiomers, which was further separated and purified by chiral HPLC using the following conditions: Column, (R,R)Whelk-O 1, 21.1×250 mm, 5 μ m; mobile phase, hexane/DCM (3:1) and EtOH (with 8 mM NH_4OH), isocratic 50% EtOH in 16 min; Detector, UV, 254 nm. N-[(5S)-2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-3-(1-methylcyclopropyl)-1,2,4-oxadiazole-5-carboxamide was obtained as a white solid (92 mg, 67.6%). HPLC: 99.2% purity. MS: $m/z=565.2$ $[M+H].sup.+$. 1H -NMR (300 MHz, $DMSO-d_6$, ppm): δ 12.76 (br s, 1H), 9.90 (d, $J=8.0$ Hz, 1H), 8.29 (d, $J=5.2$ Hz, 1H), 8.22 (d, $J=1.9$ Hz, 1H), 8.17 (d, $J=7.9$ Hz, 1H), 7.51 (d, $J=5.3$ Hz, 1H), 7.33 (d, $J=8.1$ Hz, 1H), 5.28-5.23 (m, 1H), 4.59-4.55 (m, 1H), 2.96-2.90 (m, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.97-1.94 (m, 3H), 1.84-1.80 (m, 2H), 1.52 (s, 3H), 1.39-1.30 (m, 7H), 1.26-1.24 (m, 2H), 1.03-1.00 (m, 2H).

(189) ##STR00212##

Example 3

(190) ##STR00213##

(191) tert-butyl N-[2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (Step 1): In a 100-mL round-bottom flask, to a solution of tert-butyl N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-

yl]carbamate (2 g, 4.89 mmol, 1.00 equiv) and 1,5-dimethyl-1H-pyrazole-3-carbaldehyde (670 mg, 5.13 mmol, 1.05 equiv) in N,N-dimethylformamide (40 mL) was added p-toluenesulfonic acid 4-methylbenzene-1-sulfonic acid (93 mg, 0.54 mmol, 0.11 equiv) at room temperature. The resulting solution was stirred for 2 h at 120° C. After the reaction was done, the reaction mixture was concentrated under reduced pressure. The residue was purified in a silica gel column eluting with methanol in dichloromethane (5% to 10% gradient) to afford tert-butyl N-[2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate as brown oil (1.1 g, 48%). MS: m/z=473.3 [M+H].sup.+.

(192) 2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine, HCl salt (Step 2): At room temperature, tert-butyl N-[2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (1.1 g, 2.09 mmol, 1.00 equiv) was added to a solution of HCl in MeOH (4 M, 10 mL, 40 mmol). The resulting mixture was stirred for 1 h at room temperature. After the reaction was done, the reaction mixture was concentrated under reduced pressure to give 2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine, HCl salt as brown oil (800 mg, crude). MS: m/z=373.2[M+H].sup.+.

(193) 5-tert-butyl-N-[2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,3,4-oxadiazole-2-carboxamide (Step 3): In a 50-mL round-bottom flask, to a solution of 2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine, HCl salt (800 mg, ~1.93 mmol, 1.00 equiv) and TEA (1.52 g, 15.02 mmol, 7.77 equiv) in methanol (15 mL) was added methyl 5-tert-butyl-1,3,4-oxadiazole-2-carboxylate (2 g, 9.77 mmol, 5.05 equiv) at room temperature. The resulting solution was stirred for 2 h at 50° C. When the reaction was done, it was quenched by the addition of water (20 mL) and the mixture was extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by prep-HPLC with the following conditions: column, XBridge Prep C18 OBD Column, 19×150 mm 5 um; mobile phase, ACN in water (with 10 mM NH₄HCO₃ and 0.1% NH₄OH), 25% to 49% gradient in 8 min; Detector, UV 254 nm. 5-tert-butyl-N-[2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,3,4-oxadiazole-2-carboxamide was obtained as an off-white solid (250 mg, 25%). HPLC: 96.5% purity. MS: m/z=525.3 [M+H].sup.+.

.sup.1H NMR (300 MHz, DMSO-d₆, ppm): δ 13.39 (br s, 1H), 9.91 (d, J=8.1 Hz, 1H), 8.30 (d, J=5.2 Hz, 1H), 8.20 (dd, J=8.1, 1.8 Hz, 1H), 8.07 (d, J=1.9 Hz, 1H), 7.49 (d, J=5.2 Hz, 1H), 7.40 (d, J=8.1 Hz, 1H), 6.80-6.74 (m, 1H), 5.35-5.23 (m, 1H), 3.88 (s, 3H), 3.03-2.97 (m, 2H), 2.34 (s, 3H), 2.03-1.97 (m, 4H), 1.91-1.78 (m, 2H), 1.45 (s, 9H).

(194) ##STR00214##

Example 4

(195) ##STR00215##

(196) tert-butyl N-(6-[2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate (Step 1): In a 30-mL vial, tert-butyl N-[6-(2,3-diaminopyridin-4-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate (300 mg, 0.76 mmol, 1.00 equiv) and 1-(propan-2-yl)-1H-pyrazole-4-carbaldehyde (120 mg, 0.78 mmol, 1.03 equiv) were mixed in N,N-dimethylformamide (15 mL) at room temperature. The resulting solution was stirred overnight at 80° C. After the reaction was done, the reaction mixture was concentrated under reduced pressure. The residue was purified by flash chromatography eluting with methanol in DCM (1% to 10% gradient) to afford tert-butyl N-(6-[2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate as a yellow solid (220 mg, 40%). MS: m/z=473.4 [M+H].sup.+.

(197) 6-[2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine, HCl salt (Step 2): At room temperature, tert-butyl N-(6-[2-[1-

(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate (220 mg, 0.31 mmol, 1.00 equiv) was added to a solution of HCl in MeOH (4 M, 5 mL, 20 mmol). The resulting mixture was stirred for 16 h at room temperature. After the reaction was done, the reaction mixture was concentrated under reduced pressure to give 6-[2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine, HCl salt as yellow solid (180 mg, crude). MS: $m/z=373.2$ [M+H].sup.+.

(198) 3-tert-butyl-N-(6-[2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)-1,2,4-oxadiazole-5-carboxamide (Step 3): In a 50-mL round-bottom flask, to a solution of 6-[2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine, HCl salt (180 mg, 0.30 mmol, 1.00 equiv) and TEA (120 mg, 1.42 mmol, 3.93 equiv) in methanol (4 mL) was added methyl 3-tert-butyl-1,2,4-oxadiazole-5-carboxylate (225 mg, 1.47 mmol, 4.04 equiv) at 25° C. The resulting solution was stirred overnight at 25° C. When the reaction was done, it was quenched by the addition of water (20 mL) and the mixture was extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by Prep-HPLC with the following conditions: column, XBridge Prep C18 OBD Column, 19×150 mm 5 μ m; mobile phase, acetonitrile in water (with 10 mM NH₄HCO₃ and 0.1% NH₃·H₂O), 45% to 65% gradient in 10 min; Detector, UV 254 nm. 3-tert-butyl-N-(6-[2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)-1,2,4-oxadiazole-5-carboxamide was obtained as a white solid (45 mg, 24.6% for two steps). HPLC: 97.8% purity. MS: $m/z=525.2$ [M+H].sup.+.

.sup.1H-NMR (300 MHz, DMSO-d₆, ppm): δ 13.26 (br s, 1H), 9.77 (d, $J=8.7$ Hz, 1H), 8.47 (s, 1H), 8.30 (s, 1H), 8.27-8.04 (m, 3H), 7.46-7.33 (m, 2H), 5.30-5.24 (m, 1H), 4.61 (p, $J=6.6$ Hz, 1H), 2.93-2.87 (m, 2H), 2.05-1.99 (m, 3H), 1.85-1.79 (m, 1H), 1.47 (d, $J=6.6$ Hz, 6H), 1.36 (s, 9H).

(199) ##STR00216## ##STR00217##

Example 5

(200) ##STR00218##

(201) 3-methyl-1-(tetrahydro-2H-pyran-4-yl)-1H-pyrazole-4-carbaldehyde (Step 1): In a 250-mL round-bottom flask, to a solution of 3-methyl-1H-pyrazole-4-carbaldehyde (2.2 g, 18.98 mmol, 1.00 equiv) in DMF (40 mL) was added sodium hydride (60% in oil, 1.2 g, 30.0 mmol, 1.58 equiv) at 0° C. The mixture was stirred for 1 h at room temperature, and then was added by 4-bromooxane (4 g, 23.03 mmol, 1.21 equiv). The reaction mixture was then stirred overnight at 120° C. When the reaction was done, it was quenched by the addition of water (20 mL) and the mixture was extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by reverse phase flash chromatography eluting with MeOH in water (10% to 40% gradient in 60 min) to give a mixture of 3-methyl-1-(tetrahydro-2H-pyran-4-yl)-1H-pyrazole-4-carbaldehyde and 5-methyl-1-(tetrahydro-2H-pyran-4-yl)-1H-pyrazole-4-carbaldehyde as yellow oil (1.2 g, 1:1, 33%). MS: $m/z=195.2$ [M+H].sup.+.

(202) 4-[7-bromo-3H-imidazo[4,5-b]pyridin-2-yl]-3-methyl-1-(oxan-4-yl)-1H-pyrazole (Step 2): In a 25-mL round-bottom flask, to a solution of 4-bromopyridine-2,3-diamine (500 mg, 2.53 mmol, 1.00 equiv, 95%) and 3-methyl-1-(oxan-4-yl)-1H-pyrazole-4-carbaldehyde (520 mg, as a 1:1 mixture with other regioisomer, total 2.54 mmol, 1.01 equiv) in N,N-dimethylformamide (10 mL) was added p-toluenesulfonic acid (46 mg, 0.25 mmol, 0.10 equiv) at room temperature. The resulting mixture was stirred for 3 days at 100° C. After the reaction was done, the reaction mixture was concentrated under reduced pressure and the residue was purified by reverse phase flash chromatography eluting with with methanol in dichloromethane (5% to 10% gradient) to afford the desired single regioisomer 4-[7-bromo-3H-imidazo[4,5-b]pyridin-2-yl]-3-methyl-1-(oxan-4-yl)-1H-pyrazole as a white solid (200 mg, 21%). MS: $m/z=362.0$ [M+H].sup.+.

(203) Tert-butyl N-(6-[2-[3-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-

yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate (Step 3): In a 50-mL microwave vial purged and maintained with an inert atmosphere of nitrogen, to a stirred mixture of tert-butyl N-[6-(tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate (200 mg, 0.48 mmol, 1.00 equiv) and 4-[7-bromo-3H-imidazo[4,5-b]pyridin-2-yl]-3-methyl-1-(oxan-4-yl)-1H-pyrazole (160 mg, 0.40 mmol, 0.82 equiv) in dioxane (10 mL) and water (1 mL) were added Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (37 mg, 0.04 mmol, 0.09 equiv) and Cs.sub.2CO.sub.3 (578 mg, 1.69 mmol, 3.50 equiv) at room temperature. The reaction mixture was irradiated with microwave for 4 h at 110° C. The mixture was allowed to cool down to room temperature and concentrated under reduced pressure. The residue was purified by silica gel column eluting with methanol in DCM (1% to 10% gradient) to afford tert-butyl N-(6-[2-[3-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate as a yellow solid (250 mg, 88%). MS: m/z=529.3 [M+H].sup.+

(204) 6-[2-[3-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine, HCl salt (Step 4): A solution of tert-butyl N-(6-[2-[3-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate (250 mg, 0.43 mmol, 1.00 equiv) and HCl in MeOH (4 M, 5 mL, 200 mmol) was stirred for 16 h at room temperature. After the reaction was done, the reaction mixture was concentrated under reduced pressure to give 6-[2-[3-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine, HCl salt as yellow solid (200 mg, crude). MS: m/z=429.2[M+H].sup.+.

(205) 3-tert-butyl-N-(6-[2-[3-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)-1,2,4-oxadiazole-5-carboxamide (Step 5): In a 25-mL round-bottom flask, to a solution of 6-[2-[3-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine (200 mg, ~0.43 mmol, 1.00 equiv) and TEA (283 mg, 2.66 mmol, 6.18 equiv) in methanol (10 mL) was added methyl 3-tert-butyl-1,2,4-oxadiazole-5-carboxylate (500 mg, 2.58 mmol, 6 equiv) at room temperature. The resulting solution was stirred for 16 h at 50° C. When the reaction was done, it was quenched by the addition of water (20 mL) and the mixture was extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The crude product was purified by prep-HPLC with the following conditions: column, XBridge Prep C18 OBD Column, 19×150 mm 5 um; mobile phase, ACN in water (with 10 mM NH.sub.4HCO.sub.3 and 0.1% NH.sub.3.Math.H.sub.2O), 49% to 66% gradient in 9 min; detector, UV 254 nm. 3-tert-butyl-N-(6-[2-[3-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)-1,2,4-oxadiazole-5-carboxamide was obtained as an off-white solid (120 mg, 43.6% for two steps). HPLC: 99.2% purity. MS: m/z=581.4 [M+H].sup.+ .sup.1H-NMR (300 MHz, DMSO-d.sub.6, ppm): δ 13.11 (br s, 1H), 9.76 (d, J=9.0 Hz, 1H), 8.42 (s, 1H), 8.34 (s, 1H), 8.26 (d, J=4.5 Hz, 1H), 8.15 (d, J=7.8 Hz, 1H), 7.49 (d, J=4.8 Hz, 1H), 7.38 (d, J=8.1 Hz, 1H), 5.30-5.25 (m, 1H), 4.42-4.39 (m, 1H), 4.01-3.98 (m, 2H), 3.54-3.46 (m, 2H), 2.89-2.79 (m, 2H), 2.63 (s, 3H), 2.04-1.88 (m, 8H), 1.36 (s, 9H).

(206) ##STR00219##

Example 6

(207) ##STR00220##

(208) Tert-butyl N-[2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (Step 1): In a 100-mL round bottom flask purged and maintained with an inert atmosphere of nitrogen, to a solution of tert-butyl N-[2-(2,3-diaminopyridin-4-yl)-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (1.3 g, 3.18 mmol, 1.00 equiv) and 1,3,5-trimethyl-1H-pyrazole-4-carbaldehyde (488 mg, 3.36 mmol, 1.06 equiv) in N,N-dimethylformamide (30 mL) was added 4-methylbenzene-1-sulfonic acid (61 mg, 0.35 mmol, 0.11 equiv) at room temperature. The resulting mixture was stirred for 4 days at 100° C. After the reaction was done, the reaction mixture was concentrated under reduced pressure. The residue was

purified by silica gel column eluting with methanol in DCM (1% to 10% gradient) to afford tert-butyl N-[2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate as a brown solid (0.6 g, 39%). MS: $m/z=487.4$ [M+H].sup.+.

(209) 2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine (Step 2): At room temperature, tert-butyl N-[2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate (590 mg, 1.09 mmol, 1.00 equiv) was added to a solution of HCl in MeOH (4 M, 5 mL, 20 mmol). The resulting mixture was stirred for 16 h at room temperature. After the reaction was done, the reaction mixture was concentrated under reduced pressure to give tert-butyl N-[2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]carbamate, HCl salt as yellow solid (400 mg, crude). MS: $m/z=387.1$ [M+H].sup.+.

(210) 3-tert-butyl-N-[2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2,4-oxadiazole-5-carboxamide (Step 3): In a 25-mL round-bottom flask, to a solution of 2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-amine (440 mg, ~1.02 mmol, 1.00 equiv) and TEA (576 mg, 5.69 mmol, 5.56 equiv) in methanol (5 mL) was added methyl 3-tert-butyl-1,2,4-oxadiazole-5-carboxylate (1.05 g, 5.13 mmol, 5.01 equiv) at room temperature. The resulting solution was stirred for 16 h at 50° C. When the reaction was done, it was quenched by the addition of water (20 mL) and the mixture was extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The crude product was purified by Prep-HPLC with the following conditions: Column, XBridge Prep C18 OBD Column, 19×150 mm Sum; mobile phase, ACN in water (with 10 mM NH₄HCO₃ and 0.1% NH₃·H₂O), 50% to 53% gradient in 10 min; detector, uv 254 nm. 3-tert-butyl-N-[2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2,4-oxadiazole-5-carboxamide was obtained as a light brown solid. HPLC: 97.5% purity. MS: $m/z=539.4$ [M+H].sup.+.

¹H-NMR (300 MHz, DMSO-d₆, ppm): δ 12.78 (br s, 1H), 9.98 (d, J=7.8 Hz, 1H), 8.28 (d, J=5.2 Hz, 1H), 8.26 (d, J=1.9 Hz, 1H), 8.17 (dd, J=8.1, 1.9 Hz, 1H), 7.53 (d, J=5.2 Hz, 1H), 7.36 (d, J=8.1 Hz, 1H), 5.32-5.23 (m, 1H), 3.76 (s, 3H), 2.98-2.96 (m, 2H), 2.54 (s, 3H), 2.41 (s, 3H), 1.97-1.83 (m, 6H), 1.42 (s, 9H).

(211) ##STR00221## ##STR00222##

Example 7

(212) ##STR00223##

(213) 6-bromo-1,2,3,4-tetrahydronaphthalen-1-amine 1-(oxan-4-yl)-1H-pyrazole-4-carbaldehyde (Step 1): In a 250-mL round bottom flask purged and maintained with an inert atmosphere of nitrogen, to a suspension of 6-bromo-1,2,3,4-tetrahydronaphthalen-1-one (4.5 g, 19.99 mmol, 1.00 equiv) in 2-propanol (55 mL) was added NaBH₃CN (6.3 g, 100.25 mmol, 5.01 equiv) and ammonium acetate (30.6 g, 396.98 mmol, 19.86 equiv) in sequence at room temperature. The reaction mixture was then stirred at room temperature for 4 hour, then heated up to reflux and stirred overnight under reflux. The reaction mixture was then cooled down to room temperature and poured into cold H₂O (500 ml). The pH value of the resulting mixture was adjusted to 12 with sodium hydroxide solution (1 N) and the mixture were extracted with DCM (3×30 mL). The organic layers were combined, dried over anhydrous sodium sulfate and concentrated under to give 6-bromo-1,2,3,4-tetrahydronaphthalen-1-amine as yellow oil (4.5 g, crude). MS: $m/z=226.0$ [M+H].sup.+.

(214) Tert-butyl N-(6-bromo-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate (Step 3): In a 50-mL round-bottom flask, to a solution of 6-bromo-1,2,3,4-tetrahydronaphthalen-1-amine (4.34 g, 19.17 mmol, 1.00 equiv) and TEA (3.07 g, 30.33 mmol, 1.58 equiv) in dichloromethane (80 mL) was added (Boc)₂O (6.65 g, 30.47 mmol, 1.59 equiv) at 0° C. The resulting solution was stirred for 16 h at 25° C. After the reaction was done, it was quenched by the addition of water (10 mL) and

the mixture was extracted with dichloromethane (3×20 mL). The combined organic layers were washed by brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified in a silica gel column eluting with dichloromethane in petroleum ether (5% to 20% gradient) to afford tert-butyl N-(6-bromo-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate as white oil (6.2 g, 95% for two steps). MS: $m/z=269.9[M+H-56]$.sup.-

(215) Tert-butyl N-[6-(tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate (Step 3): In a 100-mL round-bottom flask purged and maintained with an inert atmosphere of nitrogen, to a solution of tert-butyl N-(6-bromo-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate (6.20 g, 17.10 mmol, 1.00 equiv) and 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (5.21 g, 20.52 mmol, 1.20 equiv) in 1,4-dioxane (62 mL) were added KOAc (3.71 g, 34.02 mmol, 1.99 equiv), Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (1552.03 mg, 1.71 mmol, 0.10 equiv) at room temperature. The resulting mixture was stirred at 110° C. for 16 h. T After the reaction was done, the reaction mixture was filtered through a Celite pad and the filtrate was concentrated under reduced pressure. The residue was purified by silica gel column eluting with ethyl acetate in petroleum ether (5% to 20% gradient) to afford tert-butyl N-[6-(tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate as a yellow solid (5.2 g, 81%). MS: $m/z=396.1[M+H]$.sup.+.

(216) Tert-butyl N-[6-(2-amino-3-nitropyridin-4-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate (Step 4): (The following reaction was done in 4 equivalent batches but combined for purification) In each of the four 50-mL microwave vials purged and maintained with an inert atmosphere of nitrogen, 4-chloro-3-nitropyridin-2-amine (479.7 mg, 2.76 mmol, 0.80 equiv), tert-butyl N-[6-(tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate (1.50 g, 3.46 mmol, 1.00 equiv), sodium bicarbonate (580.6 mg, 6.91 mmol, 2.00 equiv) and Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (313.56 mg, 0.35 mmol, 0.10 equiv) were mixed in tetrahydrofuran (15.00 mL) and water (1.50 mL) were added at room temperature. The reaction mixture was irradiated with microwave for 30 min at 80° C. under N.sub.2. The reaction was then quenched by the addition of 20 mL water. The mixture was allowed to cool down to room temperature. The reaction mixture was concentrated under reduced pressure. The residue of the four batches of reactions were combined and purified by silica gel column eluting with methanol in DCM (1% to 10% gradient) to afford tert-butyl N-[6-(2-amino-3-nitropyridin-4-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate as a yellow solid (3.5 g, 75%). MS: $m/z=385.1[M+H]$.sup.+.

(217) Tert-butyl N-[6-(2,3-diaminopyridin-4-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate (Step 5): In a 250-mL round-bottom flask, to a solution of N-[6-(2-amino-3-nitropyridin-4-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate (3.50 g, 7.74 mmol, 1.00 equiv) in ethanol (26 mL) and water (13 mL) were added iron powder (4.32 g, 77.36 mmol, 10.00 equiv) and NH.sub.4Cl (8.28 g, 154.79 mmol, 20.00 equiv) at room temperature. The resulting mixture was stirred at 80° C. for 1 h. After the reaction was done, the reaction mixture was filtered through a Celite pad and the filtrate was concentrated under reduced pressure. The residue was diluted with water (100 mL) and the mixture was extracted with dichloromethane (3×200 mL). The combined organic layers were washed with brine (3×100 mL), dried over sodium sulfate and concentrated under reduced pressure to afford tert-butyl N-[6-(2,3-diaminopyridin-4-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate as a yellow solid (2.8 g, 92%). MS: $m/z=355.1[M+H]$.sup.+.

(218) Tert-butyl N-(6-[2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate (Step 6): In a 20-mL vial, to a solution of tert-butyl N-[6-(2,3-diaminopyridin-4-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]carbamate (1 g, 2.54 mmol, 1.00 equiv) in N,N-dimethylformamide (11 mL) was added 5-methyl-1-(propan-2-yl)-1H-pyrazole-4-carbaldehyde (433.68 mg as a 1:1 mixture with other regioisomer, total 2.56 mmol, 1.01 equiv) at room temperature. The resulting mixture was stirred for 3 days at 100° C. After the reaction was done, the reaction mixture was concentrated under reduced pressure and the residue

was purified. The residue was purified by silica gel column eluting with methanol in DCM (1% to 10% gradient) to afford the desired single regioisomer tert-butyl N-(6-[2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate as a yellow solid (350 mg, 29%). MS: m/z=487.2[M+H].sup.+

(219) 6-[2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine, HCl salt (Step 7): A solution of tert-butyl N-(6-[2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)carbamate (180 mg, 0.33 mmol, 1.0 eq) and HCl in MeOH (5 mL, 4M) was stirred for 16 h at room temperature. After the reaction was done, the reaction mixture was concentrated under reduced pressure to give 6-[2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine, HCl salt as yellow solid, which was used in next step without further purification (150 mg, crude). MS: m/z=387.1[M+H].sup.+

(220) 3-tert-butyl-N-(6-[2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)-1,2,4-oxadiazole-5-carboxamide (Step 8): In a 100-mL vial, to a solution of 6-[2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-amine, HCl salt (150 mg, ~0.33 mmol, 1.00 equiv) and TEA (180 mg, 1.78 mmol, 5.30 equiv) in methanol (3 mL) was added methyl 3-tert-butyl-1,2,4-oxadiazole-5-carboxylate (357 mg, 1.74 mmol, 5.27 equiv) at room temperature. The resulting solution was stirred for 2 days at 50° C. When the reaction was done, it was quenched by the addition of water (20 mL) and the mixture was extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over sodium sulfate and concentrated under reduced pressure. The crude product was purified by Prep-HPLC with the following conditions: column, XBridge Prep C18 OBD Column, 19×150 mm 5 um; mobile phase, ACN in water (with 10 mM NH₄HCO₃ and 0.1% NH₄OH), 55% to 67% gradient in 10 min; detector, UV 254 nm. 3-tert-butyl-N-(6-[2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl)-1,2,4-oxadiazole-5-carboxamide was obtained as a white solid (100 mg, 56% for 2 steps). HPLC: 100% purity. MS: m/z=539.2 [M+H].sup.+ .sup.1H-NMR (300 MHz, DMSO-d₆, ppm): δ 13.19 (br s, 1H), 9.77 (d, J=8.7 Hz, 1H), 8.28-8.25 (m, 2H), 8.16-8.11 (m, 2H), 7.48 (d, J=5.2 Hz, 1H), 7.38 (d, J=8.2 Hz, 1H), 5.28-5.27 (m, 1H), 4.66-4.62 (m, 1H), 2.91-2.85 (m, 2H), 2.81 (s, 3H), 2.06-1.95 (m, 3H), 1.84-1.82 (m, 1H), 1.43-1.36 (m, 15H).

(221) ##STR00224##

Example 8

(222) ##STR00225##

(223) 7-Bromo-2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (Step 1) To a reaction vial with stirbar was added a mixture of 2,3-diamino-4-bromopyridine (1.10 g; 5.85 mmol; 1.00 eq.), 1,3,5-Trimethyl-1H-pyrazole-4-carboxylic acid (0.99 g; 6.44 mmol; 1.10 eq.) and polyphosphoric acid (5.34 mL). The reaction mixture was heated at 130° C. overnight. Ice was added into the vial with stirring until a clear brown solution was obtained. The solution was transferred to a 250 mL flask with some water rinses (~25 mL) and cooled to room temperature. NaOH (21.06 g; 263.26 mmol; 45.00 eq.) solid was added to neutralize the reaction mixture to pH 8. Sodium hydroxide was added portion wise so that the temperature of the aqueous solution did not exceed 50° C. The resulting suspension was stirred in an icebath for 30 min then the precipitate was collected by filtration, washed with water (2×20 mL), and dried under high vacuum at 35° C. overnight to yield 967 mg (54%) of the title compound. LC/MS: 306 (M+H).

(224) {2-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-carbamic acid tert-butyl ester (Step 2). In a microwave vial with a stirbar was added 7-Bromo-2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (360.00 mg; 1.18 mmol; 1.00 eq.), [2-(4,4,5,5-Tetramethyl-[1,3,2]dioxaborolan-2-yl)-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl]-carbamic acid tert-butyl ester (500.97 mg; 1.29 mmol; 1.10

eq.), Sodium carbonate (373.88 mg; 3.53 mmol; 3.00 eq.), and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(ii), complex with dichloromethane (144.04 mg; 0.18 mmol; 0.15 eq.). The vial was sealed then evacuated/N.sub.2-backfilled (3×). EtOH (5.40 ml; 15.00 V) and water (0.54 ml; 1.50 V) were added via syringe and the vial was evacuated/N.sub.2-backfilled (3×) again. The mixture was stirred in a 80 degrees C. heating block overnight. LCMS-5 after 3 hr indicated the reaction was complete. The reaction was diluted with DCM (10 mL); filtered through a celite plug; and concentrated on SiO2. Purified via flash chromatography (Biotage): 25 g column using 50% EA/Hex isocratic for 5 min then ramped to 100% EA over 10 min. Ran 100% EA isocratic for 10 min and then ramped 0-10% MeOH/EA over 10 min at a flow rate of 25 mL/min. The product began eluting ~5% MeOH/EA. Combined and concentrated product fractions to yield 513 mg (90%) of the title compound as an orange solid foam. LCMS: 487 (M+H).

(225) 2-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (Step 3) In a round bottom flask with magnetic stirbar {2-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-carbamic acid tert-butyl ester (510.00 mg; 0.94 mmol; 1.00 eq.) was dissolved in methanol, anhydrous (7.65 ml; 15.00 V). 2.0M hydrogen chloride in diethyl ether (4.72 ml; 9.43 mmol; 10.00 eq.) was then added to the stirring solution. The reaction was stirred at room temperature overnight. The reaction was diluted with Et2O (50 mL) and the resulting suspension quickly became an oil. Concentrated to dryness and placed under high vacuum overnight. Assumed 100% yield of the title compound. LCMS: 387 (M+H).

(226) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (Step 4) In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (466.12 mg; 0.94 mmol; 1.00 eq.), 5-tert-butyl-1,2,4-oxadiazole-3-carboxylic acid (319.91 mg; 1.88 mmol; 2.00 eq.), DIPEA (0.82 ml; 4.70 mmol; 5.00 eq.), DCM (9.32 ml; 20.00 V), and DMF (0.60 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (1.40 ml; 2.35 mmol; 2.50 eq.). The reaction mixture was stirred at room temperature overnight. The mixture was concentrated, re-dissolved in EA (75 mL) and water (50 mL). The layers were separated, and the EA layer was washed with water (2×30 mL) and brine (1×30 mL); dried (Na.sub.2SO.sub.4); filtered; and concentrated on KPNH silica. Purified via flash chromatography (Biotage): 28 g KPNH column using 50% EA/Hex isocratic for 5 min then ramped to 100% EA over 5 min at a flow rate of 30 mL/min. Yielded 59 mg (11%) of the title compound as a white solid. LCMS: 539 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.54 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.25 (d, J=1.9 Hz, 1H), 8.16 (dd, J=8.1, 1.9 Hz, 1H), 7.52 (d, J=5.3 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.6 Hz, 1H), 3.75 (s, 3H), 2.98 (d, J=9.6 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.97 (d, J=12.1 Hz, 3H), 1.83 (q, J=9.5 Hz, 2H), 1.46 (s, 9H), 1.41-1.30 (m, 1H).

Example 9

(227) ##STR00226##

(228) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide: The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 18 mg (31%) of the title compound. LCMS: 539 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.54 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.25 (d, J=1.9 Hz, 1H), 8.16 (dd, J=8.1, 1.9 Hz, 1H), 7.52 (d, J=5.3 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.6 Hz, 1H), 3.75 (s, 3H), 2.98 (d, J=9.6 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.97 (d, J=12.1 Hz, 3H), 1.83 (q, J=9.5 Hz, 2H), 1.46 (s, 9H), 1.41-1.30 (m, 1H).

Example 10

(229) ##STR00227##

(230) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide: The desired compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 19 mg (32%) of the title compound. LCMS: 539 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.54 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.25 (d, J=1.9 Hz, 1H), 8.16 (dd, J=8.1, 1.9 Hz, 1H), 7.52 (d, J=5.3 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.6 Hz, 1H), 3.75 (s, 3H), 2.98 (d, J=9.6 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.97 (d, J=12.1 Hz, 3H), 1.83 (q, J=9.5 Hz, 2H), 1.46 (s, 9H), 1.41-1.30 (m, 1H).

Example 11

(231) ##STR00228##

(232) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide: In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (466.12 mg; 0.94 mmol; 1.00 eq.), 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid (316.12 mg; 1.88 mmol; 2.00 eq.), DIPEA (818.66 μ l; 4.70 mmol; 5.00 eq.), DCM (9.32 ml; 20.00 V), and DMF (0.96 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (1.40 ml; 2.35 mmol; 2.50 eq.). The reaction mixture was stirred at room temperature overnight. The reaction was diluted with DCM (50 mL) and washed with water (1×50 mL) and brine (1×25 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. The crude was dissolved in DMSO (3 mL) and purified via prep HPLC (Interchim P4250; 30×150 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 20-70% B over 20 min. Combined product fractions and lyophilized overnight to provide 420 mg (83%) of the title compound as a white solid. LCMS: 537 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.48 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.24 (d, J=1.9 Hz, 1H), 8.15 (dd, J=8.0, 1.9 Hz, 1H), 7.51 (d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.6 Hz, 1H), 3.75 (s, 3H), 2.97 (d, J=10.4 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.96 (d, J=11.0 Hz, 3H), 1.82 (q, J=9.6 Hz, 2H), 1.57 (s, 3H), 1.42 (q, J=4.1 Hz, 2H), 1.36 (d, J=7.2 Hz, 1H), 1.19 (q, J=4.2 Hz, 2H).

Example 12

(233) ##STR00229##

(234) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 150 mg (38%) of the title compound. LCMS: 537 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.48 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.24 (d, J=1.9 Hz, 1H), 8.15 (dd, J=8.0, 1.9 Hz, 1H), 7.51 (d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.6 Hz, 1H), 3.75 (s, 3H), 2.97 (d, J=10.4 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.96 (d, J=11.0 Hz, 3H), 1.82 (q, J=9.6 Hz, 2H), 1.57 (s, 3H), 1.42 (q, J=4.1 Hz, 2H), 1.36 (d, J=7.2 Hz, 1H), 1.19 (q, J=4.2 Hz, 2H).

Example 13

(235) ##STR00230##

(236) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 152 mg (38%) of the title compound. LCMS: 537 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.48 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.24 (d, J=1.9 Hz, 1H), 8.15 (dd, J=8.0, 1.9 Hz, 1H), 7.51

(d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.6 Hz, 1H), 3.75 (s, 3H), 2.97 (d, J=10.4 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.96 (d, J=11.0 Hz, 3H), 1.82 (q, J=9.6 Hz, 2H), 1.57 (s, 3H), 1.42 (q, J=4.1 Hz, 2H), 1.36 (d, J=7.2 Hz, 1H), 1.19 (q, J=4.2 Hz, 2H).

(237) ##STR00231##

Example 14

(238) ##STR00232##

(239) {8-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-carbamic acid tert-butyl ester (Step 1) In a microwave vial with a stirbar was added 7-Bromo-2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (440.00 mg; 1.15 mmol; 1.00 eq.), [8-(4,4,5,5-Tetramethyl-[1,3,2]dioxaborolan-2-yl)-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl]-carbamic acid tert-butyl ester (492.34 mg; 1.26 mmol; 1.10 eq.), Sodium carbonate (365.57 mg; 3.45 mmol; 3.00 eq.), and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(ii), complex with dichloromethane (140.84 mg; 0.17 mmol; 0.15 eq.). The vial was sealed then evacuated/N.sub.2-backfilled (3×). EtOH (8.80 ml; 20.00 V) and water (0.88 ml; 2.00 V) were added via syringe and the vial was evacuated/N.sub.2-backfilled (3×) again. The mixture was stirred in an 80° C. heating block overnight. The reaction contents were concentrated onto a sample of KP-NH silica gel. The sample was then loaded onto a 50 g KP-NH column and the Biotage was run 50% to 100% EtOAc in Hexanes. The pure fractions were combined and concentrated to obtain the title compound as a yellow solid. LC/MS: 489 (M+H).

(240) 8-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-ylamine hydrochloride (Step 2) In a reaction vial with magnetic stirbar {8-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-carbamic acid tert-butyl ester (350.00 mg; 0.72 mmol; 1.00 eq.) was dissolved in methanol, anhydrous (5.25 ml; 15.00 V) and 4.0M hydrogen chloride in dioxane (2.69 ml; 10.75 mmol; 15.00 eq.). The golden homogeneous reaction mixture was stirred at room temperature overnight. The reaction was diluted with Et₂O (~25 mL) and stirred at room temperature for 30 min. The reaction suspension was then filtered and the solids were rinsed with Et₂O (2×10 mL). The solid was dried under high vacuum overnight to provide 308 mg (86%) of the title compound as a white solid. LCMS: 389 (M+H).

(241) Step 3 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {8-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (Step 3): In a reaction vial with magnetic stirbar a stirring solution of 8-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-ylamine hydrochloride (3) (150.00 mg; 0.30 mmol; 1.00 eq.), 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid (102.54 mg; 0.60 mmol; 2.00 eq.), DIPEA (262.40 µl; 1.51 mmol; 5.00 eq.), DCM (3.00 ml; 20.00 V), and DMF (0.75 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (448.40 µl; 0.75 mmol; 2.50 eq.). The reaction mixture was stirred at room temperature overnight. The reaction was diluted with DCM (50 mL) and washed with water (1×50 mL) and brine (1×25 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (4 mL) and purified via prep HPLC (Interchim P4250; 30×250 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 25-75% B over 20 min. The product fractions were combined, froze, and lyophilized overnight. Yielded 132 mg (81%) of the title compound as a white solid. LCMS: 541 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.43 (s, 1H), 9.41 (d, J=8.4 Hz, 1H), 8.32 (d, J=5.2 Hz, 1H), 8.02 (s, 2H), 7.97 (d, J=2.4 Hz, 1H), 7.48 (s, 1H), 7.39 (dd, J=8.4, 0.8 Hz, 1H), 6.96 (d, J=2.4 Hz, 1H), 5.34 (t, J=5.8 Hz, 1H), 4.66 (hept, J=6.7 Hz, 1H), 4.31 (d, J=12.1 Hz, 1H), 3.85 (dd, J=11.8, 7.7 Hz, 1H), 2.09-1.96 (m, 4H), 1.53 (s, 3H), 1.51 (s, 3H), 1.46 (s, 8H).

Example 15

(242) ##STR00233##

(243) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-8-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 55 mg (46%) of the title compound. LCMS: 541 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.43 (s, 1H), 9.41 (d, J=8.4 Hz, 1H), 8.32 (d, J=5.2 Hz, 1H), 8.02 (s, 2H), 7.97 (d, J=2.4 Hz, 1H), 7.48 (s, 1H), 7.39 (dd, J=8.4, 0.8 Hz, 1H), 6.96 (d, J=2.4 Hz, 1H), 5.34 (t, J=5.8 Hz, 1H), 4.66 (hept, J=6.7 Hz, 1H), 4.31 (d, J=12.1 Hz, 1H), 3.85 (dd, J=11.8, 7.7 Hz, 1H), 2.09-1.96 (m, 4H), 1.53 (s, 3H), 1.51 (s, 3H), 1.46 (s, 8H).

Example 16

(244) ##STR00234##

(245) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-8-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 56 mg (46%) of the title compound. LCMS: 541 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.43 (s, 1H), 9.41 (d, J=8.4 Hz, 1H), 8.32 (d, J=5.2 Hz, 1H), 8.02 (s, 2H), 7.97 (d, J=2.4 Hz, 1H), 7.48 (s, 1H), 7.39 (dd, J=8.4, 0.8 Hz, 1H), 6.96 (d, J=2.4 Hz, 1H), 5.34 (t, J=5.8 Hz, 1H), 4.66 (hept, J=6.7 Hz, 1H), 4.31 (d, J=12.1 Hz, 1H), 3.85 (dd, J=11.8, 7.7 Hz, 1H), 2.09-1.96 (m, 4H), 1.53 (s, 3H), 1.51 (s, 3H), 1.46 (s, 8H).

Example 17

(246) ##STR00235##

(247) 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {8-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. In a reaction vial with magnetic stirbar a stirring solution of 8-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-ylamine hydrochloride (3) (150.00 mg; 0.30 mmol; 1.00 eq.), 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid (102.54 mg; 0.60 mmol; 2.00 eq.), DIPEA (262.40 μl; 1.51 mmol; 5.00 eq.), DCM (3.00 ml; 20.00 V), and DMF (0.75 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (448.40 μl; 0.75 mmol; 2.50 eq.). The reaction mixture was stirred at room temperature overnight. The reaction was diluted with DCM (50 mL) and washed with water (1×50 mL) and brine (1×25 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (4 mL) and purified via prep HPLC (Interchim P4250; 30×250 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 25-75% B over 20 min. The product fractions were combined, froze, and lyophilized overnight. Yielded 33 mg (20%) of the title compound as a white solid. LCMS: 541 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.42 (s, 1H), 9.77 (d, J=8.4 Hz, 1H), 8.31 (d, J=5.2 Hz, 1H), 8.02 (s, 2H), 7.96 (d, J=2.4 Hz, 1H), 7.47 (s, 1H), 7.40 (d, J=7.9 Hz, 1H), 6.95 (d, J=2.4 Hz, 1H), 5.32 (d, J=7.8 Hz, 1H), 4.66 (p, J=6.7 Hz, 1H), 4.36 (d, J=12.1 Hz, 1H), 3.79 (t, J=9.8 Hz, 1H), 2.03 (s, 4H), 1.52 (d, J=6.7 Hz, 6H), 1.43 (s, 9H).

Example 18

(248) ##STR00236##

(249) 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {(R)-8-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 11 mg (41%) of the title compound. LCMS: 541 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.42 (s, 1H), 9.77 (d, J=8.4 Hz, 1H), 8.31 (d, J=5.2 Hz, 1H), 8.02 (s, 2H), 7.96 (d, J=2.4 Hz, 1H), 7.47 (s, 1H), 7.40 (d, J=7.9 Hz, 1H), 6.95 (d, J=2.4 Hz, 1H), 5.32 (d, J=7.8 Hz, 1H), 4.66 (p, J=6.7 Hz, 1H), 4.36 (d, J=12.1 Hz, 1H), 3.79 (t, J=9.8 Hz, 1H), 2.03 (s, 4H), 1.52 (d, J=6.7 Hz, 6H), 1.43 (s, 9H).

Example 19

(250) ##STR00237##

(251) 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {(S)-8-[2-(1,3,5-trimethyl-1H-pyrazol-4-

yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 10 mg (37%) of the title compound. LCMS: 541 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.42 (s, 1H), 9.77 (d, J=8.4 Hz, 1H), 8.31 (d, J=5.2 Hz, 1H), 8.02 (s, 2H), 7.96 (d, J=2.4 Hz, 1H), 7.47 (s, 1H), 7.40 (d, J=7.9 Hz, 1H), 6.95 (d, J=2.4 Hz, 1H), 5.32 (d, J=7.8 Hz, 1H), 4.66 (p, J=6.7 Hz, 1H), 4.36 (d, J=12.1 Hz, 1H), 3.79 (t, J=9.8 Hz, 1H), 2.03 (s, 4H), 1.52 (d, J=6.7 Hz, 6H), 1.43 (s, 9H).

(252) ##STR00238##

Example 20

(253) ##STR00239##

(254) 7-Bromo-2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (Step 1) In a 250 mL RBF with stirbar was added 4-Bromo-pyridine-2,3-diamine (5.000 g; 26.592 mmol; 1.00 eq.), 1-Isopropyl-5-methyl-1H-pyrazole-4-carboxylic acid (4.920 g; 29.251 mmol; 1.10 eq.), and polyphosphoric acid (11.905 ml). The flask was heated at 120° C. overnight. Cooled to 50° C. Added crushed ice to the reaction and added. NaOH (47.862 g; 1196.649 mmol; 45.00 eq.) portion wise. The contents were poured into a sep funnel and 300 mL of EtOAc was added. The aq layer was extracted 2×300 mL of EtOAc. The combined organics were then washed with 200 mL of brine and dried over Na.sub.2SO.sub.4. The solids were dried under high vacuum overnight to provide 3.80 g (45%) of the title compound as a brown crystalline solid. LCMS: 320 (M+H).

(255) {2-[2-(1-Isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-carbamic acid tert-butyl ester (Step 2) In a microwave vial with a stirbar was added 7-Bromo-2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (1.26 g; 3.94 mmol; 1.00 eq.), [2-(4,4,5,5-Tetramethyl-[1,3,2]dioxaborolan-2-yl)-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl]-carbamic acid tert-butyl ester (1.68 g; 4.33 mmol; 1.10 eq.), Sodium carbonate (1.25 g; 11.81 mmol; 3.00 eq.), and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(ii), complex with dichloromethane (482.05 mg; 0.59 mmol; 0.15 eq.). The vial was sealed then evacuated/N.sub.2-backfilled (3×). EtOH (12.60 ml; 10.00 V) and water (1.26 ml; 1.00 V) were added via syringe and the vial was evacuated/N.sub.2-backfilled (3×) again. The mixture was stirred in a 80° C. heating block overnight. The reaction was diluted with DCM (10 mL); filtered through a celite plug and concentrated on KPNH silica. Purified via flash chromatography (Biotage): 50 g KPNH column using 50% EA/Hex isocratic for 5 min then ramped to 100% EA over 20 min at 50 mL/min. The product fractions were combined and concentrated to provide 1.52 g (75%) of the title compound as a peach solid. LCMS: 501 (M+H).

(256) 2-[2-(1-Isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (Step 3) In a round bottom flask with magnetic stirbar {2-[2-(1-Isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-carbamic acid tert-butyl ester (1.60 g; 3.20 mmol; 1.00 eq.) was dissolved in methanol, anhydrous (8.00 ml; 5.00 V) and 2.0M hydrogen chloride in diethyl ether (15.98 ml; 31.96 mmol; 10.00 eq.). The reaction was stirred at room temperature overnight. The reaction was diluted with Et2O (100 mL) and stirred for 1 hr. The resulting suspension was filtered and the solids were washed with Et2O (2×25 mL). The solids were dried under high vacuum to yield 1.47 g (90%) of the title compound as a peach solid. LCMS: 401 (M+H).

(257) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (Step 4) In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1-Isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine (212.28 mg; 0.53 mmol; 1.00 eq.), 5-tert-butyl-1,2,4-oxadiazole-3-carboxylic acid (180.38 mg; 1.06 mmol; 2.00 eq.), DIPEA (0.28 ml; 1.59 mmol; 3.00 eq.), DCM (4.25 ml; 20.00 V), and DMF

(0.60 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (0.79 ml; 1.33 mmol; 2.50 eq.). The milky reaction mixture was stirred at room temperature overnight. The mixture was diluted with DCM (75 mL) and water (75 mL). The mixture was shaken in a separatory funnel and the layers were separated. The aqueous layer was extracted with DCM (1×25 mL), and the combined DCM layers were washed with brine (1×50 mL); dried (Na.sub.2SO.sub.4); filtered; and concentrated. Purified via flash chromatography (Biotage) on a 55 g KPNH column and eluted the product using 100% DCM isocratic for 2 min then ramped to 10% MeOH/DCM over 20 min at a flow rate of 50 mL/min. The product fractions were concentrated and dried under high vacuum to yield 50 mg (17%) of the title compound as a pale yellow solid. LCMS: 553 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.55 (d, J=8.1 Hz, 1H), 8.35-8.06 (m, 4H), 7.49 (d, J=5.2 Hz, 1H), 7.36 (d, J=8.1 Hz, 1H), 5.30 (t, J=8.6 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (s, 2H), 2.82 (s, 3H), 1.98 (d, J=12.2 Hz, 3H), 1.82 (p, J=10.3, 9.9 Hz, 2H), 1.47 (s, 7H), 1.43 (d, J=6.5 Hz, 6H).

Example 21

(258) ##STR00240##

(259) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 17 mg (39%) of the title compound. LCMS: 553 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.55 (d, J=8.1 Hz, 1H), 8.35-8.06 (m, 4H), 7.49 (d, J=5.2 Hz, 1H), 7.36 (d, J=8.1 Hz, 1H), 5.30 (t, J=8.6 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (s, 2H), 2.82 (s, 3H), 1.98 (d, J=12.2 Hz, 3H), 1.82 (p, J=10.3, 9.9 Hz, 2H), 1.47 (s, 7H), 1.43 (d, J=6.5 Hz, 6H).

Example 22

(260) ##STR00241##

(261) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 16 mg (36%) of the title compound. LCMS: 553 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.55 (d, J=8.1 Hz, 1H), 8.35-8.06 (m, 4H), 7.49 (d, J=5.2 Hz, 1H), 7.36 (d, J=8.1 Hz, 1H), 5.30 (t, J=8.6 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (s, 2H), 2.82 (s, 3H), 1.98 (d, J=12.2 Hz, 3H), 1.82 (p, J=10.3, 9.9 Hz, 2H), 1.47 (s, 7H), 1.43 (d, J=6.5 Hz, 6H).

Example 23

(262) ##STR00242##

(263) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1-Isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (364.00 mg; 0.71 mmol; 1.00 eq.), 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid (240.07 mg; 1.43 mmol; 2.00 eq.), DIPEA (621.71 μl; 3.57 mmol; 5.00 eq.), DCM (7.28 ml; 20.00 V), and DMF (0.75 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (1.06 ml; 1.78 mmol; 2.50 eq.). The reaction mixture was stirred at room temperature overnight. The reaction was diluted with DCM (50 mL) and washed with water (1×50 mL) and brine (1×25 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (2 mL) and purified via prep HPLC (Interchim P4250; 30×150 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 30-80% B over 20 min. The product fractions were combined and lyophilized overnight to provide 110 mg (28%) of the title compound as a white solid. LCMS: 551 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.19 (s, 1H), 9.49 (d, J=8.1 Hz, 1H), 8.26 (d,

J=7.0 Hz, 2H), 8.17 (s, 2H), 7.55-7.42 (m, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.8 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (d, J=8.6 Hz, 2H), 2.81 (s, 3H), 1.96 (d, J=11.4 Hz, 3H), 1.81 (p, J=11.5, 10.6 Hz, 2H), 1.58 (s, 3H), 1.42 (d, J=6.5 Hz, 9H), 1.19 (q, J=4.3 Hz, 2H).

Example 24

(264) ##STR00243##

(265) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide: The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 36 mg (36%) of the title compound. LCMS: 551 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.19 (s, 1H), 9.49 (d, J=8.1 Hz, 1H), 8.26 (d, J=7.0 Hz, 2H), 8.17 (s, 2H), 7.55-7.42 (m, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.8 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (d, J=8.6 Hz, 2H), 2.81 (s, 3H), 1.96 (d, J=11.4 Hz, 3H), 1.81 (p, J=11.5, 10.6 Hz, 2H), 1.58 (s, 3H), 1.42 (d, J=6.5 Hz, 9H), 1.19 (q, J=4.3 Hz, 2H).

Example 25

(266) ##STR00244##

(267) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 32 mg (32%) of the title compound. LCMS: 551 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.19 (s, 1H), 9.49 (d, J=8.1 Hz, 1H), 8.26 (d, J=7.0 Hz, 2H), 8.17 (s, 2H), 7.55-7.42 (m, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.8 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (d, J=8.6 Hz, 2H), 2.81 (s, 3H), 1.96 (d, J=11.4 Hz, 3H), 1.81 (p, J=11.5, 10.6 Hz, 2H), 1.58 (s, 3H), 1.42 (d, J=6.5 Hz, 9H), 1.19 (q, J=4.3 Hz, 2H).

Example 26

(268) ##STR00245##

(269) 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a round bottom flask with magnetic stirbar a stirring solution of 2-[2-(1-Isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (330.00 mg; 0.65 mmol; 1.00 eq.), 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid (870.59 mg; 5.18 mmol; 8.00 eq.), DIPEA (563.64 μl; 3.24 mmol; 5.00 eq.), DCM (25.00 ml; 75.76 V), and DMF (3.00 ml; 9.09 V) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (0.96 ml; 1.62 mmol; 2.50 eq.) dropwise via syringe over 5 min. The reaction mixture was stirred at room temperature overnight. The reaction was diluted with DCM (50 mL) and washed with water (1×50 mL) and brine (1×25 mL). The organic layer was then dried (Na₂SO₄); filtered; and concentrated. Dissolved in DMSO (3 mL); filtered through a syringe filter; and purified via prep HPLC (Interchim P4250; 30×250 mm C-18 column; 0.1% NH₄OH modified mobile phases (A=water, B=ACN); gradient 30-80% B over 20 min. The product fractions were combined and lyophilized overnight to yield 225 mg (63%) of the title compound as a white solid. LCMS: 551 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.89 (d, J=8.0 Hz, 1H), 8.26 (dd, J=6.4, 3.5 Hz, 2H), 8.15 (d, J=7.3 Hz, 2H), 7.49 (d, J=5.2 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.27 (t, J=8.5 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (d, J=6.7 Hz, 2H), 2.82 (s, 3H), 1.97 (s, 3H), 1.81 (d, J=11.8 Hz, 2H), 1.52 (s, 3H), 1.42 (d, J=6.5 Hz, 6H), 1.38 (d, J=11.2 Hz, 1H), 1.25 (dt, J=4.2, 2.0 Hz, 2H), 1.06-0.98 (m, 2H).

Example 27

(270) ##STR00246##

(271) 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1-isopropyl-5-

methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 96 mg (44%) of the title compound. LCMS: 551 (M+H). ¹H NMR (400 MHz, DMSO-d.sub.6) δ 13.18 (s, 1H), 9.89 (d, J=8.0 Hz, 1H), 8.26 (dd, J=6.4, 3.5 Hz, 2H), 8.15 (d, J=7.3 Hz, 2H), 7.49 (d, J=5.2 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.27 (t, J=8.5 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (d, J=6.7 Hz, 2H), 2.82 (s, 3H), 1.97 (s, 3H), 1.81 (d, J=11.8 Hz, 2H), 1.52 (s, 3H), 1.42 (d, J=6.5 Hz, 6H), 1.38 (d, J=11.2 Hz, 1H), 1.25 (dt, J=4.2, 2.0 Hz, 2H), 1.06-0.98 (m, 2H).

Example 28

(272) ##STR00247##

(273) 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(S)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 100 mg (46%) of the title compound. LCMS: 551 (M+H). ¹H NMR (400 MHz, DMSO-d.sub.6) δ 13.18 (s, 1H), 9.89 (d, J=8.0 Hz, 1H), 8.26 (dd, J=6.4, 3.5 Hz, 2H), 8.15 (d, J=7.3 Hz, 2H), 7.49 (d, J=5.2 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.27 (t, J=8.5 Hz, 1H), 4.65 (p, J=6.6 Hz, 1H), 2.98 (d, J=6.7 Hz, 2H), 2.82 (s, 3H), 1.97 (s, 3H), 1.81 (d, J=11.8 Hz, 2H), 1.52 (s, 3H), 1.42 (d, J=6.5 Hz, 6H), 1.38 (d, J=11.2 Hz, 1H), 1.25 (dt, J=4.2, 2.0 Hz, 2H), 1.06-0.98 (m, 2H).

(274) ##STR00248##

Example 29

(275) ##STR00249##

(276) 7-Bromo-2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (Step 1) To a 50 mL RBF with stirbar was added a mixture of 2,3-diamino-4-bromopyridine (2850.00 mg; 15.16 mmol; 1.00 eq.), 1-isopropyl-3,5-dimethyl-1h-pyrazole-4-carboxylic acid (3038.21 mg; 16.67 mmol; 1.10 eq.) and polyphosphoric acid (13.59 ml). The reaction mixture was heated at 130° C. overnight. Ice was added into the vial with stirring until a clear brown solution was obtained. The solution was transferred to a 1000 mL flask and diluted to 500 mL with water. NaOH (27.2 g; 682.09 mmol; 45.00 eq.) was added to neutralize the reaction mixture to pH 8. The sodium hydroxide was added portion wise so that the temperature of the aqueous solution did not exceed 50° C. The product precipitated at pH=7-8 (litmus paper). The precipitate was collected by filtration, washing with H.sub.2O (3×50 mL), and dried under vacuum. Loaded compound onto celite and purified by Biotage chromatography (100 g KP-NH; 75 mL/min; 50-100% EtOAc/hex 15 col. vol.). Concentrated clean product fractions and dried overnight to afford 2.67 g (48%) of the title compound as a tan solid. LCMS: 334 (M+H).

(277) {2-[2-(1-Isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-carbamic acid tert-butyl ester (Step 2) In a microwave vial with a stirbar was added 7-Bromo-2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (750.00 mg; 2.24 mmol; 1.00 eq.), [2-(4,4,5,5-Tetramethyl-[1,3,2]dioxaborolan-2-yl)-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl]-carbamic acid tert-butyl ester (956.09 mg; 2.47 mmol; 1.10 eq.), Sodium carbonate (713.53 mg; 6.73 mmol; 3.00 eq.), and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(ii), complex with dichloromethane (274.89 mg; 0.34 mmol; 0.15 eq.). The vial was sealed then evacuated/N.sub.2-backfilled (3×). EtOH (11.25 ml; 15.00 V) and water (1.13 ml; 1.50 V) were added via syringe and the vial was evacuated/N.sub.2-backfilled (3×) again. The mixture was stirred in a 80° C. heating block overnight. The reaction was diluted with EA (15 mL); filtered through a celite plug; and concentrated. Purified via flash chromatography (Biotage): 50 g SiO₂ column using 20% EA/Hex isocratic for 3 min then ramped to 70% EA over 10 min at 100 mL/min. The product fractions were combined, concentrated, and dried under high vacuum to afford 793 mg (69%) of the title compound as a peach solid foam. LCMS: 515 (M+H).

(278) 2-[2-(1-Isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (Step 3) In a round bottom flask with magnetic stirbar {2-[2-(1-Isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-carbamic acid tert-butyl ester (793.00 mg; 1.39 mmol; 1.00 eq.) was dissolved in methanol, anhydrous (11.90 ml; 15.00 V). 2.0M hydrogen chloride in diethyl ether (6.93 ml; 13.87 mmol; 10.00 eq.) was then added to the stirring solution. The reaction was stirred at room temperature overnight. The reaction was diluted with Et₂O (50 mL) and the resulting white suspension was stirred for 30 min. The solids were filtered and dried under high vacuum to afford 726 mg (99.9%) of the title compound as an off-white solid. LCMS: 415 (M+H).

(279) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (Step 4): In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1-Isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (364.00 mg; 0.69 mmol; 1.00 eq.), 5-tert-butyl-1,2,4-oxadiazole-3-carboxylic acid (236.45 mg; 1.39 mmol; 2.00 eq.), DIPEA (605.07 μ l; 3.47 mmol; 5.00 eq.), DCM (7.28 ml; 20.00 V), and DMF (0.75 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (1.03 ml; 1.74 mmol; 2.50 eq.). The reaction mixture was stirred at room temperature overnight. The reaction was diluted with DCM (25 mL) and washed with water (1 $\times$ 25 mL) and brine (1 $\times$ 25 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (3 mL) and purified via prep HPLC: Interchim P4250; 30 $\times$ 150 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 30-80% B over 15 min. The product fractions were combined and lyophilized overnight to afford 275 mg (70%) of the title compound as a white solid. LCMS: 567 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.55 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.24 (d, J=1.9 Hz, 1H), 8.16 (dd, J=8.0, 1.9 Hz, 1H), 7.52 (d, J=5.2 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.6 Hz, 1H), 4.57 (hept, J=6.5 Hz, 1H), 2.98 (d, J=8.5 Hz, 2H), 2.56 (s, 3H), 2.43 (s, 3H), 1.97 (d, J=12.7 Hz, 3H), 1.83 (q, J=9.6 Hz, 2H), 1.46 (s, 9H), 1.40 (d, J=6.5 Hz, 7H).

Example 30

(280) ##STR00250##

(281) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21 $\times$ 250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 105 mg (40%) of the title compound. LCMS: 567 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.55 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.24 (d, J=1.9 Hz, 1H), 8.16 (dd, J=8.0, 1.9 Hz, 1H), 7.52 (d, J=5.2 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.6 Hz, 1H), 4.57 (hept, J=6.5 Hz, 1H), 2.98 (d, J=8.5 Hz, 2H), 2.56 (s, 3H), 2.43 (s, 3H), 1.97 (d, J=12.7 Hz, 3H), 1.83 (q, J=9.6 Hz, 2H), 1.46 (s, 9H), 1.40 (d, J=6.5 Hz, 7H).

Example 31

(282) ##STR00251##

(283) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21 $\times$ 250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 110 mg (42%) of the title compound. LCMS: 567 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.55 (d, J=8.1 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.24 (d, J=1.9 Hz, 1H), 8.16 (dd, J=8.0, 1.9 Hz, 1H), 7.52 (d, J=5.2 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.29 (t, J=8.6 Hz, 1H), 4.57 (hept, J=6.5 Hz, 1H), 2.98 (d, J=8.5 Hz, 2H), 2.56 (s, 3H), 2.43 (s, 3H), 1.97 (d, J=12.7 Hz, 3H), 1.83 (q, J=9.6 Hz, 2H), 1.46

(s, 9H), 1.40 (d, J=6.5 Hz, 7H).

Example 32

(284) ##STR00252##

(285) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1-Isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (364.00 mg; 0.69 mmol; 1.00 eq.), 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid (233.65 mg; 1.39 mmol; 2.00 eq.), DIPEA (605.07 μ l; 3.47 mmol; 5.00 eq.), DCM (7.28 ml; 20.00 V), and DMF (0.75 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (1.03 ml; 1.74 mmol; 2.50 eq.). The reaction mixture was stirred at room temperature overnight. The reaction was diluted with DCM (25 mL) and washed with water (1 $\times$ 25 mL) and brine (1 $\times$ 25 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (3 mL) and purified via prep HPLC: Interchim P4250; 30 $\times$ 150 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 35-80% B over 15 min. The product fractions were combined and lyophilized overnight to provide 245 mg (63%) of the title compound as a white solid. LCMS: 565 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 12.73 (s, 1H), 9.49 (d, J=8.1 Hz, 1H), 7.50 (s, 1H), 7.33 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.7 Hz, 1H), 4.58 (h, J=6.5 Hz, 1H), 2.97 (d, J=8.0 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.96 (d, J=11.4 Hz, 3H), 1.77 (d, J=9.4 Hz, 2H), 1.57 (s, 3H), 1.46-1.30 (m, 9H), 1.22-1.14 (m, 2H).

Example 33

(286) ##STR00253##

(287) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21 $\times$ 250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 93 mg (40%) of the title compound. LCMS: 565 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 12.73 (s, 1H), 9.49 (d, J=8.1 Hz, 1H), 7.50 (s, 1H), 7.33 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.7 Hz, 1H), 4.58 (h, J=6.5 Hz, 1H), 2.97 (d, J=8.0 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.96 (d, J=11.4 Hz, 3H), 1.77 (d, J=9.4 Hz, 2H), 1.57 (s, 3H), 1.46-1.30 (m, 9H), 1.22-1.14 (m, 2H).

Example 34

(288) ##STR00254##

(289) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21 $\times$ 250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 63 mg (27%) of the title compound. LCMS: 565 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 12.73 (s, 1H), 9.49 (d, J=8.1 Hz, 1H), 7.50 (s, 1H), 7.33 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.7 Hz, 1H), 4.58 (h, J=6.5 Hz, 1H), 2.97 (d, J=8.0 Hz, 2H), 2.55 (s, 3H), 2.42 (s, 3H), 1.96 (d, J=11.4 Hz, 3H), 1.77 (d, J=9.4 Hz, 2H), 1.57 (s, 3H), 1.46-1.30 (m, 9H), 1.22-1.14 (m, 2H).

Example 35

(290) ##STR00255##

(291) 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. 2-[2-(1-Isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (300.00 mg; 0.57 mmol; 1.00 eq.) was added to a reaction vial with EtOH (4.00 mL) and TEA (399.05 μ l; 2.86 mmol; 5.00 eq.). To the stirring solution was then added 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid ethyl ester (337.04 mg; 1.72 mmol; 3.00 eq.) dissolved in

EtOH (0.5 mL). The reaction was placed under N.sub.2 atmosphere and then the reaction was stirred in a preheated 75° C. heating block overnight. The reaction was concentrated and purified via flash chromatography (Biotage): 55 g KPNH column using 50-100% EA/Hex over 5 min then isocratic 100% EA at a flow rate of 50 mL/min. The product fractions were combined, concentrated, and placed under high vacuum overnight to provide 179 mg (55%) of the title compound as an off-white foam solid. LCMS: 565 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.89 (d, J=8.0 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.23 (d, J=1.9 Hz, 1H), 8.16 (dd, J=8.1, 1.9 Hz, 1H), 7.51 (d, J=5.3 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.26 (t, J=8.5 Hz, 1H), 4.57 (p, J=6.5 Hz, 1H), 2.97 (d, J=8.0 Hz, 2H), 2.56 (s, 3H), 2.43 (s, 3H), 2.03-1.89 (m, 3H), 1.81 (d, J=11.7 Hz, 2H), 1.52 (s, 3H), 1.40 (d, J=6.5 Hz, 7H), 1.25 (dt, J=4.1, 1.8 Hz, 2H), 1.04-0.98 (m, 2H).

Example 36

(292) ##STR00256##

(293) 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(S)-2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 60 mg (35%) of the title compound. LCMS: 565 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.89 (d, J=8.0 Hz, 1H), 8.29 (d, J=5.2 Hz, 1H), 8.23 (d, J=1.9 Hz, 1H), 8.16 (dd, J=8.1, 1.9 Hz, 1H), 7.51 (d, J=5.3 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.26 (t, J=8.5 Hz, 1H), 4.57 (p, J=6.5 Hz, 1H), 2.97 (d, J=8.0 Hz, 2H), 2.56 (s, 3H), 2.43 (s, 3H), 2.03-1.89 (m, 3H), 1.81 (d, J=11.7 Hz, 2H), 1.52 (s, 3H), 1.40 (d, J=6.5 Hz, 7H), 1.25 (dt, J=4.1, 1.8 Hz, 2H), 1.04-0.98 (m, 2H).

(294) ##STR00257##

Example 37

(295) ##STR00258##

(296) 7-Bromo-2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (Step 1) To a 500 mL RBF with stirbar was added a mixture of 4-Bromo-pyridine-2,3-diamine (35.00 g; 186.15 mmol; 1.00 eq.), 1-methyl-1h-pyrazole-4-carboxylic acid (25.82 g; 204.76 mmol; 1.10 eq.) and polyphosphoric acid (160.00 ml). The reaction mixture was heated at 130° C. overnight. Ice was added into the vial with stirring until a clear brown solution was obtained. The solution was transferred to a 2 L flask and then ammonium chloride (19914.21 mg; 372.29 mmol; 2.00 eq.) and NaOH (335036.70 mg; 8376.55 mmol; 45.00 eq.) were added to neutralize reaction mixture to pH 7. Sodium hydroxide was added portion wise so that the temperature of the aqueous solution did not exceed 50° C. (cooling by adding ice pellets from time to time). The product precipitated at pH=7. The suspension was chilled at 4° C. in fridge overnight. The precipitate was collected by filtration, washing with H.sub.2O (3×100 mL), and dried under high vacuum to afford 44.5 g (82%) of the title compound as a brown solid. LCMS: 278 (M+H).

(297) {8-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-carbamic acid tert-butyl ester (Step 2) In a microwave vial with a stirbar was added 7-Bromo-2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine (275.00 mg; 0.79 mmol; 1.00 eq.), [8-(4,4,5,5-Tetramethyl-[1,3,2]dioxaborolan-2-yl)-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl]-carbamic acid tert-butyl ester (400.34 mg; 1.03 mmol; 1.30 eq.), Sodium carbonate (251.53 mg; 2.37 mmol; 3.00 eq.), and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(ii), complex with dichloromethane (96.90 mg; 0.12 mmol; 0.15 eq.). The vial was sealed then evacuated/N.sub.2-backfilled (3×). EtOH (5.50 ml; 20.00 V) and water (0.55 ml; 2.00 V) were added via syringe and the vial was evacuated/N.sub.2-backfilled (3×) again. The mixture was stirred in a 80° C. heating block overnight. The reaction was diluted with DCM (10 mL); filtered through a celite plug; and concentrated on KPNH silica. Purified via flash chromatography (Biotage): 28 g KPNH column using 50-100% EA/Hex over 5 min followed by a ramp of 0-10%

MeOH/EA was ran to elute the product. The product fractions were combined, concentrated, and placed under high vacuum overnight to afford 330 mg (91%) of the title compound as a yellow solid. LCMS: 461 (M+H).

(298) 8-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-ylamine hydrochloride (Step 3) In a reaction vial with magnetic stirbar {8-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-carbamic acid tert-butyl ester (330.00 mg; 0.72 mmol; 1.00 eq.) was dissolved in methanol, anhydrous (5.00 ml; 123.43 mmol; 15.15 V). 4.0M hydrogen chloride in dioxane (5.00 ml; 10.00 mmol; 13.96 eq.) was then added to the stirring solution. The reaction was stirred at room temperature for 30 min. The reaction was concentrated to a dark yellow solid and placed under high vacuum overnight. LCMS: 361 (M+H).

(299) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (Step 4) In a reaction vial with magnetic stirbar a stirring solution of 8-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-ylamine hydrochloride (3) (169.13 mg; 0.36 mmol; 1.00 eq.), 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid (122.52 mg; 0.72 mmol; 2.00 eq.), DIPEA (313.53 μ l; 1.80 mmol; 5.00 eq.), DCM (3.38 ml; 20.00 V), and DMF (0.96 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (0.54 ml; 0.90 mmol; 2.50 eq.). The reaction mixture (pH 9) was stirred at room temperature for 90 min. The reaction was diluted with DCM (25 mL) and washed with water (1 $\times$ 20 mL) and brine (1 $\times$ 20 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (3 mL) and purified via prep HPLC (Interchim P4250; 30 $\times$ 150 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 20-80% B over 15 min. The product fractions were combined and lyophilized overnight to afford 140 mg (76%) of the title compound as a white solid. LCMS: 513 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.33 (s, 1H), 9.40 (d, J=8.4 Hz, 1H), 8.43 (s, 1H), 8.27 (d, J=5.3 Hz, 1H), 8.13 (s, 1H), 8.04 (d, J=8.1 Hz, 1H), 7.98 (s, 1H), 7.46 (s, 1H), 7.37 (d, J=8.1 Hz, 1H), 5.33 (d, J=7.3 Hz, 1H), 4.31 (d, J=12.0 Hz, 1H), 3.95 (s, 3H), 3.90-3.78 (m, 1H), 2.02 (s, 4H), 1.46 (s, 9H).

Example 38

(300) ##STR00259##

(301) 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. In a reaction vial with magnetic stirbar a stirring solution of 8-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-ylamine hydrochloride (3) (169.13 mg; 0.36 mmol; 1.00 eq.), 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid (122.52 mg; 0.72 mmol; 2.00 eq.), DIPEA (313.53 μ l; 1.80 mmol; 5.00 eq.), DCM (3.38 ml; 20.00 V), and DMF (0.96 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (0.54 ml; 0.90 mmol; 2.50 eq.). The reaction mixture (pH 9) was stirred at room temperature for 90 min. The reaction was diluted with DCM (25 mL) and washed with water (1 $\times$ 20 mL) and brine (1 $\times$ 20 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (3 mL) and purified via prep HPLC (Interchim P4250; 30 $\times$ 150 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 20 m—80% B over 15 min. The product fractions were combined and lyophilized overnight to afford 100 (54%) of the title compound as a white solid. LCMS: 513 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.32 (s, 1H), 9.85 (d, J=8.3 Hz, 1H), 8.42 (s, 1H), 8.27 (d, J=5.2 Hz, 1H), 8.12 (s, 1H), 8.04 (d, J=8.1 Hz, 1H), 7.98 (s, 1H), 7.47 (d, J=5.2 Hz, 1H), 7.39 (d, J=8.1 Hz, 1H), 5.31 (d, J=7.9 Hz, 1H), 4.34 (d, J=12.0 Hz, 1H), 3.95 (s, 3H), 3.83 (d, J=11.5 Hz, 1H), 2.04 (s, 4H), 1.40 (s, 9H).

Example 39

(302) ##STR00260##

(303) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-

imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a reaction vial with magnetic stirbar was added 2-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (168.00 mg; 0.36 mmol; 1.00 eq.), 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid (122.22 mg; 0.72 mmol; 2.00 eq.), DCM (3.36 ml; 20.00 V), DMF (0.96 ml) and DIPEA (625.51 μ l; 3.59 mmol; 10.00 eq.). After ~10 min the solids had not completely dissolved, additional DMF (3 mL) was added. The suspension was then treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (0.53 ml; 0.90 mmol; 2.50 eq.). The reaction mixture (pH 10) was stirred at room temperature for 90 min. The reaction was diluted with DCM (25 mL) and washed with water (1 $\times$ 20 mL) and brine (1 $\times$ 20 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (3 mL) and purified via prep HPLC (Interchim P4250); 30 $\times$ 150 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 30-80% B over 18 min. The product fractions were combined and lyophilized overnight to afford 110 mg (59%) of the title compound a white solid. LCMS: 511 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.29 (s, 1H), 9.56 (d, J=7.9 Hz, 1H), 8.43 (s, 1H), 8.27 (d, J=4.7 Hz, 1H), 8.16 (d, J=15.6 Hz, 2H), 8.06 (s, 1H), 7.46 (s, 1H), 7.36 (d, J=8.0 Hz, 1H), 5.30 (t, J=8.7 Hz, 1H), 3.95 (s, 3H), 2.99 (d, J=6.6 Hz, 2H), 1.99 (d, J=14.4 Hz, 3H), 1.84 (q, J=10.2 Hz, 2H), 1.47 (s, 9H), 1.40 (d, J=15.4 Hz, 1H).

Example 40

(304) ##STR00261##

(305) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (168.00 mg; 0.36 mmol; 1.00 eq.), 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid (120.77 mg; 0.72 mmol; 2.00 eq.), DIPEA (625.51 μ l; 3.59 mmol; 10.00 eq.), and DMF (4.00 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (0.53 ml; 0.90 mmol; 2.50 eq.). The reaction mixture (pH 10) was stirred at room temperature for 1 hr. The reaction was diluted with DCM (25 mL) and washed with water (1 $\times$ 20 mL) and brine (1 $\times$ 20 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (3 mL) and purified via prep HPLC (Interchim P4250); 30 $\times$ 150 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 30-80% B over 18 min. The product fractions were combined and lyophilized overnight to afford 96 mg (51%) of the title compound as a white solid. LCMS: 509 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.29 (s, 1H), 9.49 (d, J=8.0 Hz, 1H), 8.43 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.21-8.15 (m, 1H), 8.13 (s, 1H), 8.06 (s, 1H), 7.47 (d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.7 Hz, 1H), 3.95 (s, 3H), 2.99 (d, J=6.9 Hz, 2H), 1.97 (s, 3H), 1.90-1.74 (m, 2H), 1.58 (s, 3H), 1.43 (q, J=4.0 Hz, 2H), 1.38 (s, 1H), 1.23-1.16 (m, 2H).

Example 41

(306) ##STR00262##

(307) 5-(1-Methyl-cyclopropyl)-isoxazole-3-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (168.00 mg; 0.36 mmol; 1.00 eq.), 5-(1-Methyl-cyclopropyl)-isoxazole-3-carboxylic acid (120.06 mg; 0.72 mmol; 2.00 eq.), DIPEA (625.51 μ l; 3.59 mmol; 10.00 eq.), and DMF (4.00 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (0.53 ml; 0.90 mmol; 2.50 eq.). The reaction mixture (pH 10) was stirred at room temperature for 1 hr. The reaction was diluted with DCM (25 mL) and washed with water (1 $\times$ 20 mL) and brine (1 $\times$ 20 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (3 mL) and purified

via prep HPLC (Interchim P4250); 30×150 mm C-18 column; 0.1% NH₄OH modified mobile phases (A=water, B=ACN); gradient 30-80% B over 18 min. The product fractions were combined and lyophilized overnight to afford 145 mg (80%) of the title compound as a white solid. LCMS: 508 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.29 (s, 1H), 9.27 (d, J=8.1 Hz, 1H), 8.42 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.21-8.15 (m, 1H), 8.13 (s, 1H), 8.09-8.01 (m, 1H), 7.47 (d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 6.63 (s, 1H), 5.27 (t, J=8.9 Hz, 1H), 3.95 (s, 3H), 2.98 (d, J=6.7 Hz, 2H), 1.96 (d, J=10.1 Hz, 3H), 1.81 (dt, J=20.6, 10.9 Hz, 2H), 1.48 (s, 3H), 1.34 (d, J=17.8 Hz, 1H), 1.19 (q, J=4.1 Hz, 2H), 1.02-0.95 (m, 2H).

Example 42

(308) ##STR00263##

(309) 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (168.00 mg; 0.36 mmol; 1.00 eq.), 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid (122.22 mg; 0.72 mmol; 2.00 eq.), DIPEA (625.51 μl; 3.59 mmol; 10.00 eq.), and DMF (4.00 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (0.53 ml; 0.90 mmol; 2.50 eq.). The reaction mixture (pH 10) was stirred at room temperature for 1 hr. The reaction was diluted with DCM (25 mL) and washed with water (1×20 mL) and brine (1×20 mL). The organic layer was then dried (Na₂SO₄); filtered; and concentrated. Dissolved in DMSO (3 mL) and purified via prep HPLC (Interchim P4250); 30×150 mm C-18 column; 0.1% NH₄OH modified mobile phases (A=water, B=ACN); gradient 30-80% B over 18 min. The product fractions were combined and lyophilized overnight to afford 130 mg (70%) of the title compound as a white solid. LCMS: 511 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.30 (s, 1H), 9.89 (d, J=8.0 Hz, 1H), 8.42 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.23-8.15 (m, 1H), 8.13 (s, 1H), 8.09-8.03 (m, 1H), 7.47 (d, J=5.2 Hz, 1H), 7.38 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.5 Hz, 1H), 3.94 (s, 3H), 2.99 (d, J=7.6 Hz, 2H), 1.98 (s, 3H), 1.92-1.76 (m, 2H), 1.44 (s, 9H), 1.37 (s, 1H).

Example 43

(310) ##STR00264##

(311) 3-tert-Butoxy-azetidine-1-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a reaction vial with magnetic stirbar 2-[2-(1-Methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (190.00 mg; 0.41 mmol; 1.00 eq.) was added followed by DMF (4.00 ml; 51.88 mmol; 21.05 V) and Ethyl-diisopropyl-amine (403.77 μl; 2.44 mmol; 6.00 eq.). To the stirring solution was then added Di-imidazol-1-yl-methanone (98.78 mg; 0.61 mmol; 1.50 eq.). After stirring for 2 h at room temperature, 3-tert-Butoxy-azetidine hydrochloride (80.74 mg; 0.49 mmol; 1.20 eq.) was added and the reaction was stirred overnight. The reaction was partially concentrated and then purified directly via prep HPLC (Interchim P4250); 30×150 mm C-18 column; 0.1% formic acid modified mobile phases (A=water, B=ACN); gradient 20-75% B over 20 min at 60 mL/min. The product fractions were combined and lyophilized overnight to afford 142 mg (67%) of the title compound as a white solid. LCMS: 514 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.17 (s, 1H), 8.43 (s, 1H), 8.26 (d, J=5.0 Hz, 1H), 8.14 (s, 2H), 8.00 (s, 1H), 7.44 (s, 1H), 7.36 (d, J=8.1 Hz, 1H), 6.83 (d, J=8.1 Hz, 1H), 4.91 (t, J=9.1 Hz, 1H), 4.50 (ddd, J=11.6, 6.6, 5.0 Hz, 1H), 4.17-4.04 (m, 2H), 3.95 (s, 3H), 3.66 (ddd, J=17.3, 8.6, 5.0 Hz, 2H), 2.92 (d, J=11.2 Hz, 2H), 1.91 (s, 3H), 1.79 (q, J=11.7, 9.2 Hz, 1H), 1.70-1.55 (m, 1H), 1.43-1.26 (m, 1H), 1.15 (s, 9H).

Example 44

(312) ##STR00265##

(313) 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. The title compound

was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 28 mg (30%) of the title compound. LCMS: 513 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.84 (d, J=7.9 Hz, 1H), 8.39 (s, 1H), 8.24 (d, J=5.1 Hz, 1H), 8.10 (s, 1H), 7.98 (d, J=23.8 Hz, 2H), 7.38 (d, J=8.0 Hz, 2H), 5.31 (s, 1H), 4.33 (d, J=12.0 Hz, 1H), 3.94 (s, 3H), 3.83 (d, J=10.7 Hz, 1H), 2.04 (s, 4H), 1.40 (s, 9H).

Example 45

(314) ##STR00266##

(315) 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(S)-8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 29 mg (31%) of the title compound. LCMS: 513 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.84 (d, J=7.9 Hz, 1H), 8.39 (s, 1H), 8.24 (d, J=5.1 Hz, 1H), 8.10 (s, 1H), 7.98 (d, J=23.8 Hz, 2H), 7.38 (d, J=8.0 Hz, 2H), 5.31 (s, 1H), 4.33 (d, J=12.0 Hz, 1H), 3.94 (s, 3H), 3.83 (d, J=10.7 Hz, 1H), 2.04 (s, 4H), 1.40 (s, 9H).

Example 46

(316) ##STR00267##

(317) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 36 mg (27%) of the title compound. LCMS: 513 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.40 (d, J=8.4 Hz, 1H), 8.41 (s, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.12 (d, J=0.7 Hz, 1H), 7.98 (d, J=23.1 Hz, 2H), 7.42 (s, 1H), 7.37 (d, J=8.1 Hz, 1H), 5.33 (q, J=6.7, 6.3 Hz, 1H), 4.31 (d, J=12.0 Hz, 1H), 3.94 (s, 3H), 3.84 (dd, J=11.9, 7.9 Hz, 1H), 2.02 (s, 4H), 1.46 (s, 9H).

Example 47

(318) ##STR00268##

(319) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 33 mg (25%) of the title compound. LCMS: 513 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.40 (d, J=8.4 Hz, 1H), 8.41 (s, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.12 (d, J=0.7 Hz, 1H), 7.98 (d, J=23.1 Hz, 2H), 7.42 (s, 1H), 7.37 (d, J=8.1 Hz, 1H), 5.33 (q, J=6.7, 6.3 Hz, 1H), 4.31 (d, J=12.0 Hz, 1H), 3.94 (s, 3H), 3.84 (dd, J=11.9, 7.9 Hz, 1H), 2.02 (s, 4H), 1.46 (s, 9H).

Example 48

(320) ##STR00269##

(321) 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 36 mg (30%) of the title compound. LCMS: 511 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.29 (s, 1H), 9.88 (d, J=8.0 Hz, 1H), 8.42 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.1, 1.8 Hz, 1H), 8.13 (s, 1H), 8.06 (d, J=1.9 Hz, 1H), 7.46 (d, J=5.2 Hz, 1H), 7.38 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.5 Hz, 1H), 3.94 (s, 3H), 2.99 (t, J=5.2 Hz, 2H), 1.98 (s, 3H), 1.84 (t, J=9.6 Hz, 2H), 1.43 (s, 9H), 1.38 (d, J=5.6 Hz, 1H).

Example 49

(322) ##STR00270##

(323) 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {(S)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 42 mg (34%) of the title compound. LCMS:

511 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.29 (s, 1H), 9.88 (d, J=8.0 Hz, 1H), 8.42 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.1, 1.8 Hz, 1H), 8.13 (s, 1H), 8.06 (d, J=1.9 Hz, 1H), 7.46 (d, J=5.2 Hz, 1H), 7.38 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.5 Hz, 1H), 3.94 (s, 3H), 2.99 (t, J=5.2 Hz, 2H), 1.98 (s, 3H), 1.84 (t, J=9.6 Hz, 2H), 1.43 (s, 9H), 1.38 (d, J=5.6 Hz, 1H).

Example 50

(324) ##STR00271##

(325) 3-tert-Butoxy-azetidine-1-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 44 mg (33%) of the title compound. LCMS: 514 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 8.42 (s, 1H), 8.25 (d, J=5.1 Hz, 1H), 8.04 (d, J=74.9 Hz, 3H), 7.38 (dd, J=13.9, 6.5 Hz, 2H), 6.83 (d, J=8.1 Hz, 1H), 4.91 (t, J=9.2 Hz, 1H), 4.50 (p, J=5.9 Hz, 1H), 4.10 (dt, J=16.4, 7.7 Hz, 2H), 3.95 (s, 3H), 3.66 (ddd, J=17.0, 8.6, 5.1 Hz, 2H), 2.92 (d, J=11.2 Hz, 2H), 1.92 (d, J=10.3 Hz, 3H), 1.78 (d, J=10.5 Hz, 1H), 1.62 (d, J=11.0 Hz, 1H), 1.33 (d, J=13.7 Hz, 1H), 1.15 (s, 9H).

Example 51

(326) ##STR00272##

(327) 3-tert-Butoxy-azetidine-1-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 44 mg (33%) of the title compound. LCMS: 514 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 8.42 (s, 1H), 8.25 (d, J=5.1 Hz, 1H), 8.04 (d, J=74.9 Hz, 3H), 7.38 (dd, J=13.9, 6.5 Hz, 2H), 6.83 (d, J=8.1 Hz, 1H), 4.91 (t, J=9.2 Hz, 1H), 4.50 (p, J=5.9 Hz, 1H), 4.10 (dt, J=16.4, 7.7 Hz, 2H), 3.95 (s, 3H), 3.66 (ddd, J=17.0, 8.6, 5.1 Hz, 2H), 2.92 (d, J=11.2 Hz, 2H), 1.92 (d, J=10.3 Hz, 3H), 1.78 (d, J=10.5 Hz, 1H), 1.62 (d, J=11.0 Hz, 1H), 1.33 (d, J=13.7 Hz, 1H), 1.15 (s, 9H).

Example 52

(328) ##STR00273##

(329) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 41 mg (40%) of the title compound. LCMS: 511 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.29 (s, 1H), 9.56 (d, J=7.9 Hz, 1H), 8.43 (s, 1H), 8.27 (d, J=4.7 Hz, 1H), 8.16 (d, J=15.6 Hz, 2H), 8.06 (s, 1H), 7.46 (s, 1H), 7.36 (d, J=8.0 Hz, 1H), 5.30 (t, J=8.7 Hz, 1H), 3.95 (s, 3H), 2.99 (d, J=6.6 Hz, 2H), 1.99 (d, J=14.4 Hz, 3H), 1.84 (q, J=10.2 Hz, 2H), 1.47 (s, 9H), 1.40 (d, J=15.4 Hz, 1H).

Example 53

(330) ##STR00274##

(331) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amid. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 39 mg (38%) of the title compound. LCMS: 511 (M+H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.29 (s, 1H), 9.56 (d, J=7.9 Hz, 1H), 8.43 (s, 1H), 8.27 (d, J=4.7 Hz, 1H), 8.16 (d, J=15.6 Hz, 2H), 8.06 (s, 1H), 7.46 (s, 1H), 7.36 (d, J=8.0 Hz, 1H), 5.30 (t, J=8.7 Hz, 1H), 3.95 (s, 3H), 2.99 (d, J=6.6 Hz, 2H), 1.99 (d, J=14.4 Hz, 3H), 1.84 (q, J=10.2 Hz, 2H), 1.47 (s, 9H), 1.40 (d, J=15.4 Hz, 1H).

Example 54

(332) ##STR00275##

(333) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-

amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 38 mg (41%) of the title compound. LCMS: 509 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.29 (s, 1H), 9.49 (d, J=8.0 Hz, 1H), 8.43 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.21-8.15 (m, 1H), 8.13 (s, 1H), 8.06 (s, 1H), 7.47 (d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.7 Hz, 1H), 3.95 (s, 3H), 2.99 (d, J=6.9 Hz, 2H), 1.97 (s, 3H), 1.90-1.74 (m, 2H), 1.58 (s, 3H), 1.43 (q, J=4.0 Hz, 2H), 1.38 (s, 1H), 1.23-1.16 (m, 2H).

Example 55

(334) ##STR00276##

(335) 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 31 mg (34%) of the title compound. LCMS: 509 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.29 (s, 1H), 9.49 (d, J=8.0 Hz, 1H), 8.43 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.21-8.15 (m, 1H), 8.13 (s, 1H), 8.06 (s, 1H), 7.47 (d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 5.28 (t, J=8.7 Hz, 1H), 3.95 (s, 3H), 2.99 (d, J=6.9 Hz, 2H), 1.97 (s, 3H), 1.90-1.74 (m, 2H), 1.58 (s, 3H), 1.43 (q, J=4.0 Hz, 2H), 1.38 (s, 1H), 1.23-1.16 (m, 2H).

Example 56

(336) ##STR00277##

(337) 5-(1-Methyl-cyclopropyl)-isoxazole-3-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 59 mg (43%) of the title compound. LCMS: 508 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.29 (s, 1H), 9.27 (d, J=8.1 Hz, 1H), 8.42 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.21-8.15 (m, 1H), 8.13 (s, 1H), 8.09-8.01 (m, 1H), 7.47 (d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 6.63 (s, 1H), 5.27 (t, J=8.9 Hz, 1H), 3.95 (s, 3H), 2.98 (d, J=6.7 Hz, 2H), 1.96 (d, J=10.1 Hz, 3H), 1.81 (dt, J=20.6, 10.9 Hz, 2H), 1.48 (s, 3H), 1.34 (d, J=17.8 Hz, 1H), 1.19 (q, J=4.1 Hz, 2H), 1.02-0.95 (m, 2H).

Example 57

(338) ##STR00278##

(339) 5-(1-Methyl-cyclopropyl)-isoxazole-3-carboxylic acid {(S)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 59 mg (43%) of the title compound. LCMS: 508 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.29 (s, 1H), 9.27 (d, J=8.1 Hz, 1H), 8.42 (s, 1H), 8.26 (d, J=5.2 Hz, 1H), 8.21-8.15 (m, 1H), 8.13 (s, 1H), 8.09-8.01 (m, 1H), 7.47 (d, J=5.2 Hz, 1H), 7.34 (d, J=8.1 Hz, 1H), 6.63 (s, 1H), 5.27 (t, J=8.9 Hz, 1H), 3.95 (s, 3H), 2.98 (d, J=6.7 Hz, 2H), 1.96 (d, J=10.1 Hz, 3H), 1.81 (dt, J=20.6, 10.9 Hz, 2H), 1.48 (s, 3H), 1.34 (d, J=17.8 Hz, 1H), 1.19 (q, J=4.1 Hz, 2H), 1.02-0.95 (m, 2H).

Example 58

(340) ##STR00279##

(341) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. In a reaction vial with magnetic stirbar a stirring solution of 2-[2-(1-Isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride (3) (160.00 mg; 0.31 mmol; 1.00 eq.), 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid (106.79 mg; 0.63 mmol; 2.00 eq.), DIPEA (273.28 μl; 1.57 mmol; 5.00 eq.), DCM (3.20 ml; 20.00 V), and DMF (0.50 ml) was treated with 2,4,6-Tripropyl-[1,3,5,2,4,6]trioxatriphosphinane 2,4,6-trioxide (466.98 μl; 0.78 mmol; 2.50 eq.). The reaction mixture was stirred at room

temperature overnight. The reaction was diluted with DCM (50 mL) and washed with water (1×50 mL) and brine (1×25 mL). The organic layer was then dried (Na.sub.2SO.sub.4); filtered; and concentrated. Dissolved in DMSO (2 mL) and purified via prep HPLC (Interchim P4250); 30×250 mm C-18 column; 0.1% NH.sub.4OH modified mobile phases (A=water, B=ACN); gradient 35-65% B over 20 min. The product fractions were combined and lyophilized to afford 149 mg (86%) of the title compound as a white solid. LCMS: 553 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.04 (s, 1H), 9.55 (d, J=8.0 Hz, 1H), 8.39 (s, 1H), 8.30 (d, J=1.9 Hz, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.1, 1.9 Hz, 1H), 7.50 (d, J=5.3 Hz, 1H), 7.36 (d, J=8.1 Hz, 1H), 5.30 (t, J=8.6 Hz, 1H), 4.51 (p, J=6.6 Hz, 1H), 2.99 (d, J=9.3 Hz, 2H), 2.62 (s, 3H), 1.96 (s, 3H), 1.83 (q, J=9.6 Hz, 2H), 1.46 (d, J=3.1 Hz, 16H), 1.38 (s, 1H).

Example 59

(342) ##STR00280##

(343) 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 30 mg (37%) of the title compound. LCMS: 551 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.04 (s, 1H), 9.88 (d, J=7.9 Hz, 1H), 8.39 (s, 1H), 8.29 (d, J=1.9 Hz, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.2, 1.9 Hz, 1H), 7.49 (d, J=5.3 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.27 (t, J=8.5 Hz, 1H), 4.51 (p, J=6.6 Hz, 1H), 2.98 (d, J=7.8 Hz, 2H), 2.62 (s, 3H), 1.97 (s, 3H), 1.81 (d, J=13.5 Hz, 2H), 1.52 (s, 3H), 1.47 (d, J=6.6 Hz, 6H), 1.36 (s, 1H), 1.25 (dt, J=4.1, 1.9 Hz, 2H), 1.08-0.96 (m, 2H).

Example 60

(344) ##STR00281##

(345) 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(S)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 28 mg (34%) of the title compound. LCMS: 551 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.04 (s, 1H), 9.88 (d, J=7.9 Hz, 1H), 8.39 (s, 1H), 8.29 (d, J=1.9 Hz, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.2, 1.9 Hz, 1H), 7.49 (d, J=5.3 Hz, 1H), 7.35 (d, J=8.1 Hz, 1H), 5.27 (t, J=8.5 Hz, 1H), 4.51 (p, J=6.6 Hz, 1H), 2.98 (d, J=7.8 Hz, 2H), 2.62 (s, 3H), 1.97 (s, 3H), 1.81 (d, J=13.5 Hz, 2H), 1.52 (s, 3H), 1.47 (d, J=6.6 Hz, 6H), 1.36 (s, 1H), 1.25 (dt, J=4.1, 1.9 Hz, 2H), 1.08-0.96 (m, 2H).

Example 61

(346) ##STR00282##

(347) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 48 mg (34%) of the title compound. LCMS: 553 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.04 (s, 1H), 9.55 (d, J=8.0 Hz, 1H), 8.39 (s, 1H), 8.30 (d, J=1.9 Hz, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.1, 1.9 Hz, 1H), 7.50 (d, J=5.3 Hz, 1H), 7.36 (d, J=8.1 Hz, 1H), 5.30 (t, J=8.6 Hz, 1H), 4.51 (p, J=6.6 Hz, 1H), 2.99 (d, J=9.3 Hz, 2H), 2.62 (s, 3H), 1.96 (s, 3H), 1.83 (q, J=9.6 Hz, 2H), 1.46 (d, J=3.1 Hz, 16H), 1.38 (s, 1H).

Example 62

(348) ##STR00283##

(349) 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 50 mg (36%) of the title compound. LCMS: 553 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.04 (s, 1H), 9.55 (d,

J=8.0 Hz, 1H), 8.39 (s, 1H), 8.30 (d, J=1.9 Hz, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.1, 1.9 Hz, 1H), 7.50 (d, J=5.3 Hz, 1H), 7.36 (d, J=8.1 Hz, 1H), 5.30 (t, J=8.6 Hz, 1H), 4.51 (p, J=6.6 Hz, 1H), 2.99 (d, J=9.3 Hz, 2H), 2.62 (s, 3H), 1.96 (s, 3H), 1.83 (q, J=9.6 Hz, 2H), 1.46 (d, J=3.1 Hz, 16H), 1.38 (s, 1H).

Example 63

(350) ##STR00284##

(351) 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 22 mg (33%) of the title compound. LCMS: 553 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.04 (s, 1H), 9.95 (d, J=8.0 Hz, 1H), 8.39 (s, 1H), 8.30 (d, J=1.9 Hz, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.2, 1.9 Hz, 1H), 7.50 (d, J=5.3 Hz, 1H), 7.37 (d, J=8.2 Hz, 1H), 5.29 (t, J=8.5 Hz, 1H), 4.51 (p, J=6.6 Hz, 1H), 2.99 (d, J=8.3 Hz, 2H), 2.62 (s, 3H), 2.00 (d, J=20.4 Hz, 3H), 1.84 (t, J=9.4 Hz, 2H), 1.47 (d, J=6.7 Hz, 6H), 1.41 (s, 9H), 1.37 (d, J=9.7 Hz, 1H).

Example 64

(352) ##STR00285##

(353) 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 21 mg (32%) of the title compound. LCMS: 553 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.04 (s, 1H), 9.95 (d, J=8.0 Hz, 1H), 8.39 (s, 1H), 8.30 (d, J=1.9 Hz, 1H), 8.25 (d, J=5.2 Hz, 1H), 8.18 (dd, J=8.2, 1.9 Hz, 1H), 7.50 (d, J=5.3 Hz, 1H), 7.37 (d, J=8.2 Hz, 1H), 5.29 (t, J=8.5 Hz, 1H), 4.51 (p, J=6.6 Hz, 1H), 2.99 (d, J=8.3 Hz, 2H), 2.62 (s, 3H), 2.00 (d, J=20.4 Hz, 3H), 1.84 (t, J=9.4 Hz, 2H), 1.47 (d, J=6.7 Hz, 6H), 1.41 (s, 9H), 1.37 (d, J=9.7 Hz, 1H).

Example 65

(354) ##STR00286##

(355) 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(5-ethyl-1-methyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 6 mg (27%) of the title compound. LCMS: 539 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.32 (s, 1H), 9.97 (d, J=7.6 Hz, 1H), 8.27 (d, J=5.1 Hz, 1H), 8.18 (s, 1H), 8.02 (s, 1H), 7.42 (s, 1H), 7.38 (d, J=8.1 Hz, 1H), 6.75 (s, 1H), 5.28 (s, 1H), 3.86 (s, 3H), 2.99 (d, J=7.1 Hz, 2H), 2.75-2.65 (m, 2H), 2.00 (d, J=22.5 Hz, 3H), 1.85 (t, J=9.5 Hz, 2H), 1.41 (s, 9H), 1.27 (t, J=7.5 Hz, 3H).

Example 66

(356) ##STR00287##

(357) 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(S)-2-[2-(5-ethyl-1-methyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide. The title compound was obtained via chiral SFC using ChiralPak AD-H 21×250 mm and a mobile phase of 30% 2-Propanol+0.5% DMEA in CO.sub.2. Yielded 6 mg (27%) of the title compound. LCMS: 539 (M+H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.32 (s, 1H), 9.97 (d, J=7.6 Hz, 1H), 8.27 (d, J=5.1 Hz, 1H), 8.18 (s, 1H), 8.02 (s, 1H), 7.42 (s, 1H), 7.38 (d, J=8.1 Hz, 1H), 6.75 (s, 1H), 5.28 (s, 1H), 3.86 (s, 3H), 2.99 (d, J=7.1 Hz, 2H), 2.75-2.65 (m, 2H), 2.00 (d, J=22.5 Hz, 3H), 1.85 (t, J=9.5 Hz, 2H), 1.41 (s, 9H), 1.27 (t, J=7.5 Hz, 3H).

(358) Methods for IAP Fused Analogs in Exemplification Table 1.

(359) Method A:

(360) ##STR00288## ##STR00289##

Method B:

(361) ##STR00290##

Method C:

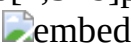
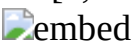
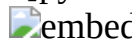
(362) ##STR00291##

Method D

(363) ##STR00292## ##STR00293##

Example Table 1

(364) TABLE-US-00003 Example # and Preparation IUPAC Name Structure Method Starting Material Example 67 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1,3-dimethyl-1H-pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 7-Bromo-2-(1,3- dimethyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridine LCMS: 511 (M + H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) d 13.13 (s, 1H), 9.75 (d, J = 8.7 Hz, 1H), 8.42-8.21 (m, 3H), 8.15 (dd, J = 8.2, 1.9 Hz, 1H), 7.60-7.30 (m, 2H), 5.28 (d, J = 4.9 Hz, 1H), 3.88 (s, 3H), 2.89 (d, J = 6.3 Hz, 2H), 2.61 (s, 3H), 2.20-1.71 (m, 4H), 1.37 (s, 9H). Example 68 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1,3,5-trimethyl-1H-pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 7-Bromo-2-(1,3,5- trimethyl-1H-pyrazol- 4-yl)-3H-imidazo[4,5- b]pyridine LCMS: 525 (M + H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) 12.76 (s, 1H), 9.76 (d, J = 8.7 Hz, 1H), 8.52-8.22 (m, 2H), 8.22-8.03 (m, 1H), 7.51 (d, J = 5.2 Hz, 1H), 7.38 (d, J = 8.2 Hz, 1H), 5.44-5.14 (m, 1H), 3.76 (s, 3H), 2.89 (s, 2H), 2.55 (s, 3H), 2.43 (s, 3H), 2.13-1.68 (m, 3H), 1.37 (s, 9H). Example 69 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {1-[4- (4,4,5,5-tetramethyl- [1,3,2]dioxaborolan- 2-yl)-phenyl]- cyclopropyl}-amide  Method A 2-[2-(1,3,5-Trimethyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine LCMS: 524 (M + H). 1H NMR (400 MHz, DMSO-d6) d 9.29 (d, J = 8.2 Hz, 1H), 8.20 (d, J = 43.1 Hz, 3H), 7.34 (d, J = 8.1 Hz, 1H), 6.64 (d, J = 0.9 Hz, 1H), 5.28 (t, J = 8.6 Hz, 1H), 3.75 (s, 3H), 3.21-3.09 (m, 1H), 2.56 (s, 3H), 2.43 (s, 3H), 1.89 (dd, J = 62.3, 13.7 Hz, 4H), 1.30 (d, J = 6.9 Hz, 6H). Example 70 5-tert-butyl-N-12-[2- (1,3,5-trimethyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl}-1,2- oxazole-3- carboxamide  Method B 2-[2-(1,3,5-Trimethyl- 1H-pyrazol-4- yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine LCMS: 538 (M + H). NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.54 (d, J = 8.1 Hz, 1H), 8.29 (d, J = 5.2 Hz, 1H), 8.25 (d, J = 1.9 Hz, 1H), 7.91-8.13 (m, 2H), 7.52 (d, J = 5.3 Hz, 1H), 7.35 (d, J = 8.1 Hz, 1H), 5.29 (t, J = 8.6 Hz, 1H), 3.75 (s, 3H), 2.96 (d, J = 9.4 Hz, 2H), 2.55 (s, 3H), 2.43 (s, 3H), 1.96 (d, J = 12.1 Hz, 3H), 1.83 (q, J = 9.5 Hz, 2H), 1.46 (s, 9H). Example 71 5-tert-butyl-N-[(5S)- 2-[2-(1,3,5-trimethyl- 1H-pyrazol-4-yl)-1H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,2-oxazole-3- carboxamide  Method Chiral SFC 5-tert-butyl-N- {2-[2- (1,3,5-trimethyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl}- 1,2-oxazole-3- carboxamide LCMS: 538 (M + H). NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.54 (d, J = 8.1 Hz, 1H), 8.29 (d, J = 5.2 Hz, 1H), 8.25 (d, J = 1.9 Hz, 1H), 7.91-8.13 (m, 2H), 7.52 (d, J = 5.3 Hz, 1H), 7.35 (d, J = 8.1 Hz, 1H), 5.29 (t, J = 8.6 Hz, 1H), 3.75 (s, 3H), 2.96 (d, J = 9.4 Hz, 2H), 2.55 (s, 3H), 2.43 (s, 3H), 1.97 (d, J = 12.1 Hz, 3H), 1.83 (q, J = 9.5 Hz, 2H), 1.44 (s, 9H). Example 72 5-tert-butyl-N-[(5R)- 2-[2-(1,3,5-trimethyl- 1H-pyrazol-4-yl)-1H- imidazo[4,5- b]pyridin-7-yl]-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,2-oxazole-3- carboxamide  Method Chiral SFC 5-tert-butyl-N-{2-[2- (1,3,5-trimethyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl}- 1,2-oxazole-3- carboxamide LCMS: 538 (M + H). NMR (400 MHz, DMSO-d.sub.6) δ 12.75 (s, 1H), 9.54 (d, J = 8.1 Hz, 1H), 8.29 (d, J = 5.2 Hz, 1H), 8.25 (d, J = 1.9 Hz, 1H), 7.91-8.13 (m, 2H), 7.52 (d, J = 5.3 Hz, 1H), 7.35 (d, J = 8.1 Hz, 1H), 5.29 (t, J = 8.6 Hz, 1H), 3.75 (s, 3H), 2.96 (d, J = 9.4 Hz, 2H), 2.55 (s, 3H), 2.43 (s, 3H), 1.96 (d, J = 12.1 Hz, 3H), 1.83 (q, J = 9.5 Hz, 2H), 1.46 (s, 9H). Example 73 3-tert-butyl-N-[(1S)- 6-[2-(1,3,5-trimethyl- 1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4- tetrahydronaphthalen- 1-yl]-1,2,4-

oxadiazole-5- carboxamide 00  embedded image Chiral SFC 3-tert-butyl-N-[6-[2- (1,3,5-trimethyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-1,2,3,4- tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5- carboxamide LC/MS: 525 (M + H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) 12.76 (s, 1H), 9.76 (d, J = 8.7 Hz, 1H), 8.52-8.22 (m, 2H), 8.22-8.03 (m, 1H), 7.51 (d, J = 5.2 Hz, 1H), 7.38 (d, J = 8.2 Hz, 1H), 5.44-5.14 (m, 1H), 3.76 (s, 3H), 2.89 (s, 2H), 2.55 (s, 3H), 2.43 (s, 3H), 2.13-1.68 (m, 3H), 1.37 (s, 9H). Example 74 3-tert-butyl-N-[(1R)- 6-[2-(1,3,5-trimethyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4- tetrahydronaphthalen- 1-yl]-1,2,4- oxadiazole-5- carboxamide 01  embedded image Chiral SFC 3-tert-butyl-N-[6-[2-(1,3,5-trimethyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-1,2,3,4- tetrahydronaphthalen-1- yl]-1,2,4-oxadiazole-5- carboxamide LC/MS: 525 (M + H). .sup.1H NMR (400 MHz, DMSO-d.sub.6) 12.76 (s, 1H), 9.76 (d, J = 8.7 Hz, 1H), 8.52-8.22 (m, 2H), 8.22-8.03 (m, 1H), 7.51 (d, J = 5.2 Hz, 1H), 7.38 (d, J = 8.2 Hz, 1H), 5.44-5.14 (m, 1H), 3.76 (s, 3H), 2.89 (s, 2H), 2.55 (s, 3H), 2.43 (s, 3H), 2.13-1.68 (m, 3H), 1.37 (s, 9H). Example 75 3-tert-butyl-N-[(1R)- 6-[2-(1-cyclopropyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4- tetrahydronaphthalen- 1-yl]-1,2,4- oxadiazole-5- carboxamide 02  embedded image Chiral SFC Chiral SFC 3-tert-butyl-N-6-[2-(1- cyclopropyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-1,2,3,4- tetrahydronaphthalen-1- yl]-1,2,4-oxadiazole-5- carboxamide MS: 523 (M + H). .sup.1H-NMR (400 MHz, DMSO-d.sub.6, ppm): δ 13.26 (brs, 1H), 9.77 (d, J = 8.7 Hz, 1H), 8.47 (s, 1H), 8.30 (s, 1H), 8.27-8.04 (m, 3H), 7.46-7.33 (m, 2H), 5.30-5.24 (m, 1H), 4.61 (p, J = 6.6 Hz, 1H), 2.93-2.87 (m, 2H), 2.05-1.99 (m, 3H), 1.85-1.79 (m, 1H), 1.47 (d, J = 6.6 Hz, 6H), 2.11-1.36 (m, 4H). Example 76 3-tert-butyl-N-[(1S)- 6-[2-(1-cyclopropyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4- tetrahydronaphthalen- 1-yl]-1,2,4- oxadiazole-5- carboxamide 03  embedded image Chiral SFC Chiral SFC 3-tert-butyl-N-6-[2-(1- cyclopropyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-1,2,3,4- tetrahydronaphthalen-1- yl]-1,2,4-oxadiazole-5- carboxamide MS: 523 (M + H). .sup.1H-NMR (400 MHz, DMSO-d.sub.6, ppm): δ 13.22 (br s, 1H), 9.77 (d, J = 8.7 Hz, 1H), 8.47 (s, 1H), 8.30 (s, 1H), 8.27-8.04 (m, 3H), 7.46-7.33 (m, 2H), 5.29-5.23 (m, 1H), 4.61 (p, J = 6.6 Hz, 1H), 2.93-2.87 (m, 2H), 2.05-1.99 (m, 3H), 1.85-1.78 (m, 1H), 1.46 (d, J = 6.6 Hz, 6H), 2.11-1.36 (m, 4H). Example 77 5-tert-butyl-N-[(5S)- 2-{2-[3-methyl-1- (propan-2-yl)-1H- pyrazol-4-yl]-1H- imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro-5H-benzo[7]annulen- 5-yl]-1,3,4- oxadiazole-2- carboxamide 04  embedded image Chiral SFC 5-tert-butyl-N-[-2-{2- [3-methyl-1-(propan-2- yl)-1H-pyrazol-4-yl]- 1H-imidazo[4,5- b]pyridin-7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,3,4-oxadiazole-2- carboxamide LC/MS: (M + H) 553. HPLC: 3.74 min. 1H NMR (400 MHz, DMSO-d6) δ 13.05 (s, 1H), 9.88 (d, J = 8.2 Hz, 1H), 8.40 (s, 1H), 8.32-8.13 (m, 3H), 7.50 (d, J = 5.3 Hz, 1H), 7.39 (d, J = 8.2 Hz, 1H), 5.29 (t, J = 8.5 Hz, 1H), 4.51 (p, J = 6.6 Hz, 1H), 2.99 (d, J = 8.9 Hz, 2H), 2.62 (s, 3H), 1.98 (d, J = 10.2 Hz, 3H), 1.83 (t, J = 9.5 Hz, 2H), 1.47 (d, J = 6.7 Hz, 6H), 1.43 (s, 9H). Example 78 5-tert-butyl-N-[(5R)- 2-{2-[3-methyl-1- (propan-2-yl)-1H- pyrazol-4-yl]-1H- imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,3,4- oxadiazole-2- carboxamide 05  embedded image Chiral SFC 5-tert-butyl-N-[-2-{2- [3-methyl-1-(propan-2- yl)-1H-pyrazol-4-yl]- 1H-imidazo[4,5- b]pyridin-7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,3,4-oxadiazole-2- carboxamide LC/MS: (M + H) 553. HPLC: 3.74 min. 1H NMR (400 MHz, DMSO-d6) δ 13.01 (s, 1H), 9.87 (d, J = 8.2 Hz, 1H), 8.42 (s, 1H), 8.32-8.13 (m, 3H), 7.50 (d, J = 5.3 Hz, 1H), 7.39 (d, J = 8.2 Hz, 1H), 5.29 (t, J = 8.5 Hz, 1H), 4.51 (p, J = 6.6 Hz, 1H), 2.99 (d, J = 8.9 Hz, 2H), 2.62 (s, 3H), 1.98 (d, J = 10.2 Hz, 3H), 1.83 (t, J = 9.5 Hz, 2H), 1.48 (d, J = 6.5 Hz, 6H), 1.42 (s, 9H). Example 79 5-tert-butyl-N-[(5R)- 2-{2-[5-methyl-1- (oxan-4-yl)-1H- pyrazol-4-yl]-1H- imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,3,4- oxadiazole-2- carboxamide 06  embedded image Chiral SFC 5-tert-butyl-N-[-2-{2-[5- methyl-1-(oxan-4-yl)- 1H-pyrazol-4-yl]-1H- imidazo[4,5-b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,3,4-oxadiazole-2- carboxamide LC/MS: (M + H) 595. HPLC: 3.17 min. 1H NMR (400 MHz, DMSO-d6) δ 13.20 (br.s, 1H), 9.90 (m, 1H), 8.27 (m, 2H), 8.14 (m, 2H), 7.5 (s, 1H), 7.36 (m, 1H),

5.33 (s, 1H), 4.51 (s, 1H), 4.0 (2H), 3.50 (t, 2H), 3.0 (m, 2H), 2.82 (s, 3H), 2.0 (m, 5H), 1.88 (m, 4H), 1.25 (m, 9H). Example 80 5-tert-butyl-N-[(5S)- 2-{2-[5-methyl-1- (oxan-4-yl)-1H- pyrazol-4-yl]-1H- imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,3,4-oxadiazole-2- carboxamide 07 Chiral SFC 5-tert-butyl-N-[2-{2-[5- methyl-1- (oxan-4-yl)- 1H-pyrazol-4-yl]-1H- imidazo[4,5-b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,3,4-oxadiazole-2- carboxamide LC/MS: (M + H) 595. HPLC: 3.17 min. 1H NMR (400 MHz, DMSO-d6) δ 13.20 (br.s, 1H), 9.90 (m, 1H), 8.27 (m, 2H), 8.14 (m, 2H), 7.5 (s, 1H), 7.36 (m, 1H), 5.33 (s, 1H), 4.51 (s, 1H), 4.0 (2H), 3.50 (t, 2H), 3.0 (m, 2H), 2.82 (s, 3H), 2.0 (m, 5H), 1.88 (m, 4H), 1.25 (m, 9H). Example 81 3-tert-butyl-N-[(5S)- 2-{2-[1-(propan-2-yl)-1H-pyrazol-4-yl]- 1H-imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,2,4- oxadiazole-5- carboxamide 08 Chiral SFC 3-tert-butyl-N-[2-{2-[1- (propan-2-yl)-1H- pyrazol-4-yl]-1H- imidazo[4,5-b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,2,4-oxadiazole-5- carboxamide LC/MS: (M + H) 581. HPLC: 4.0 min. 1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.97 (d, J = 7.9 Hz, 1H), 8.48 (s, 1H), 8.31-7.96 (m, 4H), 7.42 (dd, J = 32.9, 6.7 Hz, 2H), 5.29 (s, 1H), 4.62 (p, J = 6.7 Hz, 1H), 3.01 (s, 2H), 2.12-1.77 (m, 6H), 1.49 (dd, J = 6.7, 1.7 Hz, 6H), 1.42 (d, J = 1.7 Hz, 6H). Example 82 3-tert-butyl-N-[(5R)- 2-{2-[1-(propan-2- yl)-1H-pyrazol-4-yl]- 1H-imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- oxadiazole-5- carboxamide 09 Chiral SFC 3-tert-butyl-N-[2-{2-[1- (propan-2-yl)-1H- pyrazol-4-yl]-1H- imidazo[4,5-b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,2,4-oxadiazole-5- carboxamide LC/MS: (M + H) 581. HPLC: 4.0 min. 1H NMR (400 MHz, DMSO-d6) δ 13.21 (s, 1H), 9.98 (d, J = 7.9 Hz, 1H), 8.46 (s, 1H), 8.31-7.96 (m, 4H), 7.42 (dd, J = 32.9, 6.7 Hz, 2H), 5.29 (s, 1H), 4.62 (p, J = 6.7 Hz, 1H), 3.01 (s, 2H), 2.12-1.77 (m, 6H), 1.49 (dd, J = 6.7, 1.7 Hz, 6H), 1.41 (d, J = 1.7 Hz, 6H). Example 83 3-tert-butyl-N-{2-[2- (1-cyclopropyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl}-1,2,4- oxadiazole-5- carboxamide 0 Method D 3-tert-butyl-N-[2-{2-[1- (propan-2-yl)-1H- pyrazol-4-yl]-1H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,2,4-oxadiazole-5- carboxamide LC/MS: (M + H) 536. HPLC: 4.0 min. 1H NMR (400 MHz, DMSO-d6) δ 9.98 (d, J = 7.9 Hz, 1H), 8.47 (s, 1H), 8.18 (d, J = 56.8 Hz, 2H), 5.28 (s, 1H), 3.86 (dt, J = 7.3, 3.6 Hz, 1H), 2.99 (d, J = 8.3 Hz, 2H), 1.91 (d, J = 52.1 Hz, 5H), 1.42 (s, 9H), 1.26-0.86 (m, 4H). Example 84 5-tert-butyl-N-(2-{2- [5-methyl-1-(propan- 2-yl)-1H-pyrazol-4- yl]-3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl)-1,3,4- oxadiazole-2- carboxamide  Method B 2-[2- (1-Cyclopropyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine hydrochloride LC/MS: (M + H) 553. 1H NMR (400 MHz, DMSO- d6) δ 9.89 (d, J = 8.0 Hz, 1H), 8.23 (d, J = 35.5 Hz, 3H), 7.56-7.34 (m, 2H), 5.29 (s, 1H), 4.75-4.55 (m, 1H), 3.07-2.65 (m, 5H), 2.10-1.73 (m, 4H), 1.42 (m, 15H), 1.25-1.02 (m, 2H). Example 85 3-tert-butyl-N-{2-[2- (1-ethyl-5-methyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl}-1,2,4- oxadiazole-5- carboxamide  Method B 2-[2-(1-Ethyl-5-methyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine hydrochloride LC/MS: (M + H) 538. 1H NMR (400 MHz, Chloroform-d) δ 8.45 (d, J = 5.4 Hz, 1H), 7.91 (d, J = 21.6 Hz, 2H), 7.61 (s, 1H), 7.43 (s, 1H), 7.01 (s, 1H), 5.44 (t, J = 8.2 Hz, 1H), 4.21 (d, J = 7.2 Hz, 2H), 3.05 (d, J = 9.5 Hz, 2H), 2.39 (s, 3H), 2.05 (d, J = 16.5 Hz, 4H), 1.87 (s, 2H), 1.49 (d, J = 19.2 Hz, 12H). Example 86 5-tert-butyl-N-(2-{2- [3,5-dimethyl-1- (propan-2-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl)-1,3,4- oxadiazole-2- carboxamide  Method B 2-[2-(1-Isopropyl-3,5- dimethyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine hydrochloride LC/MS: (M + H) 567. 1H NMR (400 MHz, DMSO-d6) δ 12.76 (s, 1H), 9.88 (d, J = 8.2 Hz, 1H), 8.27 (s, 3H), 7.58-7.30 (m, 3H), 5.29 (d, J = 9.1 Hz, 1H), 4.73-4.42 (m, 1H), 3.31 (s, 9H), 2.97 (s, 3H), 2.69-2.32 (m, 10H), 1.90 (d, J = 56.1 Hz, 4H), 1.53-1.29 (m, 15H). Example 87 5-tert-butyl-N-[(5R)- 2-{2-

[3,5-dimethyl-1- (propan-2-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,3,4- oxadiazole-2- carboxamide  Chiral SFC 5-tert-butyl-N-(2-{2- [3,5-dimethyl-1- (propan-2-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl)- 1,3,4-oxadiazole-2- carboxamide LC/MS: (M + H) 567. ¹H NMR (400 MHz, DMSO-d₆) δ 12.76 (s, 1H), 9.88 (d, J = 8.2 Hz, 1H), 8.27 (s, 3H), 7.58-7.30 (m, 3H), 5.29 (d, J = 9.1 Hz, 1H), 4.73-4.42 (m, 1H), 3.31 (s, 9H), 2.97 (s, 3H), 2.69-2.32 (m, 10H), 1.90 (d, J = 56.1 Hz, 4H), 1.53-1.26 (m, 15H). Example 88 5-tert-butyl-N-[(5S)- 2-{2-[3,5-dimethyl-1- (propan-2-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,3,4- oxadiazole-2- carboxamide  Chiral SFC 5-tert-butyl-N-(2-{2- [3,5-dimethyl-1- (propan-2-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl)- 1,3,4-oxadiazole-2- carboxamide LC/MS: (M + H) 567. ¹H NMR (400 MHz, DMSO-d₆) δ 12.76 (s, 1H), 9.88 (d, J = 8.2 Hz, 1H), 8.27 (s, 3H), 7.58-7.30 (m, 3H), 5.29 (d, J = 9.1 Hz, 1H), 4.73-4.42 (m, 1H), 3.31 (s, 9H), 2.97 (s, 3H), 2.69-2.32 (m, 10H), 1.90 (d, J = 56.1 Hz, 4H), 1.53-1.29 (m, 15H). Example 89 5-tert-butyl-N-[(5S)- 2-[2-(5-ethyl-1- methyl-1H-pyrazol-3- yl)-1H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,3,4- oxadiazole-2- carboxamide  Chiral SFC 5-tert-butyl-N-[2-[2-(5- ethyl-1-methyl-1H- pyrazol-3-yl)-1H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,3,4-oxadiazole-2- carboxamide LC/MS: (M + H) 539. HPLC: 4.18 (97%) min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.79 (d, 1H), 8.30 (m, 2H), 8.23 (m, 1H), 8.02 (s, 1H), 7.49 (m, 2H), 6.75 (s, 1H), 5.25 (s, 1H), 3.84 (s, 3H), 2.99 (m, 2H), 2.45 (m, 2H), 2.02 (m, 3H), 1.79 (m, 2H), 1.47 (m, 10H), 1.25 (m, 3H), 0.99 (d, 2H). Example 90 5-tert-butyl-N-[(5R)- 2-[2-(5-ethyl-1- methyl-1H-pyrazol-3- yl)-1H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,3,4- oxadiazole-2- carboxamide  Chiral SFC 5-tert-butyl-N-[2-[2-(5- ethyl-1-methyl-1H- pyrazol-3-yl)-1H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl]- 1,3,4-oxadiazole-2- carboxamide LC/MS: (M + H) 539. HPLC: 4.18 (97%) min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.79 (d, 1H), 8.30 (m, 2H), 8.23 (m, 1H), 8.01 (s, 1H), 7.49 (m, 2H), 6.75 (s, 1H), 5.25 (s, 1H), 3.83 (s, 3H), 2.99 (m, 2H), 2.45 (m, 2H), 2.03 (m, 3H), 1.79 (m, 2H), 1.47 (m, 10H), 1.23 (m, 3H), 1.00 (d, 2H). Example 91 5-tert-butyl-N-(2-{2- [1-(propan-2-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin-7-yl}- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl)-1,2- oxazole-3- carboxamide  Method B 2-[2-(1-Isopropyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine LC/MS: (M + H) 538. ¹H NMR (400 MHz, Chloroform-d) δ 8.49-7.89 (m, 6H), 7.58-7.33 (m, 2H), 6.47 (s, 1H), 5.47 (s, 1H), 4.61 (s, 0H), 3.07 (s, 1H), 2.16-1.83 (m, 4H), 1.77-1.51 (m, 4H), 1.41 (s, 15H). Example 92 5-tert-butyl-N-[(5R)- 2-{2-[1-(propan-2- yl)-1H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,2-oxazole-3- carboxamide  Chiral SFC Chiral SFC 5-tert-butyl-N-[2-{2-[1- (propan-2-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl)- 1,2-oxazole-3- carboxamide LC/MS: (M + H) 538. .sup.¹H NMR (400 MHz, Methanol-d.sub.4) δ 10.11 (d, J = 8.0 Hz, 1H), 9.30 (s, 1H), 9.10-8.92 (m, 2H), 8.82 (s, 1H), 8.33-8.12 (m, 3H), 6.09 (t, J = 8.9 Hz, 1H), 5.51-5.34 (m, 1H), 4.38-4.07 (m, 2H), 2.90-2.53 (m, 6H), 2.29 (d, J = 6.5 Hz, 6H), 2.18 (s, 9H). Example 93 5-tert-butyl-N-[(5S)- 2-{2-[1-(propan-2- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-benzo[7]annulen- 5-yl]-1,2-oxazole-3- carboxamide  Chiral SFC Chiral SFC 5-tert-butyl-N-[2-{2-[1- (propan-2-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzo[7]annulen-5-yl)- 1,2-oxazole-3- carboxamide LC/MS: (M + H) 538. .sup.¹H NMR (400 MHz, Methanol-d.sub.4) δ 10.11 (d, J = 8.0 Hz, 1H), 9.30 (s, 1H), 9.10-8.92 (m, 2H), 8.82 (s, 1H), 8.33-8.12 (m, 3H), 6.09 (t, J = 8.9 Hz, 1H), 5.51-5.34 (m, 1H), 4.38-4.07 (m, 2H), 2.90-2.53 (m, 6H), 2.29 (d, J = 6.5 Hz, 6H), 2.18 (s, 9H). Example 94 5-tert-butyl-N-{2-[2- (1,3,5-trimethyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9- tetrahydro- 5H-benzo[7]annulen- 5-yl}-1,2-oxazole-3- carboxamide  Method B

2-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine LC/MS: (M + H) 538. ¹H NMR (400 MHz, DMSO-d₆) δ 7.55 (d, J = 5.2 Hz, 1H), 7.06 (s, 2H), 6.65 (d, J = 7.9 Hz, 2H), 5.71 (s, 1H), 5.54 (s, 1H), 3.02 (s, 3H), 2.49-2.17 (m, 2H), 1.80-1.55 (m, 6H), 1.39-0.97 (m, 4H), 0.65-0.50 (m, 12H). Example 95

3-tert-butyl-N-[(1S)-6-{2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo[4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5-carboxamide  Chiral SFC 3-tert-butyl-N-[6-{2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo[4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5-carboxamide LC/MS: (M + H) 553. HPLC: 3.69 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.75 (s, 1H), 8.31-8.05 (m, 3H), 7.50 (m, 2H), 5.25 (s, 1H), 4.51 (m, 1H), 2.89 (m, 5H), 2.51 (s, 3H), 2.48 (s, 3H), 1.99 (m, 4H), 1.34 (m, 6H), 1.27 (s, 9H). Example 96

3-tert-butyl-N-[(1R)-6-{2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo[4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5-carboxamide  Chiral SFC 3-tert-butyl-N-[(1R)-6-{2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo[4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5-carboxamide LC/MS: (M + H) 553. HPLC: 3.69 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.75 (s, 1H), 8.31-8.05 (m, 3H), 7.50 (m, 2H), 5.25 (s, 1H), 4.51 (m, 1H), 2.90 (m, 5H), 2.52 (s, 3H), 2.48 (s, 3H), 2.0 (m, 4H), 1.32 (m, 6H), 1.29 (s, 9H). Example 97

3-tert-butyl-N-{2-[2-(1-ethyl-5-methyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl}-1,2,4-oxadiazole-5-carboxamide  Method B 2-[2-(1-Ethyl-5-methyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride LC/MS: (M + H) 539. ¹H NMR (400 MHz, Chloroform-d) δ 11.48 (s, 1H), 8.53-8.39 (m, 1H), 8.15-8.01 (m, 2H), 7.64-7.38 (m, 3H), 7.28 (d, J = 2.1 Hz, 1H), 6.86 (s, 1H), 5.46 (t, J = 8.3 Hz, 1H), 4.33-4.15 (m, 2H), 3.33-3.03 (m, 2H), 2.40 (s, 3H), 2.21-1.75 (m, 6H), 1.60-1.30 (m, 12H). Example 98

N-[(5R)-2-{2-[3-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-4,5,6,7-tetrahydro-1-benzothiophene-2-carboxamide  Chiral SFC N-[2-{2-[3-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-4,5,6,7-tetrahydro-1-benzothiophene-2-carboxamide LC/MS: (M + H) 565. ¹H NMR (400 MHz, Methanol-d₄) δ 8.29 (s, 1H), 8.09 (s, 1H), 7.58 (s, 1H), 7.42 (d, J = 8.0 Hz, 1H), 5.38 (d, J = 10.0 Hz, 1H), 4.76-4.62 (m, 1H), 3.18-2.93 (m, 2H), 2.81 (d, J = 10.6 Hz, 5H), 2.69 (t, J = 6.0 Hz, 2H), 2.19-1.77 (m, 10H), 1.51 (d, J = 6.6 Hz, 6H). Example 99

N-[(5R)-2-{2-[3-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-4,5,6,7-tetrahydro-1-benzothiophene-2-carboxamide  Chiral SFC N-[2-{2-[3-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-4,5,6,7-tetrahydro-1-benzothiophene-2-carboxamide LC/MS: (M + H) 565. ¹H NMR (400 MHz, Methanol-d₄) δ 8.29 (s, 1H), 8.09 (s, 1H), 7.58 (s, 1H), 7.42 (d, J = 8.0 Hz, 1H), 5.38 (d, J = 10.0 Hz, 1H), 4.76-4.61 (m, 1H), 3.19-2.92 (m, 2H), 2.80 (d, J = 10.6 Hz, 5H), 2.68 (t, J = 6.0 Hz, 2H), 2.19-1.76 (m, 10H), 1.50 (d, J = 6.6 Hz, 6H). Example 100

3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-6-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  SFC LC/MS: (M + H) 497. HPLC: 3.28 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.30 (br.s, 1H), 9.87 (d, 1H), 8.50 (s, 1H), 8.27 (m, 1H), 8.12 (m, 3H), 7.49 (m, 2H), 5.33 (s, 1H), 3.98 (s, 3H), 2.94 (m, 2H), 2.02 (m, 3H), 1.79 (m, 1H), 1.40 (s, 9H). Example 101

3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid [6-{2-[1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl]-amide  Method C 7-Bromo-2-[1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridine and 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid [6-(4,4,5,5-tetramethyl-[1,3,2]dioxaborolan-2-yl)-1,2,3,4-tetrahydro-naphthalen-1-yl]-amide LC/MS: (M +

H) 567. HPLC: 3.44 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.74 (s, 1H), 8.47 (s, 1H), 8.04-8.21 (m, 3H), 7.29 (m, 2H), 5.27 (m, 1H), 4.50 (m, 1H), 3.99 (m, 2H), 2.88 (m, 2H), 2.0 (m, 7H), 1.37 (s, 9H). Example 102 1-tert-Butyl-1H- [1,2,3]triazole-4- carboxylic acid ((R)- 6-{2-[1-(tetrahydro-pyran-4-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  SFC LC/MS: (M + H) 566. HPLC: 3.04 min. ¹H NMR (400 MHz, DMSO-d₆) δ 8.77 (s, 1H), 8.25 (d, 1H), 8.27 (s, 1H), 8.10 (m, 3H), 7.5 (m, 2H), 5.25 (m, 1H), 4.5 (m, 1H), 4.0 (m, 2H), 2.57 (m, 2H), 2.0 (m, 6H), 1.75 (m, 1H), 1.68 (s, 9H). Example 103 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1-methyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide 0  Method C -(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid [6-(4,4,5,5-tetramethyl- [1,3,2]dioxaborolan-2- yl)-1,2,3,4-tetrahydro- naphthalen-1-yl]-amide and 7-Bromo-2-(1- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridine LC/MS: (M + H) 495. HPLC: 2.99 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.67 (m, 1H), 8.38 (m, 1H), 8.24 (m, 1H), 8.15 (m, 2H), 7.44 (m, 1H), 7.31 (m, 1H), 5.25 (s, 1H), 3.92 (s, 3H), 2.81 (m, 2H), 2.0 (m, 4H), 1.5 (s, 3H), 1.22 (m, 2H), 1.0 (s, 2H). Example 104 5-tert-Butyl- [1,3,4]oxadiazole-2- carboxylic acid {6-[2- (1-methyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 5-tert-Butyl- [1,3,4]oxadiazole-2- carboxylic acid [6-(4,4,5,5-tetramethyl- [1,3,2]dioxaborolan-2- yl)-1,2,3,4-tetrahydro- naphthalen-1-yl]-amide and 7-Bromo-2-(1- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridine LC/MS: (M + H) 497. HPLC: 2.93 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.67 (m, 1H), 8.49 (s, 1H), 8.28 (s, 1H), 8.19 (s, 1H), 8.0 (m, 2H), 7.44 (m, 2H), 5.25 (m, 1H), 2.77 (m, 2H), 2.0 (m, 4H), 1.41 (s, 9H). Example 105 5-tert-Butyl- [1,3,4]oxadiazole-2- carboxylic acid {6-[2- (1-methyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method B 2-{2-[1-(Tetrahydro- pyran-4-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylaminehydrochloride LC/MS: (M + H) 581. HPLC: 4.0 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.20 (br.s, 1H), 9.99 (m, 1H), 8.51 (s, 1H), 8.0-8.25 (m, 3H), 7.20 (m, 2H), 5.30 (s, 1H), 4.51 (s, 1H), 4.0 (2H), 3.49 (m, 2H), 3.0 (m, 2H), 2.1 (m, 9H), 1.19 (s, 10H). Example 106 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(1-Isopropyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine hydrochloride LC/MS: (M + H) 581. HPLC: 4.0 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.20 (s, 1H), 9.97 (d, J = 7.9 Hz, 1H), 8.48 (s, 1H), 8.31-7.96 (m, 4H), 7.42 (dd, J = 32.9, 6.7 Hz, 2H), 5.29 (s, 1H), 4.62 (p, J = 6.7 Hz, 1H), 3.01 (s, 2H), 2.12-1.77 (m, 6H), 1.49 (dd, J = 6.7, 1.7 Hz, 6H), 1.42 (d, J = 1.7 Hz, 10H). Example 107 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1,5-dimethyl-1H- pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid [6-(4,4,5,5-tetramethyl- [1,3,2]dioxaborolan-2- yl)-1,2,3,4-tetrahydro- naphthalen-1-yl]-amide and 7-Bromo-2-(1,5- dimethyl-1H-pyrazol-3- yl)-3H-imidazo[4,5- b]pyridine LC/MS: (M + H) 511. HPLC: 3.55 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.74 (s, 1H), 8.08-8.26 (m, 2H), 7.37 (m, 2H), 6.75 (s, 1H), 5.23 (s, 1H), 3.79 (s, 3H), 2.82 (m, 2H), 2.29 (s, 3H), 2.0 (m, 3H), 1.82 (s, 1H), 1.37 (s, 9H). Example 108 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (5-ethyl-1-methyl-1H-pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C Butyl- [1,2,4]oxadiazole-5- carboxylic acid [6- (4,4,5,5-tetramethyl- [1,3,2]dioxaborolan-2- yl)-1,2,3,4-tetrahydro- naphthalen-1-yl]-amide and 7-Bromo-2-(5- ethyl-1- methyl-1H- pyrazol-3-yl)-3H- imidazo[4,5-b]pyridine LC/MS: (M + H) 525. HPLC: 3.77 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.74 (s, 1H), 8.29 (m, 1H), 8.08-8.24 (m, 2H), 7.40 (m, 2H), 6.75 (s, 1H), 5.25 (s, 1H), 3.81 (s, 3H), 2.82 (m, 2H), 2.74 (m, 2H), 2.0 (m, 3H), 1.82 (s, 1H), 1.37 (s, 9H), 1.25 (s, 3H). Example 109 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-methyl-1H- pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H-

benzocyclohepten-5-yl}-amide  embedded image Method B 2-[2-(1-Methyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride LC/MS: (M + H) 511. HPLC: 3.54 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.5 (s, 1H), 10 (s, 1H), 8.08-8.24 (m, 3H), 7.90 (s, 1H), 7.40 (m, 2H), 6.97 (s, 1H), 5.17 (s, 1H), 4.0 (s, 3H), 3.0 (s, 2H), 2.0 (m, 5H), 1.23 (s, 10H). Example 110 3-tert-Butyl- [1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide  embedded image Method A 2-[2-(1,5-Dimethyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride LC/MS: (M + H) 525. HPLC: 3.65 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.44 (s, 1H), 9.97 (s, 1H), 8.08-8.24 (m, 3H), 7.50 (m, 2H), 6.83 (s, 1H), 5.36 (s, 1H), 3.90 (s, 3H), 3.0 (s, 2H), 2.41 (s, 3H), 2.0 (m, 5H), 1.23 (s, 10H). Example 111 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {6-[2-(1,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  embedded image Method C 7-Bromo-2-(1,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridine and 3-tert-butyl-N-[6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5-carboxamide LC/MS: (M + H) 511. HPLC: 3.52 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.87 (m, 1H), 8.29 (m, 1H), 8.04 (m, 2H), 7.49 (m, 2H), 6.75 (s, 1H), 5.31 (s, 1H), 3.82 (s, 3H), 2.94 (m, 2H), 2.38 (s, 3H), 2.0 (m, 3H), 1.82 (s, 1H), 1.20 (s, 9H) Example 112 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {6-[2-(1-isopropyl-5-methyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  embedded image Method C 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid [6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-naphthalen-1-yl]-amide and 7-Bromo-2-(1-isopropyl-5-methyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridine LC/MS: (M + H) 539. HPLC: 3.89 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.87 (m, 1H), 8.20 (m, 1H), 8.04 (m, 2H), 7.49 (m, 2H), 6.63 (s, 1H), 5.33 (s, 1H), 4.55 (m, 1H), 2.94 (m, 2H), 2.38 (s, 3H), 2.0 (m, 3H), 1.82 (s, 1H), 1.25 (m, 6H), 1.20 (s, 9H) Example 113 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(S)-6-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  embedded image SFC separation LC/MS: (M + H) 539. HPLC: 4.18 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.75 (d, 1H), 8.43 (m, 2H), 8.27 (m, 1H), 8.12 (m, 1H), 7.49 (m, 2H), 5.25 (s, 1H), 4.5 (m, 1H), 2.94 (m, 2H), 2.65 (s, 3H), 2.02 (m, 3H), 1.79 (m, 1H), 1.51 (m, 6H), 1.33 (s, 9H) Example 114 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-6-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  embedded image SFC separation LC/MS: (M + H) 539. HPLC: 4.18 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.75 (d, 1H), 8.43 (m, 2H), 8.27 (m, 1H), 8.12 (m, 1H), 7.49 (m, 2H), 5.25 (s, 1H), 4.5 (m, 1H), 2.94 (m, 2H), 2.65 (s, 3H), 2.02 (m, 3H), 1.79 (m, 1H), 1.51 (m, 6H), 1.33 (s, 9H) Example 115 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid ((S)-6-{2-[3-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  embedded image SFC separation LC/MS: (M + H) 581. HPLC: 3.55 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.07 (s, 1H), 9.75 (m, 1H), 8.31-8.05 (m, 4H), 7.50 (m, 2H), 5.27 (m, 1H), 4.46 (m, 1H), 4.0 (m, 2H), 3.5 (t, 2H), 2.90 (m, 2H), 2.52 (s, 3H), 2.48 (s, 3H), 2.0 (m, 8H), 1.36 (s, 9H). Example 116 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide  embedded image Method B 2-[2-(1,3,5-Trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride LC/MS: (M + H) 539. HPLC: 3.07 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.89 (m, 1H), 8.0-8.25 (m, 3H), 7.50 (m, 1H), 7.42 (m, 1H), 5.30 (s, 1H), 3.73 (s, 3H), 3.0 (m, 2H), 2.5 (m, 6H), 2.0 (m, 5H), 1.19 (s, 10H). Example 117 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-5-methyl-1H-pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide  embedded image Method B 2-[2-(1-Isopropyl-5-methyl-1H-

pyrazol-3-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylaminehydrochloride LC/MS: (M + H) 551. HPLC: 3.74 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.5 (m, 1H), 8.0-8.45 (m, 3H), 7.5 (m, 2H), 6.75 (s, 1H), 5.26 (m, 1H), 4.61 (m, 1H), 3.0 (m, 2H), 2.29 (s, 3H), 2.0 (m, 5H), 1.5 (m, 12H), 1.24 (s, 2H). Example 118 5-tert-Butyl- [1,3,4]oxadiazole-2-carboxylic acid {2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide  Method B 2-[2-(1-Isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride LC/MS: (M + H) 553. HPLC: 3.74 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.05 (s, 1H), 9.88 (d, J = 8.2 Hz, 1H), 8.40 (s, 1H), 8.32-8.13 (m, 3H), 7.50 (d, J = 5.3 Hz, 1H), 7.39 (d, J = 8.2 Hz, 1H), 5.29 (t, J = 8.5 Hz, 1H), 4.51 (p, J = 6.6 Hz, 1H), 2.99 (d, J = 8.9 Hz, 2H), 2.62 (s, 3H), 1.98 (d, J = 10.2 Hz, 3H), 1.83 (t, J = 9.5 Hz, 2H), 1.47 (d, J = 6.7 Hz, 6H), 1.43 (s, 10H). Example 119 3-tert-Butyl- [1,2,4]oxadiazole-5-carboxylic acid {(S)-6-[2-(1-isopropyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl}-amide  SFC LC/MS: (M + H) 525. HPLC: 3.62 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.75 (m, 1H), 8.49-8.05 (m, 4H), 7.44 (m, 2H), 5.27 (m, 1H), 4.58 (m, 1H), 2.90 (m, 2H), 2.0 (m, 4H), 1.50 (m, 6H), 1.36 (s, 9H). Example 120 5-tert-Butyl- [1,3,4]oxadiazole-2-carboxylic acid (2-{2-[5-methyl-1-(tetrahydropyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl)-amide  Method B 2-[2-[5-Methyl-1-(tetrahydropyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-ylamine hydrochloride LC/MS: (M + H) 595. HPLC: 3.17 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.20 (br.s, 1H), 9.90 (m, 1H), 8.27 (m, 2H), 8.14 (m, 2H), 7.5 (s, 1H), 7.36 (m, 1H), 5.33 (s, 1H), 4.51 (s, 1H), 4.0 (2H), 3.50 (t, 2H), 3.0 (m, 2H), 2.82 (s, 3H), 2.0 (m, 5H), 1.88 (m, 4H), 1.25 (m, 10H). Example 121 5-tert-Butyl- [1,3,4]oxadiazole-2-carboxylic acid {(R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide  SFC LC/MS: (M + H) 539. HPLC: 3.07 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.89 (m, 1H), 8.0-8.25 (m, 3H), 7.50 (m, 1H), 7.42 (m, 1H), 5.30 (s, 1H), 3.73 (s, 3H), 3.0 (m, 2H), 2.5 (m, 6H), 2.0 (m, 5H), 1.19 (s, 10H). Example 122 5-tert-Butyl- [1,3,4]oxadiazole-2-carboxylic acid {(S)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide  SFC LC/MS: (M + H) 539. HPLC: 3.07 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.89 (m, 1H), 8.0-8.25 (m, 3H), 7.50 (m, 1H), 7.42 (m, 1H), 5.30 (s, 1H), 3.73 (s, 3H), 3.0 (m, 2H), 2.5 (m, 6H), 2.0 (m, 5H), 1.19 (s, 10H). Example 123 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide 0  LC/MS: (M + H) 537. HPLC: 3.33 min. ¹H NMR (400 MHz, DMSO-d₆) δ 12.83 (m, 1H), 9.95 (s, 1H), 8.31 (m, 2H), 7.50-7.24 (m, 2H), 5.34 (s, 1H), 3.74 (s, 3H), 2.99 (s, 2H), 2.0 (m, 5H), 1.52 (s, 3H), 1.31-1.21 (m, 3H), 1.02 (m, 2H). Several proton peaks Example 124 3-tert-Butyl- [1,2,4]oxadiazole-5-carboxylic acid {(R)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl}-amide  SFC LC/MS: (M + H) 539. HPLC: 4.22 min. ¹H NMR (400 MHz, DMSO-d₆) δ 8.24 (m, 1H), 8.05 (m, 2H), 7.50 (m, 2H), 5.44 (m, 1H), 4.77 (m, 1H), 3.0 (m, 2H), 2.74 (m, 3H), 2.0 (m, 4H), 1.52 (m, 6H), 1.46 (s, 9H). Example 125 3-tert-Butyl- [1,2,4]oxadiazole-5-carboxylic acid {(S)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl}-amide  SFC LC/MS: (M + H) 539. HPLC: 4.22 min. ¹H NMR (400 MHz, DMSO-d₆) δ 8.24 (m, 1H), 8.05 (m, 2H), 7.50 (m, 2H), 5.44 (m, 1H), 4.77 (m, 1H), 3.0 (m, 2H), 2.74 (m, 3H), 2.0 (m, 4H), 1.52 (m, 6H), 1.46 (s, 9H). Example 126 3-tert-Butyl- [1,2,4]oxadiazole-5-carboxylic acid {(S)-6-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl}-amide  SFC LC/MS: (M + H) 497. HPLC: 3.28 min. ¹H NMR (400 MHz,

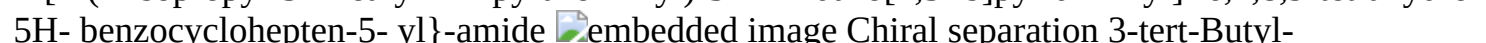
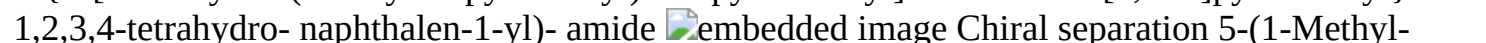
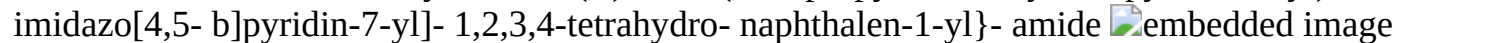
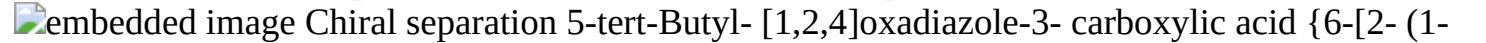
DMSO-d6) δ 13.30 (br.s, 1H), 9.87 (d, 1H), 8.50 (s, 1H), 8.27 (m, 1H), 8.12 (m, 3H), 7.49 (m, 2H), 5.33 (s, 1H), 3.98 (s, 3H), 2.94 (m, 2H), 2.02 (m, 3H), 1.79 (m, 1H), 1.40 (s, 9H) Example 127 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid (6-[2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Method C 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid [6- (4,4,5,5-tetramethyl- [1,3,2]dioxaborolan-2- yl)-1,2,3,4-tetrahydro- naphthalen-1-yl]-amide and 7-Bromo-2-[5- methyl-1-(tetrahydro- pyran-4-yl)-1H- pyrazol-4-yl]-3H- imidazo[4,5-b]pyridine LC/MS: (M + H) 581. HPLC: 3.53 min..sup.1H NMR (400 MHz, DMSO-d6) δ 1H NMR (400 MHz, DMSO-d6) δ 13.30 (s, 1H), 9.83 (s, 1H), 8.31-8.05 (m, 4H), 7.50 (m, 2H), 5.39 (s, 1H), 4.51 (m, 1H), 4.0 (m, 2H), 3.50 (t, 2H), 2.90 (m, 5H), 2.0 (m, 5H), 1.85 (m, 3H), 1.43 (s, 9H). Example 128 3-(1-Methyl-cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1,3,5-trimethyl-1H- pyrazol-4-yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(1,3,5-Trimethyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin-7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine hydrochloride LC/MS: (M + H) 537. HPLC: 3.33 min. 1H NMR (400 MHz, DMSO-d6) δ 1H NMR (400 MHz, DMSO-d6) δ 12. 83 (m, 1H), 9.95 (s, 1H), 8.31 (m, 2H), 7.50-7.24 (m, 2H), 5.34 (s, 1H), 3.74 (s, 3H), 2.99 (s, 2H), 2.0 (m, 5H), 1.52 (s, 3H), 1.31-1.21 (m, 3H), 1.02 (m, 2H). Several proton peaks overlap with solvent peak. Example 129 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1-methyl-1H-pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method B 6-[2-(1-Methyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-1,2,3,4- tetrahydro-naphthalen- 1-ylamine hydrochloride LC/MS: (M + H) 497. HPLC: 3.28 min. 1H NMR (400 MHz, DMSO-d6) δ 13.30 (br.s, 1H), 9.87 (d, 1H), 8.50 (s, 1H), 8.27 (m, 1H), 8.12 (m, 3H), 7.49 (m, 2H), 5.33 (s, 1H), 3.98 (s, 3H), 2.94 (m, 2H), 2.02 (m, 3H), 1.79 (m, 1H), 1.40 (s, 9H) Example 130 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1-isopropyl-3-methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 7-Bromo-2-(1- isopropyl-3-methyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridine and 3-tert-butyl-N-[6- (4,4,5,5-tetramethyl- 1,3,2-dioxaborolan-2- yl)-1,2,3,4- tetrahydronaphthalen-1- yl]-1,2,4-oxadiazole-5- carboxamide LC/MS: (M + H) 539. HPLC: 4.18 min. 1H NMR (400 MHz, DMSO-d6) δ 9.75 (d, 1H), 8.43 (m, 2H), 8.27 (m, 1H), 8.12 (m, 1H), 7.49 (m, 2H), 5.25 (s, 1H), 4.5 (m, 1H), 2.94 (m, 2H), 2.65 (s, 3H), 2.02 (m, 3H), 1.79 (m, 1H), 1.51 (m, 6H), 1.33 (s, 9H) Example 131 5-tert-Butyl- [1,3,4]oxadiazole-2- carboxylic acid {2-[2- (5-ethyl-1-methyl- 1H-pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(5-Ethyl-1-methyl- 1H-pyrazol-3-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylaminehydrochloride LC/MS: (M + H) 539. HPLC: 4.18 (97%) min. 1H NMR (400 MHz, DMSO-d6) δ 9.79 (d, 1H), 8.30 (m, 2H), 8.23 (m, 1H), 8.02 (s, 1H), 7.49 (m, 2H), 6.75 (s, 1H), 5.25 (s, 1H), 3.84 (s, 3H), 2.99 (m, 2H), 2.45 (m, 2H), 2.02 (m, 3H), 1.79 (m, 2H), 1.47 (m, 10H), 1.25 (m, 3H), 0.99 (d, 2H). Example 132 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-3- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(1-Isopropyl-3- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylaminehydrochloride LC/MS: (M + H) 539. HPLC: 4.18 min. 1H NMR (400 MHz, DMSO-d6) δ 13.07 (s, 1H), 9.98 (m, 1H), 8.10-8.43 (m, 4H), 7.50 (m, 2H), 5.32 (m, 1H), 4.51 (m, 1H), 3.0 (m, 2H), 2.0 (m, 3H), 1.88 (m, 2H), 1.5 (m, 6H), 1.48 (m, 10H). Example 133 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (5-ethyl-1-methyl- 1H-pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(5-Ethyl-1-methyl- 1H-pyrazol-3-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylaminehydrochloride LC/MS: (M + H) 539. HPLC: 3.86 min. 1H NMR (400 MHz, DMSO-d6) δ 13.37 (bs, 1H), 9.98 (m, 1H), 8.0-8.4 (m, 3H), 7.50 (m, 2H), 6.75 (m, 1H), 5.30 (m, 1H), 3.86 (s, 3H), 3.0 (m, 2H), 2.69 (m, 2H), 2.0 (m, 3H), 1.88 (m,

2H), 1.48 (m, 10H), 1.16 (m, 3H). Example 134 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-3- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(1- Isopropyl-3- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine hydrochloride LC/MS: (M + H) 551. HPLC: 3.52 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.00 (bs, 1H), 9.50 (m, 1H), 8.0-8.4 (m, 4H), 7.50 (m, 2H), 5.32 (m, 1H), 4.50 (m, 1H), 3.0 (m, 2H), 2.53 (s, 3H), 2.0 (m, 3H), 1.50 (s, 3H), 1.48 (m, 9H) Example 135 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-ethyl-3,5- dimethyl-1H- pyrazol- 4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}- amide  Method B 2-[2-(1-Isopropyl-3- methyl-1H-pyrazol-4- yl)-3H- imidazo[4,5- b]pyridin-7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylaminehydrochloride LC/MS: (M + H) 551. HPLC: 3.67 min. ¹H NMR (400 MHz, DMSO-d₆) δ 9.85 (m, 1H), 9.50 (m, 1H), 8.0-8.4 (m, 4H), 7.50 (m, 2H), 5.25 (m, 1H), 4.50 (m, 1H), 3.0 (m, 2H), 2.56 (s, 3H), 2.0 (m, 5H), 1.50 (s, 3H), 1.48 (m, 7H), 1.25 (m, 2H), 1.0 (m, 2H) Example 136 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(1-Isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylaminehydrochloride LC/MS: (M + H) 553. HPLC: 4.35 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.10 (br.s, 1H), 10.01 (m, 1H), 8.0- 8.4 (m, 4H), 7.50 (m, 2H), 5.31 (m, 1H), 4.55 (m, 1H), 3.0 (m, 2H), 2.77 (s, 3H), 2.0 (m, 5H), 3H), 1.48 (m, 16H) Example 137 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid ((R)- 6-{2-[5- methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4- tetrahydro- naphthalen-1-yl)- amide  SFC LC/MS: (M + H) 581. HPLC: 3.53 min. ¹H NMR (400 MHz, DMSO-d₆) δ ¹H NMR (400 MHz, DMSO-d₆) δ 13.30 (s, 1H), 9.83 (s, 1H), 8.31-8.05 (m, 4H), 7.50 (m, 2H), 5.39 (s, 1H), 4.51 (m, 1H), 4.0 (m, 2H), 3.50 (t, 2H), 2.90 (m, 5H), 2.0 (m, 5H), 1.85 (m, 3H), 1.43 (s, 9H). Example 138 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid ((S)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  SFC LC/MS: (M + H) 581. HPLC: 3.53 min. ¹H NMR (400 MHz, DMSO-d₆) δ ¹H NMR (400 MHz, DMSO-d₆) δ 13.30 (s, 1H), 9.83 (s, 1H), 8.31-8.05 (m, 4H), 7.50 (m, 2H), 5.39 (s, 1H), 4.51 (m, 1H), 4.0 (m, 2H), 3.50 (t, 2H), 2.90 (m, 5H), 2.0 (m, 5H), 1.85 (m, 3H), 1.43 (s, 9H). Example 139 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1-isopropyl-3,5- dimethyl-1H-pyrazol- 4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 7-Bromo-2-(1- isopropyl-3,5-dimethyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridine and 3-tert- Butyl[1,2,4]oxadiazole- 5-carboxylic acid [6- (4,4,5,5- tetramethyl- [1,3,2]dioxaborolan-2- yl)-1,2,3,4-tetrahydro- naphthalen-1-yl]-amide LC/MS: (M + H) 553. HPLC: 3.69 min..sup.¹H NMR (400 MHz, DMSO-d₆) δ ¹H NMR (400 MHz, DMSO-d₆) δ 9.75 (s, 1H), 8.31-8.05 (m, 3H), 7.50 (m, 2H), 5.25 (s, 1H), 4.51 (m, 1H), 2.90 (m, 5H), 2.52 (s, 3H), 2.48 (s, 3H), 2.0 (m, 4H), 1.32 (m, 6H), 1.29 (s, 9H). Example 140 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid (6-{2- [3,5-dimethyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol- 4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Method C 7-Bromo-2-[3,5- dimethyl-1-(tetrahydro- pyran-4-yl)-1H- pyrazol-4- yl]-3H- imidazo[4,5-b]pyridine and 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid [6- (4,4,5,5- tetramethyl- [1,3,2]dioxaborolan-2- yl)-1,2,3,4-tetrahydro- naphthalen-1-yl]-amide LC/MS: (M + H) 595. HPLC: 3.92 min. ¹H NMR (400 MHz, DMSO-d₆) δ ¹H NMR (400 MHz, DMSO-d₆) δ 12.77 (brs, 1H), 9.70 (m, 1H), 8.31-8.05 (m, 3H), 7.50 (m, 2H), 5.25 (s, 1H), 4.51 (m, 1H), 4.0 (m, 2H), 3.5 (t, 2H), 2.90 (m, 5H), 2.52 (s, 3H), 2.48 (s, 3H), 2.0 (m, 5H), 1.81 (m, 3H), 1.29 (s, 9H). Example 141 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {(S)- 2-[2-(1,3,5-trimethyl- 1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}- amide  SPC LC/MS: (M + H) 539. HPLC: 3.38 min. ¹H NMR (400 MHz,

DMSO-d6) δ 1H NMR (400 MHz (400 MHz, DMSO-d6) δ 12.77 (brs, 1H), 9.70 (m, 1H), 8.31-8.05 (m, 3H), 7.50 (m, 2H), 5.25 (s, 1H), 4.51 (m, 1H), 4.0 (m, 2H), 3.5 (t, 2H), 2.90 (m, 5H), 2.52 (s, 3H), 2.48 (s, 3H), 2.0 (m, 5H), 1.81 (m, 3H), 1.29 (s, 9H). Example 142 3-tert-Butyl-[1,2,4]oxadiazole-5- carboxylic acid {(R)- 2-[2-(1,3,5-trimethyl- 1H-pyrazol-4-yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  SFC LC/MS: (M + H) 539. HPLC: 3.38 min. 1H NMR (400 MHz, DMSO-d6) δ 1H NMR (400 MHz (400 MHz, DMSO-d6) δ 12.77 (brs, 1H), 9.70 (m, 1H), 8.31-8.05 (m, 3H), 7.50 (m, 2H), 5.25 (s, 1H), 4.51 (m, 1H), 4.0 (m, 2H), 3.5 (t, 2H), 2.90 (m, 5H), 2.52 (s, 3H), 2.48 (s, 3H), 2.0 (m, 5H), 1.81 (m, 3H), 1.29 (s, 9H). Example 143 3-(1-Methyl- cyclopropyl)-[1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1,3,5-trimethyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(1,3,5-Trimethyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9-tetrahydro-5H- benzocyclohepten-5- ylamine hydrochloride LC/MS: (M + H) 537. HPLC: 3.33 min. 1H NMR (400 MHz, DMSO-d6) δ 1H NMR (400 MHz, DMSO-d6) δ 12.75 (m, 1H), 9.89 (m, 1H), 8.35-8.05 (m, 3H), 7.44 (m, 2H), 5.27 (m, 1H), 3.75 (s, 3H), 2.99 (m, 2H), 2.5 (m, 6H), 2.0 (m, 5H), 1.50 (s, 3H), 1.30 (m, 3H), 1.0 (m, 2H). Example 144 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-3,5- dimethyl-1H-pyrazol- 4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Method B 2-[2-(1-Isopropyl-3,5- dimethyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- ylamine hydrochloride LC/MS: (M + H) 567. HPLC: 4.3 min. 1H NMR (400 MHz, DMSO-d6) δ 1H NMR (400 MHz, DMSO-d6) δ 12.69 (s, 1H), 10.0 (m, 1H), 8.31-8.05 (m, 3H), 7.50 (m, 2H), 5.25 (s, 1H), 4.50 (m, 1H), 3.0 (m, 2H), 2.52 (s, 3H), 2.48 (s, 3H), 2.0 (m, 5H), 1.47 (s, 16H). Example 145 3-tert-Butoxy- azetidine-1- carboxylic acid {6-[2- (1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}-amide  Method C 4-{7-bromo-3H- imidazo[4,5-b]pyridin- 2-yl}-5-methyl-1-(propan-2-yl)-1H- pyrazole LCMS: 542 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.16 (s, 1H), 8.24 (d, J = 6.8 Hz, 2H), 8.15 (s, 1H), 8.09 (d, J = 8.1 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.1 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.86 (s, 1H), 4.69-4.60 (m, 1H), 4.45 (d, J = 7.4 Hz, 1H), 4.02 (s, 2H), 3.59 (s, 2H), 2.82 (s, 5H), 1.94 (s, 3H), 1.74 (s, 2H), 1.42 (d, J = 6.6 Hz, 6H), 1.12 (s, 9H). Example 146 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 7-Bromo-2-(1- isopropyl-5-methyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridine LCMS: 537 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.18 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.31-8.21 (m, 2H), 8.16 (s, 1H), 8.14-8.10 (m, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 5.25 (s, 1H), 4.64 (p, J = 6.5 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 2H), 2.07-1.89 (m, 3H), 1.82 (s, 1H), 1.48 (s, 3H), 1.42 (d, J = 6.5 Hz, 5H), 1.18 (q, J = 3.7 Hz, 2H), 0.97 (q, J = 4.0 Hz, 2H). Example 147 3-tert-Butoxy- azetidine-1- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Method C 7-Bromo-2-[5-methyl-1-(tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridine LCMS: 584 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.19 (s, 1H), 8.27-8.22 (m, 2H), 8.17 (s, 1H), 8.08 (dd, J = 8.2, 1.9 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.90-4.80 (m, 1H), 4.59-4.40 (m, 2H), 4.00 (td, J = 12.1, 10.4, 6.2 Hz, 3H), 3.59 (dt, J = 8.7, 4.5 Hz, 2H), 3.51 (dd, J = 12.9, 10.8 Hz, 2H), 2.84 (s, 4H), 2.14-2.01 (m, 2H), 1.94 (q, J = 5.3 Hz, 2H), 1.84 (d, J = 12.8 Hz, 2H), 1.74 (q, J = 11.4, 10.2 Hz, 2H), 1.12 (s, 7H), 1.08-1.01 (m, 1H). Example 148 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Method C 7-Bromo-2-[5-methyl- 1-(tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridine LCMS: 579 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.28 (d, J = 1.9 Hz, 1H), 8.25 (d, J = 5.2

Hz, 1H), 8.17 (s, 1H), 8.10 (dd, J = 8.1, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.35 (d, J = 8.2 Hz, 1H), 5.24 (t, J = 7.0 Hz, 1H), 4.52 (td, J = 11.1, 5.4 Hz, 1H), 4.08-3.93 (m, 2H), 3.51 (dd, J = 12.8, 10.7 Hz, 2H), 2.84 (s, 4H), 2.15-1.89 (m, 5H), 1.84 (d, J = 12.7 Hz, 3H), 1.48 (s, 3H), 1.21-1.13 (m, 2H), 1.01-0.91 (m, 2H). Example 149 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid {6-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 4-{7-bromo-3H- imidazo[4,5-b]pyridin- 2- yl}-5-methyl-1- (propan-2-yl)-1H- pyrazole LCMS: 539 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.18 (s, 1H), 9.31 (d, J = 8.8 Hz, 1H), 8.28 (s, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.16 (s, 1H), 8.12 (dd, J = 8.1, 1.9 Hz, 1H), 7.48 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.29 (d, J = 5.7 Hz, 1H), 4.64 (p, J = 6.6 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 3H), 2.07-1.89 (m, 3H), 1.83 (s, 1H), 1.42 (d, J = 6.9 Hz, 15H). Example 150 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid (6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4- tetrahydro- naphthalen-1-yl)- amide  Method C 7-Bromo-2-[5-methyl- 1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridine LCMS: 581 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.30 (d, J = 8.9 Hz, 1H), 8.28 (t, J = 1.8 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.11 (dd, J = 8.3, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.28 (d, J = 4.6 Hz, 1H), 4.53 (td, J = 11.4, 5.7 Hz, 1H), 4.07-3.93 (m, 3H), 3.51 (dd, J = 12.9, 10.7 Hz, 2H), 2.84 (s, 4H), 2.07 (qd, J = 11.9, 11.5, 4.2 Hz, 4H), 1.88-1.76 (m, 3H), 1.42 (s, 8H), 1.16 (q, J = 7.3 Hz, 1H). Example 151 5-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4- yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Method C 7-Bromo-2-[5-methyl- 1-(tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridine LCMS: 579 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.21 (s, 1H), 9.25 (d, J = 8.8 Hz, 1H), 8.27 (s, 2H), 8.17 (s, 1H), 8.11 (d, J = 8.1 Hz, 1H), 7.48 (s, 1H), 7.33 (d, J = 8.5 Hz, 1H), 5.27 (s, 1H), 4.53 (s, 1H), 3.99 (d, J = 11.7 Hz, 2H), 3.50 (d, J = 12.6 Hz, 2H), 3.30 (s, 3H, overlaps with H2O peak), 2.85 (s, 4H), 2.15-1.73 (m, 7H), 1.54 (s, 2H), 1.38 (s, 2H), 1.16 (s, 2H). Example 152 3-tert-Butoxy- azetidine-1- carboxylic acid {(S)- 6-[2-(1-methyl-1H- pyrazol-4-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Chiral separation 3-tert-Butoxy- azetidine-1-carboxylic acid {6-[2-(1- isopropyl- 5-methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]-1,2,3,4- tetrahydro- naphthalen- 1-yl}-amide LCMS: 542 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.16 (s, 1H), 8.24 (d, J = 6.8 Hz, 2H), 8.15 (s, 1H), 8.09 (d, J = 8.1 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.1 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.86 (s, 1H), 4.69-4.60 (m, 1H), 4.45 (d, J = 7.4 Hz, 1H), 4.02 (s, 2H), 3.59 (s, 2H), 2.82 (s, 5H), 1.94 (s, 3H), 1.74 (s, 2H), 1.42 (d, J = 6.6 Hz, 6H), 1.12 (s, 9H). Example 153 5-tert-Butyl- [1,3,4]oxadiazole-2- carboxylic acid {(R)- 2-[2-(1,5- dimethyl- 1H-pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide 0  Chiral separation 5-tert-Butyl- [1,3,4]oxadiazole-2- carboxylic acid {2-[2- (1,5-dimethyl-1H- pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- yl}-amide LC/MS: (M + H) 525. HPLC: 3.65 min. 1H NMR (400 MHz, DMSO-d6) δ 13.40 (brs, 1H), 10.0 (s, 1H), 8.24-8.05 (m, 3H), 7.50 (m, 2H), 6.75 (s, 1H), 5.26 (m, 1H), 3.85 (s, 3H), 3.0 (m, 2H), 2.29 (s, 3H), 2.0 (m, 5H), 1.46 (m, 10H). Example 154 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid ((R)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Chiral separation 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 579 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.28 (d, J = 1.9 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.10 (dd, J = 8.1, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.35 (d, J = 8.2 Hz, 1H), 5.24 (t, J = 7.0 Hz, 1H), 4.52 (td, J = 11.1, 5.4 Hz, 1H), 4.08-3.93 (m, 2H), 3.51 (dd, J = 12.8, 10.7 Hz, 2H), 2.84 (s, 4H), 2.15-1.89 (m, 5H), 1.84

(d, J = 12.7 Hz, 3H), 1.48 (s, 3H), 1.21-1.13 (m, 2H), 1.01-0.91 (m, 2H). Example 155 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid ((S)- 6-{2-[5-methyl-1- (tetrahydro- pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}-4-yl)- 3H-imidazo[4,5- 1,2,3,4- tetrahydro- naphthalen-1-yl)- amide  Chiral separation 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 579 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.28 (d, J = 1.9 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.10 (dd, J = 8.1, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.35 (d, J = 8.2 Hz, 1H), 5.24 (t, J = 7.0 Hz, 1H), 4.52 (td, J = 11.1, 5.4 Hz, 1H), 4.08-3.93 (m, 2H), 3.51 (dd, J = 12.8, 10.7 Hz, 2H), 2.84 (s, 4H), 2.15-1.89 (m, 5H), 1.84 (d, J = 12.7 Hz, 3H), 1.48 (s, 3H), 1.21-1.13 (m, 2H), 1.01-0.91 (m, 2H). Example 156 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid ((R)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H- imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Chiral separation 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro- naphthalen- 1-yl)-amide LCMS: 581 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.30 (d, J = 8.9 Hz, 1H), 8.28 (t, J = 1.8 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.11 (dd, J = 8.3, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.28 (d, J = 4.6 Hz, 1H), 4.53 (td, J = 11.4, 5.7 Hz, 1H), 4.07-3.93 (m, 3H), 3.51 (dd, J = 12.9, 10.7 Hz, 2H), 2.84 (s, 4H), 2.07 (qd, J = 11.9, 11.5, 4.2 Hz, 4H), 1.88-1.76 (m, 3H), 1.42 (s, 8H), 1.16 (q, J = 7.3 Hz, 1H). Example 157 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid ((S)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Chiral separation 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 581 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.30 (d, J = 8.9 Hz, 1H), 8.28 (t, J = 1.8 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.11 (dd, J = 8.3, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.28 (d, J = 4.6 Hz, 1H), 4.53 (td, J = 11.4, 5.7 Hz, 1H), 4.07-3.93 (m, 3H), 3.51 (dd, J = 12.9, 10.7 Hz, 2H), 2.84 (s, 4H), 2.07 (qd, J = 11.9, 11.5, 4.2 Hz, 4H), 1.88-1.76 (m, 3H), 1.42 (s, 8H), 1.16 (q, J = 7.3 Hz, 1H). Example 158 3-tert-Butoxy- azetidine-1- carboxylic acid ((R)- 6-{2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- tetrahydro- naphthalen-1- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Chiral separation 3-tert-Butoxy- azetidine-1-carboxylic acid (6-{2-[5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 584 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.19 (s, 1H), 8.27-8.22 (m, 2H), 8.17 (s, 1H), 8.08 (dd, J = 8.2, 1.9 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.90-4.80 (m, 1H), 4.59-4.40 (m, 2H), 4.00 (td, J = 12.1, 10.4, 6.2 Hz, 3H), 3.59 (dt, J = 8.7, 4.5 Hz, 2H), 3.51 (dd, J = 12.9, 10.8 Hz, 2H), 2.84 (s, 4H), 2.14-2.01 (m, 2H), 1.94 (q, J = 5.3 Hz, 2H), 1.84 (d, J = 12.8 Hz, 2H), 1.74 (q, J = 11.4, 10.2 Hz, 2H), 1.12 (s, 7H), 1.08-1.01 (m, 1H). Example 159 3-tert-Butoxy- azetidine-1- carboxylic acid ((S)- 6-{2-[5-methyl-1- (tetrahydro- pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1- yl)- amide  Chiral separation 3-tert-Butoxy- azetidine-1-carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 584 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.19 (s, 1H), 8.27-8.22 (m, 2H), 8.17 (s, 1H), 8.08 (dd, J = 8.2, 1.9 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.90-4.80 (m, 1H), 4.59-4.40 (m, 2H), 4.00 (td, J = 12.1, 10.4, 6.2 Hz, 3H), 3.59 (dt, J = 8.7, 4.5 Hz, 2H), 3.51 (dd, J = 12.9, 10.8 Hz, 2H), 2.84 (s, 4H), 2.14-2.01 (m, 2H), 1.94 (q, J = 5.3 Hz, 2H), 1.84 (d, J = 12.8 Hz, 2H), 1.74 (q, J = 11.4, 10.2 Hz, 2H), 1.12 (s, 7H), 1.08-1.01 (m, 1H). Example 160 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid ((R)- 2-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-

yl]- 6,7,8,9-tetrahydro-5H- benzocyclohepten-5- yl}-amide  Chiral separation 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-5-methyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- yl}-amide LC/MS: (M + H) 553. HPLC: 4.35 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.10 (br.s, 1H), 10.01 (m, 1H), 8.0-8.4 (m, 4H), 7.50 (m, 2H), 5.31 (m, 1H), 4.55 (m, 1H), 3.0 (m, 2H), 2.77 (s, 3H), 2.0 (m, 5H), 3H), 1.48 (m, 16H) Example 161 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {(R)- 2-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Chiral separation 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-5-methyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- yl}-amide LC/MS: (M + H) 553. HPLC: 4.35 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.10 (br.s, 1H), 10.01 (m, 1H), 8.0-8.4 (m, 4H), 7.50 (m, 2H), 5.31 (m, 1H), 4.55 (m, 1H), 3.0 (m, 2H), 2.77 (s, 3H), 2.0 (m, 5H), 3H), 1.48 (m, 16H) Example 162 5-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-3- carboxylic acid ((R)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Chiral separation 5-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 579 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.21 (s, 1H), 9.25 (d, J = 8.8 Hz, 1H), 8.27 (s, 2H), 8.17 (s, 1H), 8.11 (d, J = 8.1 Hz, 1H), 7.48 (s, 1H), 7.33 (d, J = 8.5 Hz, 1H), 5.27 (s, 1H), 4.53 (s, 1H), 3.99 (d, J = 11.7 Hz, 2H), 3.50 (d, J = 12.6 Hz, 2H), 3.30 (s, 3H, overlaps with H₂O peak), 2.85 (s, 4H), 2.15-1.73 (m, 7H), 1.54 (s, 2H), 1.38 (s, 2H), 1.16 (s, 2H). Example 163 5-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-3- carboxylic acid ((S)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Chiral separation 5-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 579 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.21 (s, 1H), 9.25 (d, J = 8.8 Hz, 1H), 8.27 (s, 2H), 8.17 (s, 1H), 8.11 (d, J = 8.1 Hz, 1H), 7.48 (s, 1H), 7.33 (d, J = 8.5 Hz, 1H), 5.27 (s, 1H), 4.53 (s, 1H), 3.99 (d, J = 11.7 Hz, 2H), 3.50 (d, J = 12.6 Hz, 2H), 3.30 (s, 3H, overlaps with H₂O peak), 2.85 (s, 4H), 2.15-1.73 (m, 7H), 1.54 (s, 2H), 1.38 (s, 2H), 1.16 (s, 2H). Example 164 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid {(R)- 6-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Chiral separation 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid {6-[2- (1-isopropyl-5-methyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-1,2,3,4- tetrahydro-naphthalen- 1-yl}-amide LCMS: 539 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.31 (d, J = 8.8 Hz, 1H), 8.28 (s, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.16 (s, 1H), 8.12 (dd, J = 8.1, 1.9 Hz, 1H), 7.48 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.29 (d, J = 5.7 Hz, 1H), 4.64 (p, J = 6.6 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 3H), 2.07-1.89 (m, 3H), 1.83 (s, 1H), 1.42 (d, J = 6.9 Hz, 15H). Example 165 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid {(S)- 6-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Chiral separation 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid {6-[2- (1-isopropyl-5-methyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-1,2,3,4- tetrahydro- naphthalen- 1-yl}-amide LCMS: 539 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.31 (d, J = 8.8 Hz, 1H), 8.28 (s, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.16 (s, 1H), 8.12 (dd, J = 8.1, 1.9 Hz, 1H), 7.48 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.29 (d, J = 5.7 Hz, 1H), 4.64 (p, J = 6.6 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 3H), 2.07-1.89 (m, 3H), 1.83 (s, 1H), 1.42 (d, J = 6.9 Hz, 15H). Example 166 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {(R)- 6-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Chiral separation 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1-isopropyl-5-methyl- 1H-pyrazol-4-yl)-3H-

imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide LCMS: 537 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.18 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.31-8.21 (m, 2H), 8.16 (s, 1H), 8.14-8.10 (m, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 5.25 (s, 1H), 4.64 (p, J = 6.5 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 2H), 2.07-1.89 (m, 3H), 1.82 (s, 1H), 1.48 (s, 3H), 1.42 (d, J = 6.5 Hz, 5H), 1.18 (q, J = 3.7 Hz, 2H), 0.97 (q, J = 4.0 Hz, 2H). Example 167 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {(S)- 6-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Chiral separation 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1-isopropyl-5-methyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-1,2,3,4- tetrahydro-naphthalen- 1-yl}-amide LCMS: 537 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.18 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.31-8.21 (m, 2H), 8.16 (s, 1H), 8.14-8.10 (m, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 5.25 (s, 1H), 4.64 (p, J = 6.5 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 2H), 2.07-1.89 (m, 3H), 1.82 (s, 1H), 1.48 (s, 3H), 1.42 (d, J = 6.5 Hz, 5H), 1.18 (q, J = 3.7 Hz, 2H), 0.97 (q, J = 4.0 Hz, 2H). Example 168 3-tert-Butoxy- azetidine-1- carboxylic acid {6-[2- (1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 4-{7-bromo-3H- imidazo[4,5-b]pyridin- 2-yl}-5-methyl-1- (propan-2-yl)-1H- pyrazole LCMS: 542 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.16 (s, 1H), 8.24 (d, J = 6.8 Hz, 2H), 8.15 (s, 1H), 8.09 (d, J = 8.1 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.1 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.86 (s, 1H), 4.69-4.60 (m, 1H), 4.45 (d, J = 7.4 Hz, 1H), 4.02 (s, 2H), 3.59 (s, 2H), 2.82 (s, 5H), 1.94 (s, 3H), 1.74 (s, 2H), 1.42 (d, J = 6.6 Hz, 6H), 1.12 (s, 9H). Example 169 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid {6-[2- (1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 7-Bromo-2-(1- isopropyl-5-methyl-1H- pyrazol-4-yl)-3H- imidazo[4,5-b]pyridine LCMS: 537 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.18 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.31-8.21 (m, 2H), 8.16 (s, 1H), 8.14-8.10 (m, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 5.25 (s, 1H), 4.64 (p, J = 6.5 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 2H), 2.07-1.89 (m, 3H), 1.82 (s, 1H), 1.48 (s, 3H), 1.42 (d, J = 6.5 Hz, 5H), 1.18 (q, J = 3.7 Hz, 2H), 0.97 (q, J = 4.0 Hz, 2H). Example 170 3-tert-Butoxy- azetidine-1- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H- imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Method C 7-Bromo-2-[5-methyl- 1-(tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridine LCMS: 584 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.19 (s, 1H), 8.27-8.22 (m, 2H), 8.17 (s, 1H), 8.08 (dd, J = 8.2, 1.9 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.90-4.80 (m, 1H), 4.59-4.40 (m, 2H), 4.00 (td, J = 12.1, 10.4, 6.2 Hz, 3H), 3.59 (dt, J = 8.7, 4.5 Hz, 2H), 3.51 (dd, J = 12.9, 10.8 Hz, 2H), 2.84 (s, 4H), 2.14-2.01 (m, 2H), 1.94 (q, J = 5.3 Hz, 2H), 1.84 (d, J = 12.8 Hz, 2H), 1.74 (q, J = 11.4, 10.2 Hz, 2H), 1.12 (s, 7H), 1.08-1.01 (m, 1H). Example 171 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide  Method C 7-Bromo-2-[5- methyl- 1-(tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridine LCMS: 579 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.28 (d, J = 1.9 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.10 (dd, J = 8.1, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.35 (d, J = 8.2 Hz, 1H), 5.24 (t, J = 7.0 Hz, 1H), 4.52 (td, J = 11.1, 5.4 Hz, 1H), 4.08-3.93 (m, 2H), 3.51 (dd, J = 2.8, 10.7 Hz, 2H), 2.84 (s, 4H), 2.15-1.89 (m, 5H), 1.84 (d, J = 12.7 Hz, 3H), 1.48 (s, 3H), 1.21-1.13 (m, 2H), 1.01-0.91 (m, 2H). Example 172 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid {6-[2- (1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide  Method C 4-{7-bromo-3H- imidazo[4,5-b]pyridin- 2-yl}-5-methyl-1- (propan-2-yl)-1H- pyrazole LCMS: 539 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.18 (s, 1H), 9.31 (d, J = 8.8 Hz, 1H), 8.28 (s, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.16 (s, 1H), 8.12 (dd, J = 8.1, 1.9 Hz, 1H), 7.48 (d, J = 5.2 Hz, 1H), 7.34 (d, J

= 8.2 Hz, 1H), 5.29 (d, J = 5.7 Hz, 1H), 4.64 (p, J = 6.6 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 3H), 2.07-1.89 (m, 3H), 1.83 (s, 1H), 1.42 (d, J = 6.9 Hz, 15H). Example 173 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide 00 Method C 7-Bromo-2-[5-methyl- 1-(tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridine LCMS: 581 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.30 (d, J = 8.9 Hz, 1H), 8.28 (t, J = 1.8 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.11 (dd, J = 8.3, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.28 (d, J = 4.6 Hz, 1H), 4.53 (td, J = 11.4, 5.7 Hz, 1H), 4.07-3.93 (m, 3H), 3.51 (dd, J = 12.9, 10.7 Hz, 2H), 2.84 (s, 4H), 2.07 (qd, J = 11.9, 11.5, 4.2 Hz, 4H), 1.88-1.76 (m, 3H), 1.42 (s, 8H), 1.16 (q, J = 7.3 Hz, 1H). Example 174 5-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide 01 Method C 7-Bromo-2-[5-methyl- 1-(tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridine LCMS: 579 (M + H). .sup.1H NMR (400 MHz, DMSO- d6) δ 13.21 (s, 1H), 9.25 (d, J = 8.8 Hz, 1H), 8.27 (s, 2H), 8.17 (s, 1H), 8.11 (d, J = 8.1 Hz, 1H), 7.48 (s, 1H), 7.33 (d, J = 8.5 Hz, 1H), 5.27 (s, 1H), 4.53 (s, 1H), 3.99 (d, J = 11.7 Hz, 2H), 3.50 (d, J = 12.6 Hz, 2H), 3.30 (s, 3H, overlaps with H2O peak), 2.85 (s, 4H), 2.15-1.73 (m, 7H), 1.54 (s, 2H), 1.38 (s, 2H), 1.16 (s, 2H). Example 175 3-tert-Butoxy- azetidine-1- carboxylic acid {(S)- 6-[2- (1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 1,2,3,4-tetrahydro- naphthalen-1-yl}- amide 02 Chiral separation 3-tert-Butoxy- azetidine-1- carboxylic acid {6-[2-(1-isopropyl- 5-methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7- yl]-1,2,3,4- tetrahydro-naphthalen- 1-yl}-amide LCMS: 542 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.16 (s, 1H), 8.24 (d, J = 6.8 Hz, 2H), 8.15 (s, 1H), 8.09 (d, J = 8.1 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.1 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.86 (s, 1H), 4.69-4.60 (m, 1H), 4.45 (d, J = 7.4 Hz, 1H), 4.02 (s, 2H), 3.59 (s, 2H), 2.82 (s, 5H), 1.94 (s, 3H), 1.74 (s, 2H), 1.42 (d, J = 6.6 Hz, 6H), 1.12 (s, 9H). Example 176 5-tert-Butyl- [1,3,4]oxadiazole-2- carboxylic acid {(R)- 2-[2-(1,5-dimethyl- 1H-pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide 03 Chiral separation 5-tert-Butyl- [1,3,4]oxadiazole-2- carboxylic acid {2-[2- (1,5-dimethyl-1H- pyrazol-3-yl)-3H- imidazo[4,5- b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- yl}-amide LC/MS: (M + H) 525. HPLC: 3.65 min. 1H NMR (400 MHz, DMSO-d6) δ 13.40 (brs, 1H), 10.0 (s, 1H), 8.24-8.05 (m, 3H), 7.50 (m, 2H), 6.75 (s, 1H), 5.26 (m, 1H), 3.85 (s, 3H), 3.0 (m, 2H), 2.29 (s, 3H), 2.0 (m, 5H), 1.46 (m, 10H). Example 177 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid ((R)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide 04 Chiral separation 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 579 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.28 (d, J = 1.9 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.10 (dd, J = 8.1, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.35 (d, J = 8.2 Hz, 1H), 5.24 (t, J = 7.0 Hz, 1H), 4.52 (td, J = 11.1, 5.4 Hz, 1H), 4.08-3.93 (m, 2H), 3.51 (dd, J = 12.8, 10.7 Hz, 2H), 2.84 (s, 4H), 2.15-1.89 (m, 5H), 1.84 (d, J = 12.7 Hz, 3H), 1.48 (s, 3H), 1.21-1.13 (m, 2H), 1.01-0.91 (m, 2H). Example 178 3-(1- Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid ((S)- 6-{2-[5-methyl-1- (tetrahydro- pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1- yl)- amide 05 Chiral separation 3-(1-Methyl- cyclopropyl)- [1,2,4]oxadiazole-5- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 579 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.28 (d, J = 1.9 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.10 (dd, J = 8.1, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.35 (d, J = 8.2 Hz, 1H), 5.24 (t, J = 7.0 Hz, 1H), 4.52 (td, J = 11.1, 5.4 Hz, 1H), 4.08-3.93 (m, 2H), 3.51 (dd, J =

12.8, 10.7 Hz, 2H), 2.84 (s, 4H), 2.15-1.89 (m, 5H), 1.84 (d, J = 12.7 Hz, 3H), 1.48 (s, 3H), 1.21-1.13 (m, 2H), 1.01-0.91 (m, 2H). Example 179 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid ((R)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide 06  Chiral separation 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 581 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.30 (d, J = 8.9 Hz, 1H), 8.28 (t, J = 1.8 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.11 (dd, J = 8.3, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.28 (d, J = 4.6 Hz, 1H), 4.53 (td, J = 11.4, 5.7 Hz, 1H), 4.07-3.93 (m, 3H), 3.51 (dd, J = 12.9, 10.7 Hz, 2H), 2.84 (s, 4H), 2.07 (qd, J = 11.9, 11.5, 4.2 Hz, 4H), 1.88-1.76 (m, 3H), 1.42 (s, 8H), 1.16 (q, J = 7.3 Hz, 1H). Example 180 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid ((S)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide 07  Chiral separation 5-tert-Butyl- [1,2,4]oxadiazole-3- carboxylic acid (6-{2- [5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 581 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.20 (s, 1H), 9.30 (d, J = 8.9 Hz, 1H), 8.28 (t, J = 1.8 Hz, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.17 (s, 1H), 8.11 (dd, J = 8.3, 1.9 Hz, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.28 (d, J = 4.6 Hz, 1H), 4.53 (td, J = 11.4, 5.7 Hz, 1H), 4.07-3.93 (m, 3H), 3.51 (dd, J = 12.9, 10.7 Hz, 2H), 2.84 (s, 4H), 2.07 (qd, J = 11.9, 11.5, 4.2 Hz, 4H), 1.88-1.76 (m, 3H), 1.42 (s, 8H), 1.16 (q, J = 7.3 Hz, 1H). Example 181 3-tert-Butoxy- azetidine-1- carboxylic acid ((R)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide 08  Chiral separation 3-tert-Butoxy- azetidine-1- carboxylic acid (6-{2-[5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5- b]pyridin- 7-yl}-1,2,3,4- tetrahydro-naphthalen- 1-yl)-amide LCMS: 584 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.19 (s, 1H), 8.27-8.22 (m, 2H), 8.17 (s, 1H), 8.08 (dd, J = 8.2, 1.9 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.90-4.80 (m, 1H), 4.59-4.40 (m, 2H), 4.00 (td, J = 12.1, 10.4, 6.2 Hz, 3H), 3.59 (dt, J = 8.7, 4.5 Hz, 2H), 3.51 (dd, J = 12.9, 10.8 Hz, 2H), 2.84 (s, 4H), 2.14-2.01 (m, 2H), 1.94 (q, J = 5.3 Hz, 2H), 1.84 (d, J = 12.8 Hz, 2H), 1.74 (q, J = 11.4, 10.2 Hz, 2H), 1.12 (s, 7H), 1.08-1.01 (m, 1H). Example 182 3-tert-Butoxy- azetidine-1- carboxylic acid ((S)- 6-{2-[5-methyl-1- (tetrahydro-pyran-4- yl)-1H-pyrazol-4-yl]- 3H-imidazo[4,5- b]pyridin-7-yl}- 1,2,3,4-tetrahydro- naphthalen-1-yl)- amide 09  Chiral separation 3-tert-Butoxy- azetidine-1-carboxylic acid (6-{2-[5-methyl-1- (tetrahydro-pyran-4-yl)- 1H-pyrazol-4-yl]-3H- imidazo[4,5-b]pyridin- 7-yl}-1,2,3,4- tetrahydro- naphthalen- 1-yl)-amide LCMS: 584 (M + H). .sup.1H NMR (400 MHz, DMSO-d6) δ 13.19 (s, 1H), 8.27-8.22 (m, 2H), 8.17 (s, 1H), 8.08 (dd, J = 8.2, 1.9 Hz, 1H), 7.46 (d, J = 5.3 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 6.68 (d, J = 8.8 Hz, 1H), 4.90-4.80 (m, 1H), 4.59-4.40 (m, 2H), 4.00 (td, J = 12.1, 10.4, 6.2 Hz, 3H), 3.59 (dt, J = 8.7, 4.5 Hz, 2H), 3.51 (dd, J = 12.9, 10.8 Hz, 2H), 2.84 (s, 4H), 2.14-2.01 (m, 2H), 1.94 (q, J = 5.3 Hz, 2H), 1.84 (d, J = 12.8 Hz, 2H), 1.74 (q, J = 11.4, 10.2 Hz, 2H), 1.12 (s, 7H), 1.08-1.01 (m, 1H). Example 183 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {(R)- 2-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9-tetrahydro- 5H- benzocyclohepten-5- yl}-amide 0  Chiral separation 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-5-methyl- 1H-pyrazol-4-yl)-3H- imidazo[4,5-b]pyridin- 7-yl]-6,7,8,9- tetrahydro-5H- benzocyclohepten-5- yl}-amide LC/MS: (M + H) 553. HPLC: 4.35 min. .sup.1H NMR (400 MHz, DMSO-d6) δ 13.10 (br.s, 1H), 10.01 (m, 1H), 8.0-8.4 (m, 4H), 7.50 (m, 2H), 5.31 (m, 1H), 4.55 (m, 1H), 3.0 (m, 2H), 2.77 (s, 3H), 2.0 (m, 5H), 3H), 1.48 (m, 16H) Example 184 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {(S)- 2-[2-(1-isopropyl-5- methyl-1H-pyrazol-4- yl)-3H-imidazo[4,5- b]pyridin-7-yl]- 6,7,8,9- tetrahydro- 5H- benzocyclohepten-5- yl}-amide  Chiral separation 3-tert-Butyl- [1,2,4]oxadiazole-5- carboxylic acid {2-[2- (1-isopropyl-5-methyl- 1H-pyrazol-4-yl)-3H-

imidazo[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-1-methylcyclohepten-yl}-amide LC/MS: (M + H) 553. HPLC: 4.35 min. ¹H NMR (400 MHz, DMSO-d₆) δ 13.10 (br.s, 1H), 10.01 (m, 1H), 8.0-8.4 (m, 4H), 7.50 (m, 2H), 5.31 (m, 1H), 4.55 (m, 1H), 3.0 (m, 2H), 2.77 (s, 3H), 2.0 (m, 5H), 3H), 1.48 (m, 16H) Example 185 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid ((R)-6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide  Chiral separation 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid (6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide LCMS: 579 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.21 (s, 1H), 9.25 (d, J = 8.8 Hz, 1H), 8.27 (s, 2H), 8.17 (s, 1H), 8.11 (d, J = 8.1 Hz, 1H), 7.48 (s, 1H), 7.33 (d, J = 8.5 Hz, 1H), 5.27 (s, 1H), 4.53 (s, 1H), 3.99 (d, J = 11.7 Hz, 2H), 3.50 (d, J = 12.6 Hz, 2H), 3.30 (s, 3H, overlaps with H₂O peak), 2.85 (s, 4H), 2.15-1.73 (m, 7H), 1.54 (s, 2H), 1.38 (s, 2H), 1.16 (s, 2H). Example 186 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid ((S)-6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide  Chiral separation 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid (6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide LCMS: 579 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.21 (s, 1H), 9.25 (d, J = 8.8 Hz, 1H), 8.27 (s, 2H), 8.17 (s, 1H), 8.11 (d, J = 8.1 Hz, 1H), 7.48 (s, 1H), 7.33 (d, J = 8.5 Hz, 1H), 5.27 (s, 1H), 4.53 (s, 1H), 3.99 (d, J = 11.7 Hz, 2H), 3.50 (d, J = 12.6 Hz, 2H), 3.30 (s, 3H, overlaps with H₂O peak), 2.85 (s, 4H), 2.15-1.73 (m, 7H), 1.54 (s, 2H), 1.38 (s, 2H), 1.16 (s, 2H). Example 187 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  Chiral separation 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide LCMS: 539 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.31 (d, J = 8.8 Hz, 1H), 8.28 (s, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.16 (s, 1H), 8.12 (dd, J = 8.1, 1.9 Hz, 1H), 7.48 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.29 (d, J = 5.7 Hz, 1H), 4.64 (p, J = 6.6 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 3H), 2.07-1.89 (m, 3H), 1.83 (s, 1H), 1.42 (d, J = 6.9 Hz, 15H). Example 188 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  Chiral separation 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide LCMS: 539 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.31 (d, J = 8.8 Hz, 1H), 8.28 (s, 1H), 8.25 (d, J = 5.2 Hz, 1H), 8.16 (s, 1H), 8.12 (dd, J = 8.1, 1.9 Hz, 1H), 7.48 (d, J = 5.2 Hz, 1H), 7.34 (d, J = 8.2 Hz, 1H), 5.29 (d, J = 5.7 Hz, 1H), 4.64 (p, J = 6.6 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 3H), 2.07-1.89 (m, 3H), 1.83 (s, 1H), 1.42 (d, J = 6.9 Hz, 15H). Example 189 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(R)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  Chiral separation 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide LCMS: 537 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.31-8.21 (m, 2H), 8.16 (s, 1H), 8.14-8.10 (m, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 5.25 (s, 1H), 4.64 (p, J = 6.5 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 2H), 2.07-1.89 (m, 3H), 1.82 (s, 1H), 1.48 (s, 3H), 1.42 (d, J = 6.5 Hz, 5H), 1.18 (q, J = 3.7 Hz, 2H), 0.97 (q, J = 4.0 Hz, 2H). Example 190 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(S)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide  Chiral separation 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo[4,5-b]pyridin-7-yl]-1,2,3,4-

tetrahydro-naphthalen-1-yl}-amide LCMS: 537 (M + H). ¹H NMR (400 MHz, DMSO-d₆) δ 13.18 (s, 1H), 9.67 (d, J = 8.7 Hz, 1H), 8.31-8.21 (m, 2H), 8.16 (s, 1H), 8.14-8.10 (m, 1H), 7.47 (d, J = 5.2 Hz, 1H), 7.36 (d, J = 8.2 Hz, 1H), 5.25 (s, 1H), 4.64 (p, J = 6.5 Hz, 1H), 2.88 (s, 2H), 2.81 (s, 2H), 2.07-1.89 (m, 3H), 1.82 (s, 1H), 1.48 (s, 3H), 1.42 (d, J = 6.5 Hz, 5H), 1.18 (q, J = 3.7 Hz, 2H), 0.97 (q, J = 4.0 Hz, 2H).

Assay 1: BTK IC₅₀ Enzyme Assay

(365) The following describes a microfluidic, off-chip mobility shift kinase assay used to measure inherent potency of compounds against BTK enzyme. Compounds described by embodiments of the present invention were assayed using this protocol and the data from the same is recorded in Table 2 within the column labeled: "Time Dependent BTK Enzyme Assay IC₅₀". 2.5× stocks of full-length human BTK (08-080) from CarmaBio USA, Inc., Natick, MA, 1.6×ATP and appropriate kinKDR peptide substrate (FITC-AHA-EEPLYWSFPAAKKK-NH₂) were prepared in kinase reaction buffer consisting of 25 mM MgCl₂, 0.015% Brij-35 (30%), 100 mM Hepes, pH 7.5, and 10 mM DTT. 5 uL of enzyme buffer and 7.5 uL of ATP/kinKDR peptide substrate mix were added to Matrix (#115304) 384-well, sterile, polypropylene plates (Thermo Fisher Scientific, Hudson, N.H.) with 125 nL of serially diluted compounds prepared in 100% DMSO, and incubated for 90 min. at 27 C. Following the incubation period, reactions were stopped by adding 60 uL stop buffer consisting of 100 mM Hepes, pH 7.5, 0.015% Brij-35 (30%), 0.277% Coating Reagent #3 (Caliper Life Sciences, Mountain View, CA), 5% DMSO. Stopped reactions were monitored at -2 PSI, -3000 V/-700 V in a LabChip 3000 plate reader from Caliper Life Sciences, a PerkinElmer Company (Hopkinton, MA), and the activity was measured by off-chip mobility shift assay measuring the charge/mass difference between substrate and product resulting from peptide phosphorylation. IC₅₀ and efficacy were determined by plotting log [Inhibitor] vs. % Activity in GeneData Screener (Basel, Switzerland).

(366) Assay 2: PBMC IC₅₀ Cell Assay

(367) Btk is critical for mediating the signalling of B cell antigen receptor (BCR) after anti-IgM stimulation. Based on this principle, a functional cell-based assay was established to determine the potency of compounds at inhibiting anti-IgM-induced expression of CD69, a downstream BCR signaling event, in freshly isolated human peripheral blood mononuclear cells (PBMCs). In the assay, a 90 µl PBMC suspension containing 2.5×10⁵ cells was pre-treated with 10 µl of test compound at various concentrations for an hour, and then incubated overnight (approximately 16-18 hours) with 5 µl 420 µg/ml affiniPure F(ab')₂ fragment goat anti-human IgM Fc fragment per well (Dianova, Cat. No.: 109-006-129). After the incubation, the cells were washed and immunostained with an APC-labeled mouse anti-human CD69 (BD Biosciences; clone: FN50), a PerCP-Cy5.5 labelled mouse anti-human CD19 (BD Biosciences; clone: SJ25C1) and a FITC-labelled mouse anti-human CD3 (BD Biosciences; clone: HIT3a), and fixed for flow cytometric analysis of CD69 expression on CD19 positive cells (B cells). The percentage of CD69 expressing CD19 positive cells was plotted against the concentrations of test compounds to obtain a concentration response curve, and calculated an IC₅₀ value as a measure of the potency of test compounds in the assay.

(368) Assay 3: Time Dependent Human Whole Blood IC₅₀ Assay

(369) Compounds described by embodiments of the present invention were assayed using a human whole blood assay. The data from the same is recorded below within the column labeled: "Time Dependent Human Whole Blood BTK Enzyme Assay IC₅₀".

Example 191. Biological Data

(370) Assay 1: Btk Enzymatic IC₅₀ Assay 2: Btk PBMC IC₅₀ Assay 3: Human Whole Blood IC₅₀ Assay

(371) Results are given in the following table.

(372) TABLE-US-00004 Assay A Assay B Assay C Example (IC₅₀) (IC₅₀) (IC₅₀)
82 A A — 78 A B B 79 A B B 153 A B B 37 A B — 121 A B C 85 A B — 20 A B — 75 A B — 12

A B B 24 A B B 52 A B — 89 A B B 155 A B C 10 A B — 49 A B — 79 A B C 30 A B — 38 A B
 — 92 A B C 51 A B — 42 A B — 46 A B — 16 A B — 45 A B — 40 A B — 163 A B — 33 A B B
 83 A B B 74 A B C 119 A B C 123 A B B 149 A B C 8 A B — 160 A B B 54 A B — 61 A B — 86 A
 B C 41 A B — 115 A B B 23 A B — 39 A A — 21 A B — 32 A B — 84 A B — 11 A B C 43 A B —
 56 A B B 97 A — — 124 A C C 131 A B C 143 A B C 1 A B C 3 A B B 27 A B — 127 A C — 142
 A B B 72 A B C 91 A B B 63 A B — 109 A A — 107 A C C 165 A B C 14 A C — 119 A B — 87 A
 B — 59 A B — 106 A B — 151 A B — 167 A C B 2 A B B 108 A C C 110 A B — 157 A B B 6 A B
 — 29 A B — 150 A B B 133 A B B 137 A C B 120 A B B 118 A B B 148 A C C 69 A B — 58 A B
 — 100 A B — 96 A B B 135 A B B 67 A C — 114 A B C 105 A B — 162 A B — 35 A B — 134 A
 B B 140 A C D 19 A B — 144 A B C 115 A B D 136 A B C 145 A C — 117 A B — 147 A B — 156
 A B — 139 B B — 138 B B D 125 B B D 100 B B — 113 B B C 146 B B — 101 B B — 126 B B
 — 111 B B D 116 B B C 26 B B C 102 B C — 124 B B C 164 B B D 95 B B — 73 B B — 98 B —
 — 154 B C — 158 B B — 132 B B B 112 B C — 70 B B C 152 B B — 17 B C — 166 B C — 68
 B C — 47 B C — 104 B C — 44 B C — 22 B C — 62 B C — 159 B D — 81 B C C 31 B C — 25
 B C — 34 B D — 71 B C — 90 B C — 77 B C — 53 B C — 64 B C — 122 B D — 60 B C — 93
 B — — 13 B C — 36 B D — 80 B D — 57 B D — 48 B D — 55 B C — 88 B D — 15 B D — 50
 B D — 161 B D — 141 C D — 9 D — — 18 D D — 28 D D — 99 D — — A: IC₅₀ < 1 nM B:
 IC₅₀ = 1 nM-10 nM C: IC₅₀ = 10 nM-100 nM D: IC₅₀ > 100 nM

Example 192. Pharmaceutical Preparations

(373) (A) Injection vials: A solution of 100 g of an active ingredient according to the invention and 5 g of disodium hydrogen phosphate in 3 l of bidistilled water is adjusted to pH 6.5 using 2 N hydrochloric acid, sterile filtered, transferred into injection vials, is lyophilized under sterile conditions and is sealed under sterile conditions. Each injection vial contains 5 mg of active ingredient.

(374) (B) Suppositories: A mixture of 20 g of an active ingredient according to the invention is melted with 100 g of soy lecithin and 1400 g of cocoa butter, is poured into moulds and is allowed to cool. Each suppository contains 20 mg of active ingredient.

(375) (C) Solution: A solution is prepared from 1 g of an active ingredient according to the invention, 9.38 g of NaH.sub.2PO.sub.4.2H.sub.2O, 28.48 g of Na.sub.2HPO.sub.4.12H.sub.2O and 0.1 g of benzalkonium chloride in 940 ml of bidistilled water. The pH is adjusted to 6.8, and the solution is made up to 1 l and sterilized by irradiation. This solution could be used in the form of eye drops.

(376) (D) Ointment: 500 mg of an active ingredient according to the invention is mixed with 99.5 g of Vaseline under aseptic conditions.

(377) (E) Tablets: A mixture of 1 kg of an active ingredient according to the invention, 4 kg of lactose, 1.2 kg of potato starch, 0.2 kg of talc and 0.1 kg of magnesium stearate is pressed to give tablets in a conventional manner in such a way that each tablet contains 10 mg of active ingredient.

(378) (F) Coated tablets: Tablets are pressed analogously to Example E and subsequently are coated in a conventional manner with a coating of sucrose, potato starch, talc, tragacanth and dye.

(379) (G) Capsules: 2 kg of an active ingredient according to the invention are introduced into hard gelatin capsules in a conventional manner in such a way that each capsule contains 20 mg of the active ingredient.

(380) (H) Ampoules: A solution of 1 kg of an active ingredient according to the invention in 60 l of bidistilled water is sterile filtered, transferred into ampoules, is lyophilized under sterile conditions and is sealed under sterile conditions. Each ampoule contains 10 mg of active ingredient.

(381) (I) Inhalation spray: 14 g of an active ingredient according to the invention are dissolved in 10 l of isotonic NaCl solution, and the solution is transferred into commercially available spray containers with a pump mechanism. The solution could be sprayed into the mouth or nose. One spray shot (about 0.1 ml) corresponds to a dose of about 0.14 mg.

(382) While a number of embodiments of this invention are described herein, it is apparent that the

basic examples may be altered to provide other embodiments that utilize the compounds and methods of this invention. Therefore, it will be appreciated that the scope of this invention is to be defined by the appended claims rather than by the specific embodiments that have been represented by way of example.

Claims

1. A compound of formula I, ##STR00418## or a tautomer, or a pharmaceutically acceptable salt thereof, wherein: X is C(R).sub.2 or O; Y is —O—, —S—, —SO.sub.2—, —SO—, —C(O)—, —CO.sub.2—, or —NHC(O); R.sup.1 is C.sub.6-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted; Ring Z is C.sub.6-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, a 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted; each R.sup.a is independently —R, halogen, —OR, —SR, —CN, —NO.sub.2, —SO.sub.2R, —SOR, —C(O)R, —CO.sub.2R, —C(O)N(R).sub.2, —NRC(O)R, —NRC(O)N(R).sub.2, —NRSO.sub.2R, or —N(R).sub.2; or two R.sup.a groups, together with the atoms to which each is attached, forms a C.sub.6-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, a 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted; each R is independently hydrogen, C.sub.1-6 aliphatic, C.sub.6-10 aryl, a 3-8 membered saturated or partially unsaturated carbocyclic ring, a 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted; or two R groups on the same atom are taken together with the atom to which they are attached to form, a 3-8 membered saturated or partially unsaturated carbocyclic ring, a 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur, a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; a 10-15 membered fused aryl ring; a 6-15 membered saturated or partially unsaturated fused carbocyclic ring; a 10-15 membered fused heteroaryl ring; or a 6-15 membered saturated or partially unsaturated fused heterocyclic ring; each of which is optionally substituted; n is 0, 1, or 2; and p is 0, 1, 2, or 3.

2. The compound of claim 1, wherein X is CH.sub.2.

3. The compound of claim 1, wherein X is O.

4. The compound of claim 1, wherein R.sup.1 is an optionally substituted 3-7 membered heterocyclic ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or R.sup.1 is an optionally substituted 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

5. The compound of claim 4, wherein R.sup.1 is azocinyl, 2H,6H-1,5,2-dithiazinyl, dihydrofuro [2,3-b]tetrahydrofuran, furanyl, furazanyl, imidazolidinyl, imidazolinyl, imidazolyl, isothiazolyl, isoxazolyl, morpholinyl, oxadiazolyl, 1,2,3-oxadiazolyl, 1,2,4-oxadiazolyl; 1,2,5-oxadiazolyl, 1,3,4-oxadiazolyl, oxazolidinyl, oxazolyl, oxazolidinyl, pyrimidinyl, piperazinyl, piperidinyl, piperidine-one, pyranyl, pyrazinyl, pyrazolidinyl, pyrazolinyl, pyrazolyl, pyridazinyl, pyridinyl, pyridyl, pyrimidinyl, pyrrolidinyl, pyrrolinyl, 2H-pyrrolyl, pyrrolyl, tetrahydrofuranyl, tetrahydropyridine, tetrahydropyran, 6H-1,2,5-thiadiazinyl, 1,2,3-thiadiazolyl, 1,2,4-thiadiazolyl, 1,2,5-thiadiazolyl,

1,3,4-thiadiazolyl, thiazolyl, thienyl, triazinyl, 1,2,3-triazolyl, 1,2,4-triazolyl, 1,2,5-triazolyl, 1,3,4-triazolyl, oxetanyl, or azetidiny; each of which is optionally substituted.

6. The compound of claim 5, wherein R_{sup}.1 is selected from: ##STR00419##

7. The compound of claim 1, wherein Ring Z is a 3-8 membered saturated or partially unsaturated carbocyclic ring or a 5-6 membered monocyclic heteroaryl ring having 1-4 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is optionally substituted.

8. The compound of claim 7, wherein Ring Z is cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, imidazolyl, pyrazolyl, triazolyl, oxazolyl, isoxazolyl, oxadiazolyl, thiazolyl, or isothiazolyl; each of which is optionally substituted.

9. The compound of claim 8, wherein Ring Z is ##STR00420##

10. The compound of claim 9, wherein Ring Z is ##STR00421## ##STR00422##

11. The compound of claim 1, of formula II, ##STR00423## or a tautomer, or a pharmaceutically acceptable salt thereof.

12. The compound of claim 1, of formula III, ##STR00424## or a tautomer, or a pharmaceutically acceptable salt thereof.

13. The compound of claim 1, of formula IV, ##STR00425## or a tautomer, or a pharmaceutically acceptable salt thereof.

14. The compound of claim 1, selected from ##STR00426## ##STR00427## ##STR00428## ##STR00429## ##STR00430## ##STR00431## ##STR00432## ##STR00433## ##STR00434## ##STR00435## ##STR00436## ##STR00437## ##STR00438## ##STR00439## ##STR00440## ##STR00441## ##STR00442## ##STR00443## ##STR00444## ##STR00445## ##STR00446## ##STR00447## ##STR00448## ##STR00449## ##STR00450## ##STR00451## ##STR00452## ##STR00453## ##STR00454## ##STR00455## ##STR00456## ##STR00457## ##STR00458## ##STR00459## or a pharmaceutically acceptable salt thereof.

15. The compound of claim 1, wherein the compound is one of the following compounds: 3-tert-butyl-N-[(5R)-2-{2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo [4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2,4-oxadiazole-5-carboxamide (82); 5-tert-butyl-N-[(5R)-2-{2-[3-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo [4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,3,4-oxadiazole-2-carboxamide (78); 5-tert-butyl-N-[(5R)-2-{2-[5-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-1H-imidazo [4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,3,4-oxadiazole-2-carboxamide (79); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {(R)-2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (153); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (37); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {(R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (121); 3-tert-butyl-N-{2-[2-(1-ethyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl}-1,2,4-oxadiazole-5-carboxamide (85); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (20); 3-tert-butyl-N-[(1R)-6-[2-(1-cyclopropyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5-carboxamide (75); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (12); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (24); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (52); 5-tert-butyl-N-[(5S)-2-[2-(5-ethyl-1-methyl-1H-pyrazol-3-yl)-1H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,3,4-

oxadiazole-2-carboxamide (89); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid ((S)-6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (155); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (10); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {(S)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (49); 5-tert-butyl-N-[(5R)-2-{2-[5-methyl-1-(oxan-4-yl)-1H-pyrazol-4-yl]-1H-imidazo [4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,3,4-oxadiazole-2-carboxamide (79); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (30); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (38); 5-tert-butyl-N-[(5R)-2-{2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2-oxazole-3-carboxamide (92); 3-tert-Butoxy-azetidine-1-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (51); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (42); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (46); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-8-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (16); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(S)-8-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (45); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (40); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid ((S)-6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (163); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (33); 3-tert-butyl-N-{2-[2-(1-cyclopropyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl}-1,2,4-oxadiazole-5-carboxamide (83); 3-tert-butyl-N-[(1R)-6-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5-carboxamide (74); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(S)-6-[2-(1-isopropyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (119); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (123); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (149); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (8); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (160); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (54); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (61); 5-tert-butyl-N-(2-{2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl)-1,3,4-oxadiazole-2-carboxamide (86); 5-(1-Methyl-cyclopropyl)-isoxazole-3-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo

[4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (41); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid ((S)-6-{2-[3-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (115); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (23); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (39); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(R)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (21); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (32); 5-tert-butyl-N-(2-{2-[5-methyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl)-1,3,4-oxadiazole-2-carboxamide (84); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (11); 3-tert-Butoxy-azetidine-1-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (43); 5-(1-Methyl-cyclopropyl)-isoxazole-3-carboxylic acid {(R)-2-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (56); 3-tert-butyl-N-{2-[2-(1-ethyl-5-methyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl}-1,2,4-oxadiazole-5-carboxamide (97); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (124); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {2-[2-(5-ethyl-1-methyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (131); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (143); 3-tert-butyl-N-[(5R)-2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2,4-oxadiazole-5-carboxamide (1); 5-tert-butyl-N-[2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,3,4-oxadiazole-2-carboxamide (3); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (27); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid (6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (127); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (142); 5-tert-butyl-N-[(5R)-2-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-1H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2-oxazole-3-carboxamide (72); 5-tert-butyl-N-(2-{2-[1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl)-1,2-oxazole-3-carboxamide (91); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (63); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-methyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (109); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {6-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (107); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {(S)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (165); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {8-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (14); 3-tert-Butyl-

[1,2,4]oxadiazole-5-carboxylic acid {(S)-6-[2-(1-isopropyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (119); 5-tert-butyl-N-[(5R)-2-{2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,3,4-oxadiazole-2-carboxamide (87); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(R)-2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (59); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (106); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid (6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (151); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {(S)-6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (167); N-[(5S)-2-[2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-3-(1-methylcyclopropyl)-1,2,4-oxadiazole-5-carboxamide (2); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {6-[2-(5-ethyl-1-methyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (108); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1,5-dimethyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (110); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid ((S)-6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (157); 3-tert-butyl-N-[2-[2-(trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzo[7]annulen-5-yl]-1,2,4-oxadiazole-5-carboxamide (6); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (29); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid (6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (150); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(5-ethyl-1-methyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (133); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid ((R)-6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (137); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid (2-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl)-amide (120); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (118); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid (6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (148); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {1-[4-(4,4,5,5-tetramethyl-[1,3,2]dioxaborolan-2-yl)-phenyl]-cyclopropyl}-amide (69); 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid {2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (58); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-6-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (100); 3-tert-butyl-N-[(1R)-6-{2-[3,5-dimethyl-1-(propan-2-yl)-1H-pyrazol-4-yl]-1H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydronaphthalen-1-yl]-1,2,4-oxadiazole-5-carboxamide (96); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-ethyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (135); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {6-[2-(1,3-dimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (67); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {(R)-6-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (114); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {6-[2-(1-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-

7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (105); 5-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-3-carboxylic acid ((R)-6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (162); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (35); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-3-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (134); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid (6-{2-[3,5-dimethyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (140); 5-tert-Butyl-[1,3,4]oxadiazole-2-carboxylic acid {(S)-8-[2-(1,3,5-trimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-2,3,4,5-tetrahydro-benzo[b]oxepin-5-yl}-amide (19); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-3,5-dimethyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (144); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid ((S)-6-{2-[3-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (115); 3-tert-Butyl-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (136); 3-tert-Butoxy-azetidine-1-carboxylic acid {6-[2-(1-isopropyl-5-methyl-1H-pyrazol-4-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-1,2,3,4-tetrahydro-naphthalen-1-yl}-amide (145); 3-(1-Methyl-cyclopropyl)-[1,2,4]oxadiazole-5-carboxylic acid {2-[2-(1-isopropyl-5-methyl-1H-pyrazol-3-yl)-3H-imidazo [4,5-b]pyridin-7-yl]-6,7,8,9-tetrahydro-5H-benzocyclohepten-5-yl}-amide (117); 3-tert-Butoxy-azetidine-1-carboxylic acid (6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (147) or 5-tert-Butyl-[1,2,4]oxadiazole-3-carboxylic acid ((R)-6-{2-[5-methyl-1-(tetrahydro-pyran-4-yl)-1H-pyrazol-4-yl]-3H-imidazo [4,5-b]pyridin-7-yl}-1,2,3,4-tetrahydro-naphthalen-1-yl)-amide (156) or a pharmaceutically acceptable salt thereof.

16. A pharmaceutical composition comprising a compound of claim 1, and a pharmaceutically acceptable adjuvant, carrier, or vehicle.

17. A method for inhibiting BTK, or a mutant thereof, activity in a patient or in a biological sample, comprising the step of administering to said patient or contacting said biological sample with a compound of claim 1, or a physiologically acceptable salt thereof.

18. A method for treating a BTK-mediated disorder in a patient in need thereof, comprising the step of administering to said patient a compound of claim 1.

19. The method of claim 18, wherein the disorder is selected from inflammatory pelvic disease, urethritis, skin sunburn, sinusitis, pneumonitis, encephalitis, meningitis, myocarditis, nephritis, osteomyelitis, myositis, hepatitis, gastritis, enteritis, dermatitis, gingivitis, appendicitis, pancreatitis, cholecystitis, agammaglobulinemia, psoriasis, allergy, Crohn's disease, irritable bowel syndrome, ulcerative colitis, Sjogren's disease, tissue graft rejection, hyperacute rejection of transplanted organs, asthma, allergic rhinitis, chronic obstructive pulmonary disease (COPD), autoimmune polyglandular disease, autoimmune alopecia, pernicious anemia, glomerulonephritis, dermatomyositis, multiple sclerosis, scleroderma, vasculitis, autoimmune hemolytic and thrombocytopenia states, Goodpasture's syndrome, atherosclerosis, Addison's disease, Parkinson's disease, Alzheimer's disease, diabetes, septic shock, systemic lupus erythematosus (SLE), rheumatoid arthritis, psoriatic arthritis, juvenile arthritis, osteoarthritis, chronic idiopathic thrombocytopenia purpura, Waldenstrom macroglobulinemia, myasthenia gravis, Hashimoto's thyroiditis, atopic dermatitis, degenerative joint disease, vitiligo, autoimmune hypopituitarism, Guillain-Barre syndrome, Behcet's disease, scleroderma, mycosis fungoides, acute inflammatory responses, acute respiratory distress syndrome, ischemia, reperfusion injury, and Graves' disease.

20. A method for treating lupus in a subject, comprising the step of administering to said subject a compound of claim 1, or a physiologically acceptable salt thereof.

